Bayes is All You Need: Better Estimates of True Sharpe Ratios

Antoine Jacquier[1], Juha Joenväärä[2], Johannes Muhle-Karbe[1], and Joseph Mulligan[1]

[1]Department of Mathematics, Imperial College London
[2]Department of Finance, Aalto University

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Abstract

We propose a Bayesian framework for estimating the true Sharpe ratio of investments. This allows one to explicitly account for two key drivers of the disparity between historical and future performance: selection bias and alpha decay. By modelling the joint distribution of in-sample and out-of-sample Sharpe ratios, we are able to disentangle these effects and correct for the inflation present in historical track records. In a simulation study, we find that fitting the model to in-sample data alone yields poor parameter estimates, whereas a joint fitting approach successfully recovers the underlying parameters. A case study using a comprehensive dataset of hedge fund returns illustrates the substantial quantitative importance of these effects for real data. We find that our method yields accurate estimates of post-listing performance that significantly help to improve fund selection. In particular, we show that a bias-aware approach is essential for making robust allocation decisions, specifically in the post-2008 regime where selection effects and alpha decay have become more pronounced.

1 Introduction

One of the fundamental questions in finance is how to allocate capital. Investors seek to select investments that are likely to perform well in the future, and often make this choice based on attractive historical performance. The issue is that this historical performance is often not representative of future returns: the wedge between in-sample and out-of-sample performance is often dramatic, cf. Figure 1. As the disclaimer on every fund prospectus advises: "Past performance does not guarantee future results" [Sec02]. But why is this the case? And what can we do to correct for this? In the present study, we propose a Bayesian method which provides a principled solution to this problem, capturing two of the main effects that cause this disparity between historical and future performance: "selection bias" and "alpha decay". In particular, we show that a method that unifies both effects is necessary to make accurate inferences about future returns.

"Selection bias" is a general phenomenon that takes a number of forms. One prominent instance is the multiple testing problem: researchers test many hypotheses before selecting the significant ones, cf., e.g., [Bai+14; BL14; HL14; HL15; HLZ16; HLS20; HL20; HL21a; LB21; GLX21]. Another well-known example is the backfill bias present in hedge fund databases: funds opportunistically list their historical returns in commercial databases when they have performed well, causing their pre-listing returns to be inflated relative to post-listing performance, cf., e.g., [FH00; Par95; BGP99; Pv03; FH09; BGR14; JS19].

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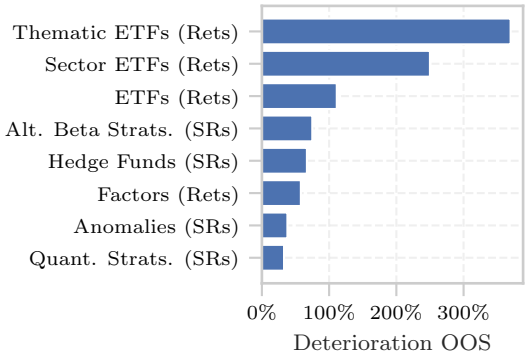


Figure 1: Performance deterioration between in-sample and out-of-sample performance in different domains. Sources are: Thematic ETFs & Sector ETFs [Ben+23] (estimated), ETFs [BL15], Alternative Beta Strategies [SLP17], Hedge Funds Appendix C.2, Factors [MP16], Anomalies Appendix E and Quantitative Strategies [CV23].

The second effect, "alpha decay", is the tendency for investments to underperform out-of-sample beyond the reduction attributable to selection bias alone. [MP16] study this by comparing returns generated between the end of the in-sample period and the publication date against post-publication returns. This decay is often significant. For example, using the open-source asset pricing dataset [CZ22], we find pre-publication Sharpe ratios are 11% lower than in-sample figures, while post-publication Sharpe ratios are 38% lower (Appendix E). Similar dynamics appear in the fund literature where increased scale (assets under management) erodes alpha [BGS22]. In practitioner research, crowding in popular trades has also been linked to performance degradation [MAN19].

In the present study, we provide a unified framework to disentangle and quantify both of these effects. This is non-trivial, as a decline in out-of-sample performance can a priori be attributed to either. To overcome this, we use an empirical Bayesian approach that explicitly accounts for selection bias and alpha decay during the fitting step. Specifically, we view the observed Sharpe ratios as the outcome of a selection mechanism: each investment has a true Sharpe ratio SR, with an in-sample estimate $\widehat{\mathrm{SR}}_{\mathrm{IS}}$, and is included in the dataset with some probability that increases in $\widehat{\mathrm{SR}}_{\mathrm{IS}}$. Conditional on inclusion, we then observe out-of-sample performance from a possibly degraded (alpha-decayed) true Sharpe ratio. Modelling this process allows us to back out the unobserved distribution of the investments that failed the filter, and thereby correct for the inflation in historical track records. Once the model has been fit, it can then be used to perform inference on either the dataset used to fit it, or new candidate investments. The resulting posterior means are then corrected for in-sample measurement error not only through standard Bayesian shrinkage to the prior, but also by explicitly controlling for selection bias.

This methodology has broad applicability across the investment industry. It can aid fund-of-funds and pension funds in selecting managers for capital allocation, assist multi-strategy firms in deciding which managers to launch, retain, or terminate, and support quantitative researchers in evaluating which strategies deserve continued inclusion in their portfolios.

While conceptually sound, this approach presents thorny numerical challenges that make it impractical to implement using naive approaches. To wit, the lack of closed-form expressions necessitates computing complex nested integrals across potentially thousands of sample Sharpe ratios. We overcome this bottleneck using a custom JIT-compiled implementation, rendering the required computations feasible for both fitting and inference across large datasets.

To demonstrate the practical applicability of our approach, we test it on a large dataset of hedge fund returns [Joe+21], covering six major commercial hedge fund databases. We find that the method yields significantly more accurate estimates of individual hedge funds' post-listing performance compared to a number of benchmarks (e.g., a 40% reduction in MSE relative to taking the pre-listing performance as an estimate for future performance) and far better calibration of

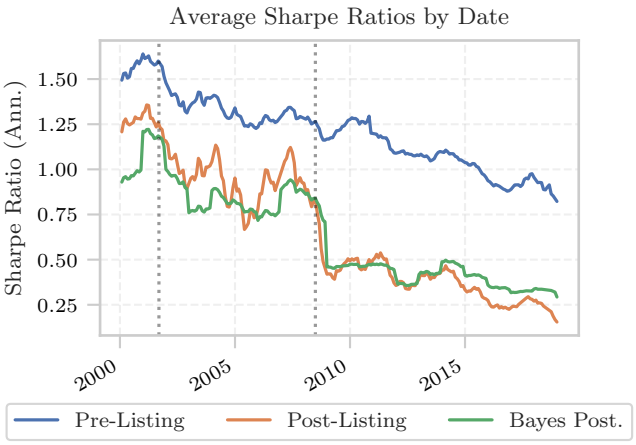


Figure 2: Average pre- and post-listing Sharpe ratios over time, with a 15-year rolling lookback, and the average Bayesian posterior mean for each date.

predictive distributions (the probability integral transforms are statistically indistinguishable from Uniform$(0, 1)$ under Kolmogorov–Smirnov tests).

We further demonstrate the model's utility in fund selection, showing that in periods where selection bias and alpha decay are prevalent, such as post-2008, a bias aware approach is essential for making well-calibrated investment decisions. In Figure 2 we see how significant the issue has become since the 2008 financial crisis, and how well the Bayesian estimated average posterior means captures this dynamic, closely tracking the average post-listing performance.

Bayesian approaches have a rich history in the finance literature [FS86; Jor86; BMW01; JKP23]. In the fund literature, they have been applied by [PS02] to estimate mutual fund alphas by incorporating information from non-benchmark passive assets, a method [KNT07] extend to hedge funds to overcome short return histories. Empirical Bayesian approaches in particular have previously been used by [HL18] in the estimation of fund alphas. More recently, they have also been applied in the estimation of publication bias in the asset pricing literature [CZ20; HL21b].[1]

In our model, we find in a simulation study that fitting to in-sample data alone yields poor parameter estimates, leading to incorrect assessments of selection bias. We address this by fitting in- and out-of-sample performance jointly, producing substantially more accurate estimates. However, the performance decline out-of-sample is not solely attributable to selection bias. If the model does not account for alpha decay, it is forced to ascribe the entire performance drop to selection bias, resulting in poor fits and unrealistic parameters. Incorporating alpha decay not only resolves this issue, but also yields insights into the sources of performance degradation and their relative importance. For example, in the hedge fund data studied in Section 3, the level of alpha decay has increased from a relatively moderate level (87% of in-sample true performance maintained out-of-sample) to quite severe (55%, in the post 2008 period).

Although we apply our method to commercial hedge fund data here, the approach has natural applications to many areas of finance, such as quantitative strategy development. The classical challenge in this domain is overfitting. Although studied from various perspectives [BL14; HLS20; CJS25; MJM25], existing frameworks often require additional inputs, such as how the strategy was constructed or an effective number of trials/tests, which are frequently unavailable in practice.

Our Bayesian approach avoids relying on these inputs, which are often unavailable, in the spirit of [LL19; PLZ26]. Instead, we parameterise a reduced-form selection mechanism and fit the model to a dataset of reasonably comparable previously launched strategies with known in- and out-of-

[1] A primary advantage of Bayesian frameworks is that, under specific conditions, they are 'immune to selection bias' [Daw94; Yek12]. However, this immunity relies on the prior's parameters being estimated accurately. If the prior is misspecified, for example, through an incorrect location or variance, any resulting inference will still be compromised. For further discussion on this, see Section 2.1 and Appendix A.

sample performance. In this way, the model can encode the estimated degree of overfitting in the parameters of the fitted prior. For example, one could fit the model to a dataset of already launched mid-frequency equity strategies and use it to assess new, similar candidate strategies.

The remainder of the paper is structured as follows: We provide a brief description of the preliminaries, followed by the model specification and a guide to application, before presenting the results of applying the model to our dataset of hedge fund returns. The appendix presents significant extensions, robustness checks, and details on the supplementary and exploratory steps required for the empirical study, alongside illustrative examples of how parameter variation affects inference and an additional application of the method to the open-source asset pricing dataset [CZ22].

2 Bayesian Model: Theory & Implementation

Our goal is to accurately estimate true Sharpe ratios, corrected for selection bias and alpha decay. In this section we set out the necessary prerequisites and describe both the model we propose to achieve this goal, and how to apply it. This methodology is applicable to a wide variety of capital allocation tasks, ranging from sovereign wealth and pension funds allocating to hedge funds, multi-strategy firms allocating to portfolio managers, or quantitative traders choosing among algorithmic trading strategies. To illustrate our framework, we use a commercially available large dataset of hedge fund returns. For clarity and parsimony, the main text develops a single, illustrative version of our Bayesian model tailored to this particular application. However, the framework is highly flexible and easily applied to a wide variety of investments; we discuss several extensions and alternative specifications in Appendix A, which we draw particular attention to for quantitative researchers wishing to model the effect of parameter tuning from a Bayesian selection perspective.

2.1 Preliminaries

Types of Sharpe Ratios

As discussed above, in-sample (historical) performance is often inflated relative to later out-of-sample (future) performance. But, in both the in- and out-of-sample periods, the performance is a noisy measurement of the investment's true, underlying performance. We therefore distinguish between three types of Sharpe ratio:

1. **The True Sharpe Ratio** SR: The true, unobserved, performance of an investment,
2. **The In-Sample Sharpe Ratio** $\widehat{\text{SR}}_{\text{IS}}$: A noisy *and* biased measurement of the true performance of an investment,
3. **The Out-of-Sample Sharpe Ratio** $\widehat{\text{SR}}_{\text{OOS}}$: A noisy but unbiased measurement of the true performance of an investment.

In theory, as the sample sizes grow, which we will denote by T_{IS} and T_{OOS} respectively, the observed Sharpe ratios $\widehat{\text{SR}}_{\text{IS}}$ and $\widehat{\text{SR}}_{\text{OOS}}$ should converge to the true Sharpe ratio SR. In practice however, the number of useful observations is typically constrained. The amount of in-sample returns for an investment is typically fixed, and out-of-sample returns are only observed gradually over time. If our goal is to estimate the true performance of an investment we must therefore have a robust method to combine the biased in-sample and unbiased out-of-sample performance measurements in a consistent manner. This is exactly what our method provides.

Types of Selection Bias

To understand how selection bias affects inference of statistics it is best to consider a clear example, such as the hedge fund dataset we use in Section 3. Selection bias arises in this dataset through funds entering commercial databases according to decisions that are correlated with in-sample performance. Think of a selection filter operating on a fund's in-sample track record: the fund has an unobserved true Sharpe ratio SR, with a corresponding in-sample estimate $\widehat{\text{SR}}_{\text{IS}}$, and is observed

in the dataset only if it passes a performance-dependent selection rule. This is defined as *random* selection bias by Yekutieli [Yek12]. In the random case, we analyse a cross-section of samples that have already passed the selection filter, so both the distribution of the latent parameter SR and the observed statistic $\widehat{\text{SR}}_{\text{IS}}$ are affected by selection. Intuitively, low-SR samples are under-represented because they are less likely to produce high $\widehat{\text{SR}}_{\text{IS}}$ and therefore less likely to be observed.[2]

2.2 Model Specification

We have two goals: to fit our model to a dataset of reasonably homogeneous investments, and then to perform inference for new individual investments. The first requires fitting the joint marginal likelihood of the in-sample and out-of-sample Sharpe ratios; the second, the posterior for the true Sharpe ratio. The ingredients for both are the same: a prior, a likelihood function, and an acceptance function.

Prior

The prior is the distribution we assume all true Sharpe ratios are drawn from, both for the observed samples in our dataset, and the unobserved samples which were not selected. We take a location-scale t prior,

$$\pi(\text{SR}) \sim lst\left(\text{SR}_0, \varsigma_0^2 \frac{\nu-2}{\nu}, \nu\right),$$

which we have parameterised such that $\mathbb{E}[\text{SR}] = \text{SR}_0$ and $\mathbb{V}[\text{SR}] = \varsigma_0^2$, with $\nu > 2$ controlling the kurtosis of the prior. We choose this prior as it strikes a good balance between flexibility – allowing us to fit to the observed data well – and relatively few parameters to fit. In particular, allowing for excess kurtosis is vital, as we observe that a few funds exhibit significantly higher Sharpe ratios than the rest. In contrast, the empirical distribution of out-of-sample Sharpe ratios is rather symmetric, and therefore we do not use an even more flexible distribution for parsimony.

Likelihood

The likelihood of the sample Sharpe ratios is determined by the assumption on the distribution of the underlying returns. When the returns are iid Gaussian, then the Sharpe ratio for a sample of T_{IS} observations follows a scaled non-central t-distribution with non-centrality parameter $T_{\text{IS}}\text{SR}$ and $T_{\text{IS}} - 1$ degrees of freedom, $T_{\text{IS}}\widehat{\text{SR}}_{\text{IS}} \sim t_{T_{\text{IS}}-1}(T_{\text{IS}}\text{SR})$, see [Pav21].

Computing non-central t-densities is slow in implementations, though. To speed up computations, we use a Normal approximation for the sampling distribution $f\left(\widehat{\text{SR}}_{\text{IS}} \,\middle|\, \text{SR}\right)$. The sampling distribution of the Sharpe ratio was first derived by Lo [Lo02], and later extended to cover cases where the returns have skewness and kurtosis [Mer02], or autocorrelation [Pav21], or both simultaneously [PLZ26], which is the form used here:

$$\sqrt{T_{\text{IS}}}\left(\widehat{\text{SR}}_{\text{IS}} - \text{SR}\right) \xrightarrow{d} \mathcal{N}\left(0, \frac{1+\phi}{1-\phi} - \frac{1+\phi+\phi^2}{1-\phi^2}\gamma_{3,r}\text{SR} + \frac{1+\phi^2}{1-\phi^2}\frac{\gamma_{4,r}-1}{4}\text{SR}^2\right).$$

Here, $\xrightarrow{d}$ denotes convergence in distribution as the sample size grows ($T_{\text{IS}} \to \infty$), ϕ is an AR(1) autocorrelation parameter, $\gamma_{3,r} = \mathbb{E}\left[\left(\frac{r_t-\mu}{\sigma}\right)^3\right]$ is the skewness of the underlying returns, and $\gamma_{4,r} = \mathbb{E}\left[\left(\frac{r_t-\mu}{\sigma}\right)^4\right]$ is the raw kurtosis of the underlying returns (not the excess kurtosis, $\gamma_{4,r} - 3$).

[2] Two related cases are useful for intuition. In *fixed* selection bias, a single fund's SR is treated as fixed and only the observed statistic $\widehat{\text{SR}}_{\text{IS}}$ is conditioned on selection, which generally requires larger posterior corrections. In *hierarchical* selection bias, selection occurs after choosing among correlated variants (e.g. parameter-tuned strategies within a family), which is particularly relevant for quantitative strategy development. We treat both cases and provide examples to aid intuition in Appendix A.2.

This likelihood is suitably flexible to allow us to consider common departures from iid Gaussian returns, but a simpler likelihood can be recovered by setting $\phi = 0, \gamma_{3,r} = 0, \gamma_{4,r} = 3$. Note that in our notation for the likelihood $f\left(\widehat{\mathrm{SR}}_{\mathrm{IS}} \,\middle|\, \mathrm{SR}\right)$, we suppress the dependence on T_{IS} (or T_{OOS}) for clarity.

Acceptance Probability

To model selection, we introduce an *acceptance function* $p_{\mathrm{acc}}(\widehat{\mathrm{SR}}_{\mathrm{IS}})$: the probability that an investment appears in our observed dataset given its in-sample Sharpe ratio. In the hedge fund setting, this captures the reduced-form effect of managers choosing whether (and when) to report returns conditional on their in-sample performance. In other applications, p_{acc} can represent publication decisions, internal capital allocation, or any performance-based gating mechanism.

Following Chen and Zimmermann [CZ20], we use a logistic specification,

$$p_{\mathrm{acc}}\left(\widehat{\mathrm{SR}}_{\mathrm{IS}}\right) = \frac{1}{1 + \exp\left(-\ell\left(\widehat{\mathrm{SR}}_{\mathrm{IS}} - \kappa\right)\right)}. \tag{2.1}$$

The midpoint $\kappa \in \mathbb{R}$ is the in-sample Sharpe ratio at which inclusion probability is 50%. The slope $\ell > 0$ controls how sharp the filter is: small ℓ corresponds to a soft threshold where non-performance considerations matter while large ℓ approximates a hard cutoff driven primarily by $\widehat{\mathrm{SR}}_{\mathrm{IS}}$.

Alpha Decay

Selection bias alone does not fully explain why out-of-sample performance is often materially worse than in-sample performance. We therefore include *alpha decay*, defined here as systematic performance deterioration out-of-sample beyond what would be implied by selection effects and estimation noise alone. We model decay by scaling the true Sharpe ratio in the out-of-sample period:

$$\mathrm{SR}_\delta := \delta\,\mathrm{SR}, \qquad \delta < 1,$$

so that out-of-sample returns are generated as if the investment's true Sharpe ratio has been shrunk toward zero.[3] Including δ is essential for identification. The empirical gap between $\widehat{\mathrm{SR}}_{\mathrm{IS}}$ and $\widehat{\mathrm{SR}}_{\mathrm{OOS}}$ can be explained by either (i) a stronger selection filter (larger inflation in $\widehat{\mathrm{SR}}_{\mathrm{IS}}$) or (ii) genuine decay (lower SR_δ). Without an explicit decay component, the model is forced to attribute the full gap to selection bias, which can lead to implausible selection parameters and poor fits. With δ included, the joint shape of in-sample and out-of-sample performance allows the model to disentangle these two sources of degradation.

2.3 Model Application

Bias Adjustment

Without selection bias, one can simply use the prior and likelihood defined in the previous section for model fitting and inference. But, as previously discussed, we must incorporate these two effects to correctly perform these tasks. Specifically, we must adjust the prior and likelihood to correct for the effect of the selection filter. These are computed by adapting the results of Yekutieli [Yek12] from hard to soft selection thresholds. The results somewhat differ depending on the kind of selection bias considered, and we collect the relevant results for all cases in Appendix A.2. For

[3] While we considered an additive form, $\mathrm{SR} - \delta$, the multiplicative form consistently provides a superior fit to observed performance patterns.

random selection bias, the bias adjusted prior $\pi_B(\mathrm{SR})$ and likelihood $f_B\left(\widehat{\mathrm{SR}}_{\mathrm{IS}} \,\middle|\, \mathrm{SR}\right)$ are given by

$$\pi_B(\mathrm{SR}) \coloneqq \frac{\mathbb{P}(\mathrm{acc} \mid \mathrm{SR})\pi(\mathrm{SR})}{\mathbb{P}\left(\mathrm{acc}\right)}, \qquad f_B\left(\widehat{\mathrm{SR}}_{\mathrm{IS}} \,\middle|\, \mathrm{SR}\right) \coloneqq \frac{p_{\mathrm{acc}}\left(\widehat{\mathrm{SR}}_{\mathrm{IS}}\right) f\left(\widehat{\mathrm{SR}}_{\mathrm{IS}} \,\middle|\, \mathrm{SR}\right)}{\mathbb{P}(\mathrm{acc} \mid \mathrm{SR})},$$

where $p_{\mathrm{acc}}(\widehat{\mathrm{SR}}_{\mathrm{IS}})$ is the pointwise acceptance probability evaluated at an observed in-sample Sharpe ratio $\widehat{\mathrm{SR}}_{\mathrm{IS}}$, while $\mathbb{P}(\mathrm{acc} \mid \mathrm{SR})$ and $\mathbb{P}(\mathrm{acc})$ are (normalising) selection probabilities obtained by integrating this acceptance function over the relevant distributions:

$$\mathbb{P}(\mathrm{acc} \mid \mathrm{SR}) \coloneqq \int_{\widehat{\mathrm{SR}}'_{\mathrm{IS}}} p_{\mathrm{acc}}\left(\widehat{\mathrm{SR}}'_{\mathrm{IS}}\right) f\left(\widehat{\mathrm{SR}}'_{\mathrm{IS}} \,\middle|\, \mathrm{SR}\right) \mathrm{d}\widehat{\mathrm{SR}}'_{\mathrm{IS}}, \tag{2.2}$$

$$\mathbb{P}\left(\mathrm{acc}\right) \coloneqq \int_{\mathrm{SR}'} \pi(\mathrm{SR}')\mathbb{P}\left(\mathrm{acc} \,\middle|\, \mathrm{SR}'\right) \mathrm{d}\mathrm{SR}'. \tag{2.3}$$

The bias adjusted prior $\pi_B(\mathrm{SR})$ is the unbiased prior $\pi(\mathrm{SR})$, adjusted by the probability that a fund with true Sharpe ratio SR is selected, $\mathbb{P}(\mathrm{acc} \mid \mathrm{SR})$, and renormalised by the total marginal probability of selection $\mathbb{P}(\mathrm{acc})$. Here, $\mathbb{P}(\mathrm{acc} \mid \mathrm{SR})$ is the expected acceptance probability for a fixed true Sharpe ratio, computed by integrating the acceptance function $p_{\mathrm{acc}}(\cdot)$ over the sampling distribution of $\widehat{\mathrm{SR}}_{\mathrm{IS}}$ given SR. The marginal probability $\mathbb{P}(\mathrm{acc})$ averages $\mathbb{P}(\mathrm{acc} \mid \mathrm{SR})$ over the unbiased prior $\pi(\mathrm{SR})$, ensuring π_B integrates to one. In general these normalising constants are not available in closed form under flexible priors and likelihoods, so they must be computed numerically; we discuss efficient and stable approximations in Appendix B.2.

Model Fitting

We fit the Bayesian model to the observed cross-section of pre- and post-listing Sharpe ratios using maximum likelihood estimation (i.e., empirical Bayes [Rob56]). This requires fitting the prior parameters, $\mathrm{SR}_0, \varsigma_0, \nu$, the acceptance function parameters, κ, ℓ, and the alpha decay parameter δ. These collectively form the parameter set $\boldsymbol{\Theta} = \{\mathrm{SR}_0, \varsigma_0, \nu, \kappa, \ell, \delta\}$.

To this end, we first compute the joint marginal likelihood of the in- and out-of-sample Sharpe ratios, $f(\widehat{\mathrm{SR}}_{\mathrm{IS}}, \widehat{\mathrm{SR}}_{\mathrm{OOS}})$. This is obtained by integrating the product of the bias-adjusted in-sample likelihood, the bias-adjusted prior, and the out-of-sample likelihood over the support of the true Sharpe ratio SR'. Note that the out-of-sample likelihood is conditional on the alpha-decayed parameter SR'_δ:

$$f\left(\widehat{\mathrm{SR}}_{\mathrm{IS}}, \widehat{\mathrm{SR}}_{\mathrm{OOS}}\right) = \int_{\mathrm{SR}'} f_B\left(\widehat{\mathrm{SR}}_{\mathrm{IS}} \,\middle|\, \mathrm{SR}'\right) f\left(\widehat{\mathrm{SR}}_{\mathrm{OOS}} \,\middle|\, \mathrm{SR}'_\delta\right) \pi_B\left(\mathrm{SR}'\right) \mathrm{d}\mathrm{SR}'. \tag{2.4}$$

Note that we must use the bias adjusted in-sample likelihood, $f_B\left(\widehat{\mathrm{SR}}_{\mathrm{IS}} \,\middle|\, \mathrm{SR}\right)$, and prior, $\pi_B\left(\mathrm{SR}\right)$. The out-of-sample Sharpe ratio however is unbiased, and we therefore use the standard likelihood but with the alpha-decayed true Sharpe ratio SR_δ rather than the standard true Sharpe ratio SR. While one could try to fit the model using only in-sample data, we find that this method often fails to recover the true parameters accurately when tested on samples simulated from a known data generating process. In particular, the alpha decay parameter δ is impossible to identify without the joint fitting approach. In Appendix B.1 we provide simulation results with known parameters demonstrating the need to fit with both in- and out-of-sample data, and how it significantly improves parameter estimates.

Next, we maximise the joint likelihood (2.4) across all pairs of in- and out-of-sample Sharpe ratios. Assuming the pairs are independent across investments, that is, $(\widehat{\mathrm{SR}}_{\mathrm{IS},i}, \widehat{\mathrm{SR}}_{\mathrm{OOS},i})$ and $(\widehat{\mathrm{SR}}_{\mathrm{IS},j}, \widehat{\mathrm{SR}}_{\mathrm{OOS},j})$ are independent for $i \neq j$, we can equivalently maximise the sum of the log-likelihoods,

$$\max_{\boldsymbol{\Theta}} \sum_i \log f\left(\widehat{\mathrm{SR}}_{\mathrm{IS},i}, \widehat{\mathrm{SR}}_{\mathrm{OOS},i} \,\middle|\, \boldsymbol{\Theta}\right).$$

The independence assumption is strong and unlikely to hold for raw returns, as they typically consist of both systematic exposure (β) and idiosyncratic performance (α). However, when comparing investments such as hedge funds, we typically wish to identify the hedge funds which deliver α rather than β, as there are much cheaper ways to buy β exposure. We therefore focus on the idiosyncratic component of the returns, which we find is sufficiently independent for empirical Bayes to yield useful parameter estimates. We study the effect of violating this assumption in Appendix B.4.

In practice, this fitting procedure is challenging due to a complex loss surface and the numerical cost of calculating the marginal likelihood (2.4). We detail how we overcome these challenges and make it feasible to apply the method in Appendix B.2. The key lies in the usage of an optimised, compiled, and GPU-enabled log likelihood calculation (using `JAX` [FJL19]), and the robust fitting methods provided by CERN's `Minuit2` library [JR75; Dem+25].

The MLE approach is extremely efficient, and the best choice when speed is required (for example, when fitting many models for walkforward analysis). But when the selection effect is relatively weak (for example $\ell < 5$), the loss surface becomes flat and a pure MLE approach tends to select extreme "non-physical" κ values. To resolve this, we suggest a hybrid approach, utilising Generalised Method of Moments to fit a subset of parameters (typically κ and δ, which are well identified by the sample moments), with the remainder being fit by MLE. This method acts as a form of regularisation, constraining the parameters to intuitive regions. Full details of this two-step approach, which we term MLE-GMM, are given in Appendix B.3. Note that a full GMM approach is inappropriate for the task at hand as some of the parameters, for example ℓ, are not well identified by the sample moments.

Inference

Once the model has been fit to the observed distribution of in- and out-of-sample Sharpe ratios, it can then be used to compute the posterior for each investment's true Sharpe ratio in a standard manner. To do so, we simply apply Bayes rule and use the bias adjusted prior and likelihood,

$$\pi_B\left(\mathrm{SR} \;\middle|\; \widehat{\mathrm{SR}}_{\mathrm{IS}}\right) = \frac{\pi_B(\mathrm{SR}) f_B\left(\widehat{\mathrm{SR}}_{\mathrm{IS}} \;\middle|\; \mathrm{SR}\right)}{\int_{\mathrm{SR}'} \pi_B(\mathrm{SR}') f_B\left(\widehat{\mathrm{SR}}_{\mathrm{IS}} \;\middle|\; \mathrm{SR}'\right) \mathrm{dSR}'}. \tag{2.5}$$

In the case of random selection bias, we observe that the bias adjustments to the prior and likelihood "cancel out", yielding the same posterior as if there was no selection bias,

$$\begin{aligned}\pi_B\left(\mathrm{SR} \;\middle|\; \widehat{\mathrm{SR}}_{\mathrm{IS}}\right) &= \frac{\pi_B(\mathrm{SR}) f_B\left(\widehat{\mathrm{SR}}_{\mathrm{IS}} \;\middle|\; \mathrm{SR}\right)}{\int_{\mathrm{SR}'} \pi_B(\mathrm{SR}') f_B\left(\widehat{\mathrm{SR}}_{\mathrm{IS}} \;\middle|\; \mathrm{SR}'\right) \mathrm{dSR}'} \\ &= \frac{\pi(\mathrm{SR}) f\left(\widehat{\mathrm{SR}}_{\mathrm{IS}} \;\middle|\; \mathrm{SR}\right)}{\int_{\mathrm{SR}'} \pi(\mathrm{SR}') f\left(\widehat{\mathrm{SR}}_{\mathrm{IS}} \;\middle|\; \mathrm{SR}'\right) \mathrm{dSR}'} = \pi\left(\mathrm{SR} \;\middle|\; \widehat{\mathrm{SR}}_{\mathrm{IS}}\right).\end{aligned}$$

Intuitively, this says that if you know the correct prior $\pi(\mathrm{SR})$, then you can perform Bayesian inference on a biased in-sample Sharpe ratio $\widehat{\mathrm{SR}}_{\mathrm{IS}}$ as if there was no selection bias (i.e., using the unadjusted likelihood $f(\widehat{\mathrm{SR}}_{\mathrm{IS}} \mid \mathrm{SR})$). But this is only true if you have the correct prior $\pi(\mathrm{SR})$. By using the bias adjusted prior $\pi_B(\mathrm{SR})$ and likelihood $f_B(\widehat{\mathrm{SR}}_{\mathrm{IS}} \mid \mathrm{SR})$ when fitting the model via MLE, we have been able to estimate the parameters of the unbiased prior $\pi(\mathrm{SR})$.

If we had naively estimated the parameters of the prior, the Bayesian approach would fail to correct for the selection bias as we would have misestimated the parameters of the prior $\pi(\mathrm{SR})$, likely leading to estimating an overly optimistic prior mean SR_0.[4]

[4] In the other (fixed and hierarchical) cases, the posterior requires further adjustment, as in Appendix A.2.

To compute the posterior for the alpha-decayed Sharpe ratio, i.e., the true Sharpe ratio in the out-of-sample period, we apply a simple change-of-variables transformation,

$$\pi\Big(\mathrm{SR}_\delta \;\Big|\; \widehat{\mathrm{SR}}_{\mathrm{IS}}\Big) = \frac{1}{\delta}\pi\Big(\mathrm{SR} \;\Big|\; \widehat{\mathrm{SR}}_{\mathrm{IS}}\Big).$$

We can also use the out-of-sample data to further update the posterior via Bayesian continuous updating. We treat the posterior derived from the in-sample data, $\pi_B\Big(\mathrm{SR} \;\Big|\; \widehat{\mathrm{SR}}_{\mathrm{IS}}\Big)$, as the "prior" for this step. The updated posterior is then proportional to the product of this and the out-of-sample likelihood:

$$\pi_B\Big(\mathrm{SR} \;\Big|\; \widehat{\mathrm{SR}}_{\mathrm{IS}}, \widehat{\mathrm{SR}}_{\mathrm{OOS}}\Big) \propto f\Big(\widehat{\mathrm{SR}}_{\mathrm{OOS}} \;\Big|\; \mathrm{SR}_\delta\Big)\pi_B\Big(\mathrm{SR} \;\Big|\; \widehat{\mathrm{SR}}_{\mathrm{IS}}\Big).$$

This allows us to use all the data available, both the biased in-sample returns and the unbiased out-of-sample returns, to estimate the true performance of a given investment. This is very useful in practice. For example, we can compute the initial posterior once for a fund when it enters the dataset using its backfilled returns, and then continuously update it as new returns arrive, refining our estimate of the true Sharpe ratio for that fund. Updating in this fashion forms a key application we explore later, using evolving estimates of the true Sharpe ratio to decide whether a fund should remain in our portfolio as we observe out-of-sample returns.

In addition to updating the posterior with out-of-sample data, we may also want to check if the out-of-sample data is in line with our expectations given the posterior computed on the in-sample data. This is done by comparing the observed out-of-sample Sharpe ratio to the posterior predictive distribution computed using the posterior from the in-sample Sharpe ratio alone. Since the in-sample and out-of-sample return series are disjoint, $\widehat{\mathrm{SR}}_{\mathrm{IS}}$ and $\widehat{\mathrm{SR}}_{\mathrm{OOS}}$ are conditionally independent given SR, thus we have

$$f\Big(\widehat{\mathrm{SR}}_{\mathrm{OOS}} \;\Big|\; \widehat{\mathrm{SR}}_{\mathrm{IS}}\Big) = \int_{\mathrm{SR}'} f\Big(\widehat{\mathrm{SR}}_{\mathrm{OOS}} \;\Big|\; \mathrm{SR}'_\delta\Big)\pi\Big(\mathrm{SR}'_\delta \;\Big|\; \widehat{\mathrm{SR}}_{\mathrm{IS}}\Big)\,\mathrm{dSR}'_\delta. \tag{2.6}$$

Here it is important to note that we must use the alpha decayed posterior to account for the possible alpha decay in the out-of-sample period. This is another application we explore later, where we can detect if a fund is performing even worse (or better than) in the out-of-sample period than we would expect given our bias adjusted estimate for their true Sharpe ratio using their in-sample returns.[5]

3 Empirical Case Study: Hedge Fund Returns

With the model fully described, we turn now to an empirical application of our approach. To do so, we utilise a dataset of hedge fund returns collected by Joenväärä et al. [Joe+21] by merging six major commercial hedge fund databases (BarclayHedge, EurekaHedge, Hedge Fund Management, Hedge Fund Research, Lipper TASS and Morningstar) into a single consistent dataset from January 1990 to December 2023. The dataset is only survivorship bias-free from 1994 onwards, so we restrict ourselves to data after this date.

The dataset consists of monthly returns net of fees, monthly assets under management and a collection of characteristics for each fund, such as style, domicile, use of leverage, etc. Importantly, we have a "listing date" for each fund, either directly given by the database or implied using the method from Jorion and Schwarz [JS19]. Using this listing date, we can split the returns into pre- and post-listing returns, which correspond to in- and out-of-sample returns in our terminology.

As the hedge fund literature documents (e.g., Bhardwaj, Gorton, and Rouwenhorst [BGR14]; Fung and Hsieh [FH09]), pre-listing returns are often inflated relative to post-listing performance. One driver is the incentive for successful funds to list and attract capital, see Appendix C.7 for

[5] This is also useful for example for answering questions such as: What is the probability that this fund will do worse out-of-sample than in-sample? We can compute this simply with the conditional CDF of the out-of-sample Sharpe ratio, evaluated at the observed in-sample Sharpe ratio: $F_{\widehat{\mathrm{SR}}_{\mathrm{OOS}}|\widehat{\mathrm{SR}}_{\mathrm{IS}}}(\widehat{\mathrm{SR}}_{\mathrm{IS}})$.

figures showing AUM change relative to listing date. Consequently, backfilled returns tend to exceed non-backfilled ones. Furthermore, funds often undergo an incubation period, deploying the manager's capital to build a track record before listing. Because funds with poor records rarely list, this induces an incubation bias. This phenomenon serves as an ideal case study for our approach.

Setup

To use the hedge fund data for our approach, some preprocessing must be done. This includes filtering the full set of funds into a clean subset, and separating the returns into systematic and non-systematic components. The details of these two steps can be found in Appendix C. The procedure yields a clean dataset of pre- and post-listing returns which have limited systematic risk exposure (as required for the independence assumptions underlying our approach).

We fit the model using the MLE-GMM approach, as described in Appendix B.3, with κ, δ as the profile set. A full discussion of why we select these two parameters for profiling is available in Appendix C.3. For the walkforward analysis, where we wish to minimise fitting time to run the procedure, we utilise the direct MLE approach, as described in Section 2.3.

In addition to studying the usage of our model across the entire cross-section of funds, we investigate how the model parameters vary across different fund types, and how this affects inference, in Appendix C.4. To conduct this analysis we cluster funds according to their style, domicile and other quantitative and qualitative factors, and find that different types of funds exhibit significantly different amounts of selection bias and alpha decay.

We divide the remainder of this section into two parts: a statistical analysis, covering all aspects independent of portfolio construction, and an application of the model to fund selection.

Statistical Analysis

Pre- and Post-listing Returns Over Time

It is well documented that hedge fund performance has broadly reduced over time, see Appendix C.2 and Bollen, Joenväärä, and Kauppila [BJK21]. But is the pre- and post-listing problem also getting worse? We find that yes, it is.

To this end, we split the full time series of returns into three periods according to the regimes identified in Appendix C.2: pre-September 2001, September 2001 to July 2008, and post-July 2008. We restrict the sample for each period to funds launched within each specific window. Without this restriction, funds would lack the pre-listing returns necessary for our method.

In Figure 3 we plot cumulative average returns as a function of months since listing. Before averaging across funds, we rescale returns so that their sample volatility over the pre-listing period equals 1, and then take cross-fund averages. This ensures that high-volatility funds do not dominate the aggregate pattern. We find that, in the post-2008 period, average returns are substantially higher before listing than after listing, as reflected by the noticeably steeper slope of the cumulative return curve in the pre-listing window. In contrast, this pre- versus post-listing difference is much weaker in the two earlier periods.

Bayesian Model Parameters

How does this materialise in the pre- and post-listing Sharpe ratios? In Figure 4 we plot the empirical distributions of the pre- and post-listing Sharpe ratios, along with the fitted marginal densities from our Bayesian model. To fit the model here we have simply taken the observed pre- and post-listing Sharpe ratios for each period (and pre- and post-listing period lengths, see Figure 36 for the distribution of period lengths in each period), and run the MLE-GMM fitting procedure. The fitted parameters are reported in Table 1.

We see that there is a reduction in risk-adjusted performance post-listing in all three periods, with the effect being most pronounced in the post 2008 period, where we observe that the mean annualised Sharpe ratio post-listing is just 0.1 in our sample of funds. We remind the reader that these are the non-systematic returns only, i.e., the performance hedged against an array of commonly used risk factors (see Appendix C.2 for further details).

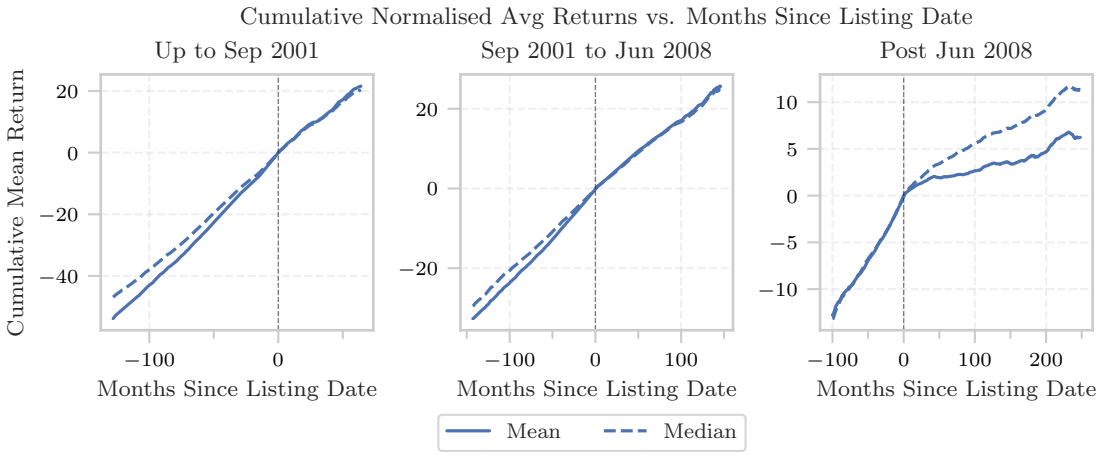


Figure 3: Cumulative (normalised) average returns vs months pre- and post-listing split into the three identified performance periods.

From the fitted parameters, we see that the location of the prior SR_0 is similar in the first two periods, but there is little to no selection bias in the second period ($\kappa \approx 0$), while in the first period the selection effect inflates the observed pre-listing Sharpe ratios. In the most recent period, we see that the prior mean for the true Sharpe ratios is essentially zero, with the positive average performance being driven strongly by selection bias (characterised by a larger ℓ value than in the first two periods). Additionally, we see that the alpha decay has become stronger over time, with δ decreasing from 0.87 to 0.68 and finally 0.55. Finally, we see that the uncertainty in the prior is much larger in the 2001 to 2008 period (large ς_0 and small ν). From this analysis we see that the bias correction has become more and more important over time.

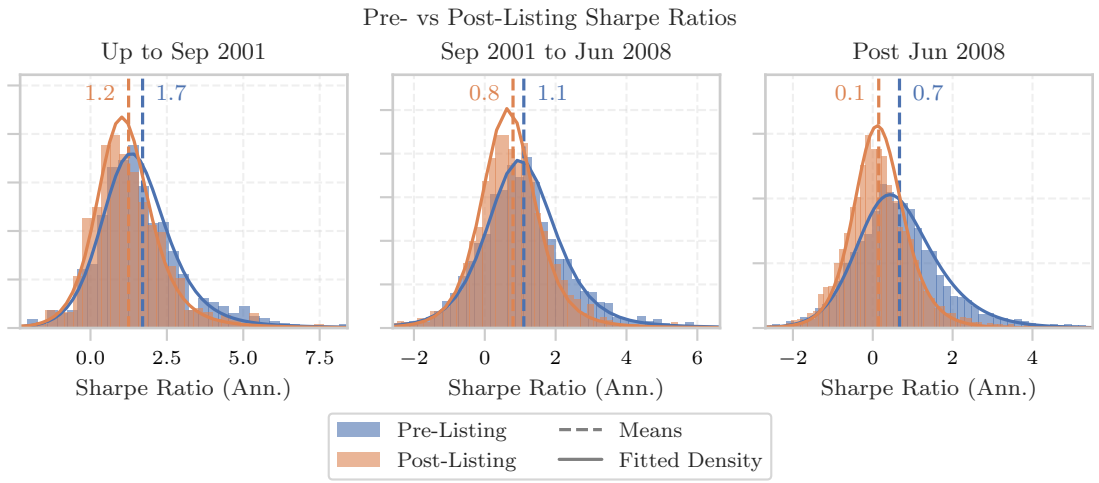


Figure 4: Empirical distribution of pre- and post-listing Sharpe ratios by period. Empirical means are marked with dashed lines and annotated with their value, and the fitted marginal likelihood from the Bayesian model is shown with a solid line.

Decomposition of Selection and Decay Effects

By examining the distribution of the prior $\pi(\text{SR})$, bias adjusted prior $\pi_B(\text{SR})$, and bias & decay adjusted prior $\pi_B(\text{SR}_\delta)$ (calculated using the expressions in Section 2.3), we can see how the selec-

	Up to Sep 2001	Sep 2001 to July 2008	Post July 2008
SR_0	0.30 (1.04)	0.27 (0.94)	-0.01 (-0.03)
ς_0	0.30 (1.04)	1.79 (6.20)	0.21 (0.73)
ν	3.26	2.01	5.21
κ	2.33 (8.07)	-0.07 (-0.24)	1.19 (4.12)
ℓ	1.24	0.86	2.50
δ	0.87	0.68	0.55

Table 1: Fitted parameters for the Bayesian model by period. Annualised equivalents are added in (brackets) for SR_0, ς_0 and κ.

tion and decay effects influence the distribution of true Sharpe ratios, which helps us understand why the post-listing Sharpe ratios are lower than their pre-listing counterparts.

In Figure 5 we plot these densities and in Table 2 we provide their means and modes. In line with the discussion above, the middle period has little difference between the unadjusted and bias adjusted prior, while the first and last period have large adjustments. Additionally, by noting the difference between the mean and modal values for the first and last period, we observe how selection bias introduces skewness into the conditional distribution of true Sharpe ratios. While the underlying unadjusted prior $\pi(SR)$ is symmetric, the population of funds that pass the selection filter is not. As discussed in Section 2.1, the selection filter applies jointly to the in-sample and true parameters, which causes this skewness in the selection adjusted true Sharpe ratios. Funds with low true Sharpe ratios are less likely to have high in-sample Sharpe ratios and are therefore less represented in the observed sample. Finally, we see that the multiplicative alpha decay has a dual effect: by shrinking values towards zero, it affects not only the location of the prior but also its variance. In essence, it enforces a stronger reversion to the mean.

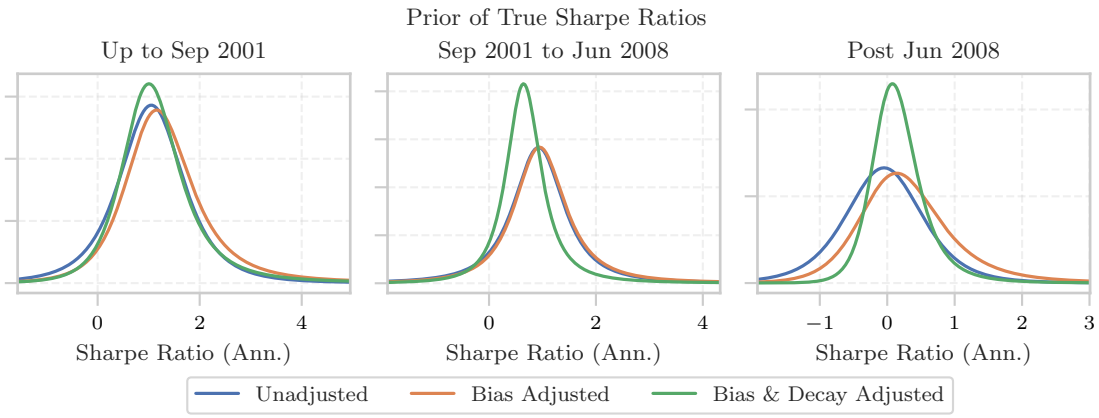


Figure 5: Unadjusted, bias adjusted, and bias & decay adjusted prior densities for the three periods.

	Up to Sep 2001		Sep 2001 to July 2008		Post July 2008	
	Mean	Mode	Mean	Mode	Mean	Mode
Unadjusted	1.06	1.07	0.94	0.94	-0.05	-0.05
Bias Adjusted	1.36	1.16	1.03	0.94	0.32	0.12
Bias & Decay Adj.	1.18	1.00	0.70	0.64	0.18	0.07

Table 2: Unadjusted, bias adjusted and bias & decay adjusted expected true Sharpe ratios $\mathbb{E}[SR]$ for the three periods.

Analysis of Posterior Estimates and Performance Haircuts

Finally, we can look at the primary quantity of interest: the posteriors of the true Sharpe ratio. To compute the posterior we take the parameters for each period from Table 1 and compute the expected posterior Sharpe ratio $\mathbb{E}[\text{SR}_i \mid \widehat{\text{SR}}_{\text{IS},i}]$, and decayed equivalent $\delta\,\mathbb{E}[\text{SR}_i \mid \widehat{\text{SR}}_{\text{IS},i}]$ for each fund which launched in that period. For comparison, we also compute the posterior using both the pre- and post-listing Sharpe ratios, $\mathbb{E}[\text{SR}_i \mid \widehat{\text{SR}}_{\text{IS},i}, \widehat{\text{SR}}_{\text{OOS},i}]$. We are interested in how much shrinkage the Bayesian method applies to the pre-listing Sharpe ratio, and how this is split between Bayesian shrinkage/selection effects and alpha decay.

To visualise this, we can look at the "haircut" estimated by the Bayesian model, i.e., the percentage reduction from the pre-listing performance to our best guess for the true Sharpe ratio. To do so, we compute the ratio of the posterior estimate (undecayed or decayed) to the in-sample Sharpe ratio $\widehat{\text{SR}}_{\text{IS},i}$, and define the "haircut" as one minus this quantity, e.g. $1 - \mathbb{E}[\text{SR}_i \mid \widehat{\text{SR}}_{\text{IS},i}]/\widehat{\text{SR}}_{\text{IS},i}$.

In Figure 6 we plot the median of these haircuts across funds, decomposed into the quantity attributable to Bayesian shrinkage and selection bias[6] and attributable to alpha decay. We observe that in the post 2008 period, the shrinkage has grown significantly in comparison to the earlier two periods. We also note that when comparing the first two periods, alpha decay was the primary cause for the reduction in post-listing performance, indicating that the selection effect was relatively consistent in the two periods, but post-listing alpha decay was more severe in the middle period. Finally, we note that the median haircut is slightly larger when the model is conditioned only on in-sample data $\pi(\text{SR}|\widehat{\text{SR}}_{\text{IS}})$ compared to the full data $\pi(\text{SR}|\widehat{\text{SR}}_{\text{IS}}, \widehat{\text{SR}}_{\text{OOS}})$. As with all Bayesian approaches, the arrival of new evidence (the out-of-sample returns) diminishes the influence of the prior, leading to a posterior that is slightly closer to the observed data and less dominated by the initial shrinkage.

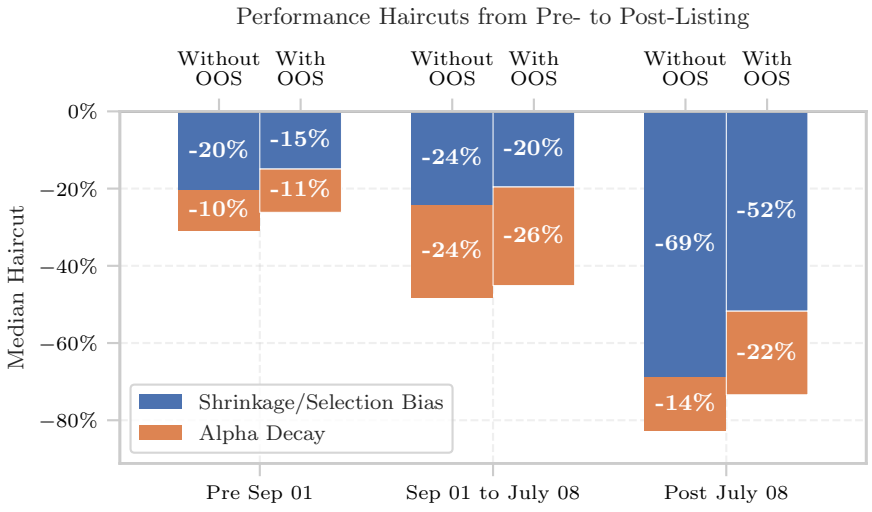


Figure 6: Bayesian estimated haircuts by period, with the posterior estimated either with pre-listing data alone, or with pre- and post-listing data.

Model Evaluation and Comparative Performance

Having established the changing nature of bias and decay over time, we now turn to a formal evaluation of our approach against baseline methods: a simple James–Stein shrinkage estimator [JS60; EM75]; our Bayesian model fit without accounting for selection bias (both with and without alpha

[6] Note that it is difficult to distinguish between Bayesian shrinkage and selection bias purely from the posterior, as the bias adjusted posterior is equal to the unbiased posterior in the random case, see the discussion in Section 2.3. This is why inspecting the prior (which does change with selection bias) can be useful for insights.

decay); and the bias aware model fit to pre-listing Sharpe ratios alone, or fit to pre- and post-listing Sharpe ratios split by the clusters described in Appendix C.4. We provide a more extensive description of these comparators in Appendix C.5.2, and we report the parameters for each model and each period in Table 12. The following analysis serves to validate that our complex Bayesian model provides a strictly superior fit to the data compared to simpler alternatives, a necessary prerequisite before assessing its economic utility in portfolio construction.

We evaluate both point prediction accuracy and model calibration following the recommendations of [Gel+13, Ch. 6 & 7]. Specifically, we compute the mean squared error (MSE) and log pointwise predictive density (LPPD) as measures of predictive accuracy, and probability integral transforms (PITs) together with Kolmogorov–Smirnov (KS) statistics as diagnostics of calibration and model fit. We evaluate these diagnostics relative to the naive baseline (percentage reduction in MSE, increase in MLPPD, and reduction in KS distance), so that moving to the right corresponds to better performance for all metrics. We provide a full description of the calculation of these statistics in Appendix C.5.1.[7]

Figure 7 shows the results, where we have ordered the models (broadly) by their efficacy from worst to best. In the two periods with relatively low selection effects (as shown in the analysis above), the unbiased model with alpha decay performs almost as well as the full bias aware approach. And when looking purely at the point predictive metrics, it continues to perform well in the post 2008 period, with the bias adjusted method providing a modest improvement in performance. However, the Kolmogorov–Smirnov statistic (which measures how closely the model's predicted probabilities line up with realised outcomes) clearly shows that accounting for selection bias is essential for accurately matching the true distribution of out-of-sample Sharpe ratios. This is important if using these probabilities to make investment decisions, as we do in the next section below.

Additionally, we observe that the model fit to the in-sample data alone does not perform as well as the model fit to the in- and out-of-sample data. This is due to two reasons: 1. As discussed in Appendix B.1, fitting to the in- and out-of-sample data yields improved parameter estimates; 2. Capturing alpha decay (which we cannot do when fitting to in-sample data alone) is vital (as demonstrated by the generally good performance of the unbiased model which does have alpha decay). This analysis confirms that:

1. Fitting to in- and out-of-sample data is key for good model fits,
2. Alpha decay is a vital model component,
3. Inclusion of selection bias is required for well calibrated predictive distributions.

Dynamic Fund Selection

We now demonstrate the economic utility of our approach in guiding fund selection. A primary advantage of the Bayesian framework is the generation of full densities, enabling probabilistic decision-making. For instance, an investor might choose to allocate capital only to funds with a 95% probability of a positive Sharpe ratio. To implement this rule, we compute the posterior density for each fund and, similar to [PLZ26], select those satisfying the condition

$$\mathbb{P}(\mathrm{SR}_{\delta,i} \geq 0 \mid \widehat{\mathrm{SR}}_{\mathrm{IS},i}, \widehat{\mathrm{SR}}_{\mathrm{OOS},i}) \geq 0.95.$$

Another useful question we often like to answer is: Is this fund performing as well as expected, given its pre-listing performance? We address this by computing the predictive distribution defined in Equation (2.6) using the posterior computed using only the pre-listing performance,

$$f\left(\widehat{\mathrm{SR}}_{\mathrm{OOS}} \,\middle|\, \widehat{\mathrm{SR}}_{\mathrm{IS}}\right) = \int_{\mathrm{SR}'_\delta} f\left(\widehat{\mathrm{SR}}_{\mathrm{OOS}} \,\middle|\, \mathrm{SR}'_\delta\right) \pi\left(\mathrm{SR}'_\delta \,\middle|\, \widehat{\mathrm{SR}}_{\mathrm{IS}}\right) \mathrm{d}\mathrm{SR}'_\delta,$$

[7] We acknowledge the in-sample bias inherent in using the full dataset to calibrate Θ for this comparative exercise. However, this step is necessary to validate that the model can adequately capture the observed effects in the hedge fund data, which simpler models fail to do even with full information. In Appendix C.6, we conduct a strict walkforward analysis to test predictive utility, though that approach faces its own limitations regarding smaller calibration windows and parameter stability.

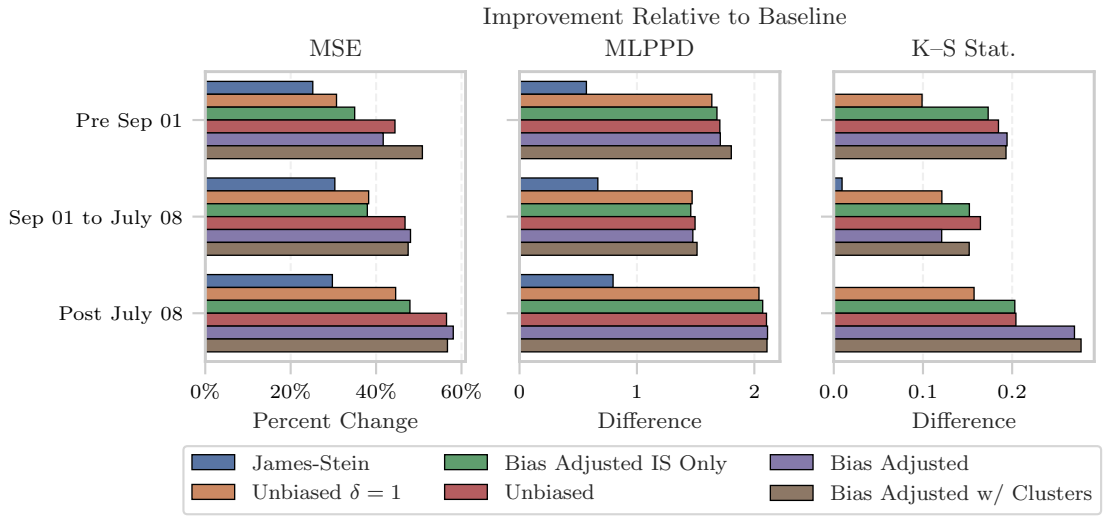


Figure 7: Improvement in statistical metrics relative to a naive baseline for the models described in Appendix C.5.2.

and then compare the observed out-of-sample Sharpe ratio $\widehat{\mathrm{SR}}_{\mathrm{OOS}}$ to this distribution. We then select funds where the quantile of the post-listing performance exceeds some threshold, for example,

$$F\left(\widehat{\mathrm{SR}}_{\mathrm{OOS}} \,\middle|\, \widehat{\mathrm{SR}}_{\mathrm{IS}}\right) \geq a,$$

where $F(\cdot)$ denotes the CDF of the predictive distribution. This serves as a safeguard to filter out funds performing significantly worse than the model's conservative forecast, identifying strategies that may be suffering from a fundamental breakdown.

We examine the profitability of these trading rules by comparing our proposed "Bias Adjusted" estimator (hereafter Bayesian, as defined in Section 2.2) against two benchmarks: the naive and James–Stein [JS60] estimators. We exclude other intermediate variants for clarity. Consistent with our previous analysis, we employ a walkforward approach where parameters are refit annually, strictly using returns observed up to time $t-1$ to compute posteriors at time t. For example, to compute the posterior for January 2002, we use the data up to and including December 2001.

We use the estimated non-systematic returns for both posterior calculation and performance computation. As detailed in Appendix C.2, we estimate the non-systematic returns using the full history for each fund. Similarly, we assume the strategy can invest in any fund, at any size and at any time. While these idealised conditions yield elevated Sharpe ratios, they allow us to isolate our approach's comparative performance against naive benchmarks which also benefit from these conditions. The results demonstrate that the Bayesian model can be successfully fitted, and is applicable to real data exhibiting autocorrelation, heavy tails and other statistical effects. Consequently, one should view the reported Sharpe ratios as metrics of relative predictive power rather than realistic live trading returns.

Having selected the funds for each month, we allocate risk equally so that high volatility funds do not dominate the portfolio; the weight for each fund is

$$\mathbf{w}_{i,t} = \frac{\mathbb{1}_{i \in \mathrm{sel}_t} \sigma_{i,t}^{-1}}{\sum_{j=1}^{N} \mathbb{1}_{j \in \mathrm{sel}_t} \sigma_{j,t}^{-1}},$$

where sel_t denotes the selected funds for month t and $\sigma_{i,t}$ is the volatility estimate for fund i using an expanding window of returns starting from Jan 1994 up to (but not including) month t. We restrict the selection universe to funds live in month t, allocating zero weight to delisted funds.

Consider first the posterior-based selection rule. We parameterise the rule by the Sharpe ratio

cutoff SR_C and the probability threshold p_C, selecting funds where

$$\mathbb{P}(\mathrm{SR}_{\delta,i} \geq \mathrm{SR}_C \mid \widehat{\mathrm{SR}}_{\mathrm{IS},i}, \widehat{\mathrm{SR}}_{\mathrm{OOS},i}) \geq p_C.$$

We run this trading strategy for the Naive, James–Stein[8] and Bayesian approaches, analysing performance for three time variants:

1. From 1996 to 2008, using parameters estimated since 1996,
2. From 2009 to 2020, using parameters estimated since 1996,
3. From 2009 to 2020, using parameters estimated since 2009,

where we allow a 24 month burn-in period from 1994 to 1996 to have sufficient history for parameter estimation. This enables us to isolate the effect of the post-2008 step change from the parameter estimates (see Figure 37 for the impact of start dates on fitted parameters).

Figure 8 shows the annualised Sharpe ratios of the portfolios constructed by the three models, with the probability threshold fixed at 50%. In the 1996–2008 period, the naive model outperforms for most cutoffs, with the Bayesian model being best only when the cutoff is raised to 1, a quite restrictive level. This is expected: pre-2008, hedge fund performance was generally strong, and selection effects were modest. However, in the later period, the Bayesian model dominates. The naive and James–Stein approaches are overoptimistic regarding fund Sharpe ratios and fail to exclude under-performers. In contrast, the Bayesian model identifies and cuts these funds aggressively. Notably, the Bayesian model performs similarly for both start dates, whereas the naive and James–Stein estimators deteriorate with the 1996 start date, demonstrating a slower reaction to the regime change.

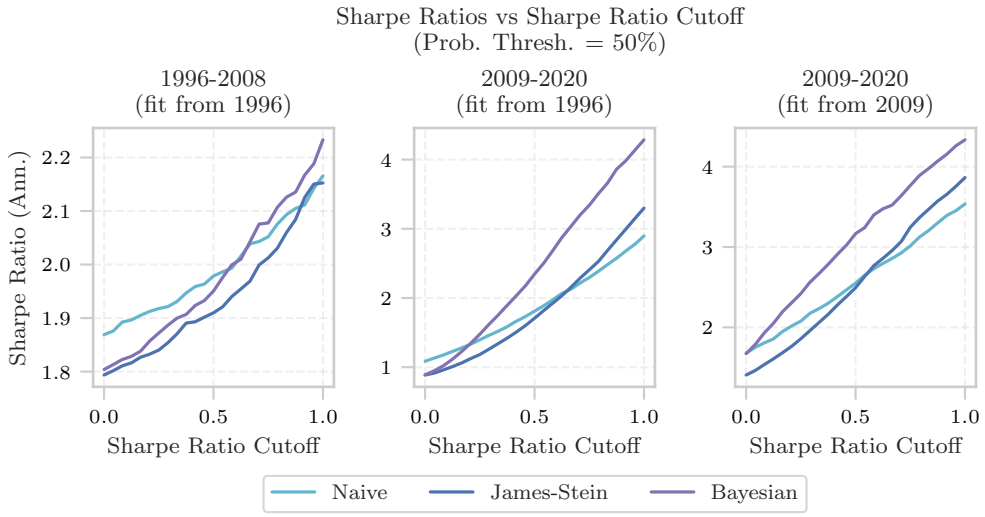


Figure 8: Sharpe ratios for the posterior based trading strategy for the Naive, James–Stein and Bayesian models by Sharpe ratio cutoff, with the probability threshold set at 50%.

Figure 9 presents a similar analysis, fixing the Sharpe ratio cutoff at 0.5 and varying the probability threshold. The results echo the previous plot. The Naive model performs marginally better in the first period, where selection bias is lower, but the Bayesian model outperforms in the latter period, where selection bias is prevalent. At very high thresholds (e.g., 95%), the models converge, selecting the same funds where confidence in the Sharpe ratio is high. The Bayesian model's outperformance stems primarily from marginal cases where the signal is less clear.

[8] For the Naive and James–Stein estimators we use the fund's entire pre- and post-listing history (truncated at the corresponding start date) to estimate the Sharpe ratio, and use the standard sampling distribution to calculate the probability of exceeding the threshold.

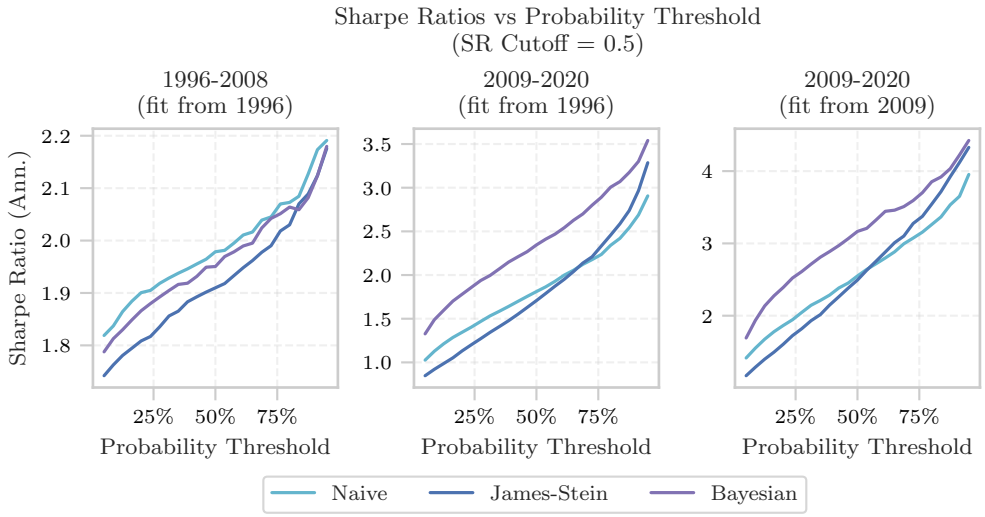


Figure 9: Sharpe ratios for the posterior based trading strategy for the Naive, James–Stein and Bayesian models by probability threshold, with the Sharpe ratio cutoff set at 0.5.

To visualise both parameters simultaneously, Figure 10 plots the improvement in Sharpe ratio against the Naive and James–Stein models, taking the maximum of the two as the benchmark. In the pre-2008 period, the Bayesian model slightly underperforms the Naive approach; in the post-2008 period, it strongly outperforms, particularly as the cutoff increases. This outperformance arises because the Bayesian model estimates future fund performance more accurately (consistent with the prior statistical analysis), effectively pruning underperforming funds.

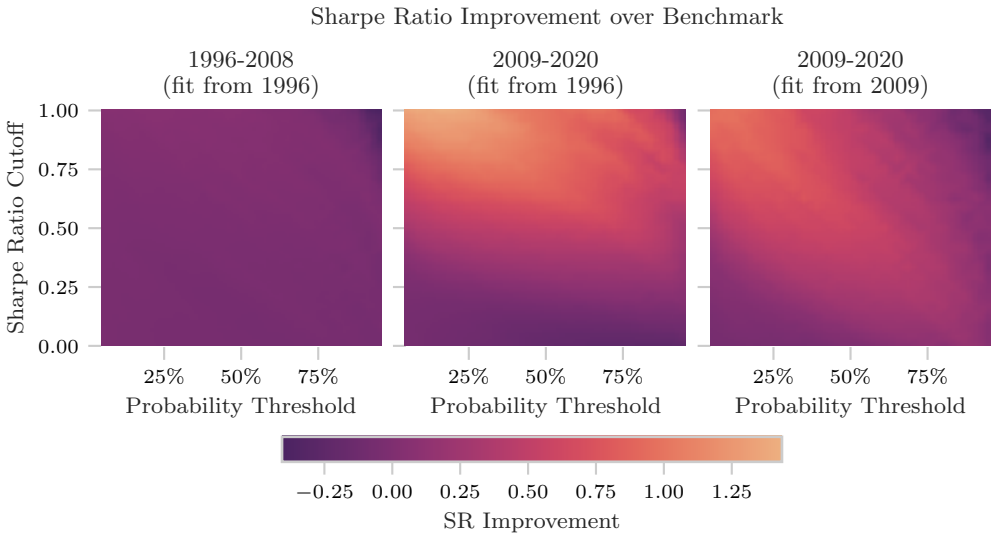


Figure 10: Improvement over the benchmark for the Bayesian strategy by Sharpe ratio cutoff and probability threshold.

Finally, in Figure 11, we examine the predictive-based strategy as a function of the quantile threshold. We observe that the James–Stein estimator performs best in the 1996 to 2008 period, where the Bayesian approach exhibits instability, potentially due to the smaller sample size of funds. Post-2008 however, the results are more interesting. For the model fit using data since 1996, the Bayesian model outperforms the James–Stein approach from $a \approx 0.6$; but for the model fit using data from 2009, this transition occurs at exactly $a = 0.5$, the point where funds switch from underperforming to outperforming expectations. This implies that funds outperforming post-listing relative to pre-listing expectations tend to persist in their outperformance, and the Bayesian model is best at identifying this.

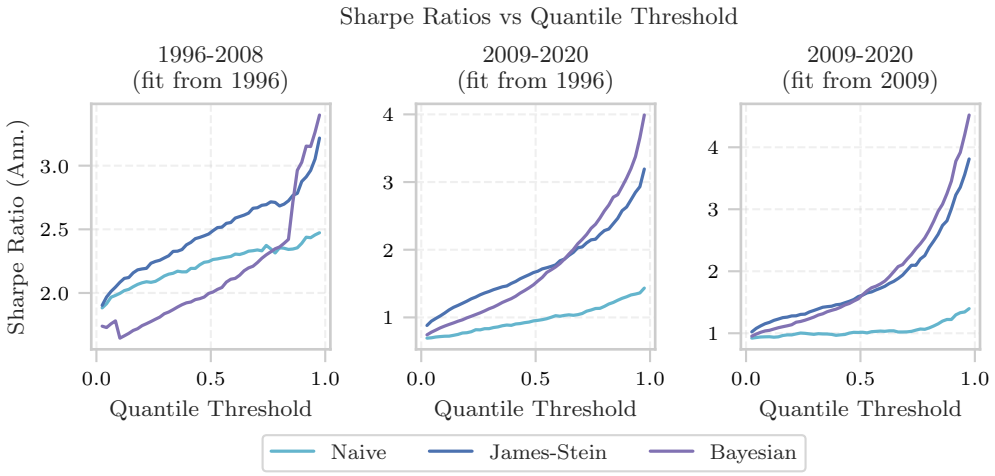


Figure 11: Sharpe ratios for the predictive based trading strategy for the Naive, James–Stein and Bayesian models by quantile threshold.

This empirical study validates our proposed framework as beneficial for risk management and investment selection. In particular, in regimes where selection effects and alpha decay are strong (such as hedge fund returns post-2008), naive approaches fail to accurately forecast the performance of candidate investments, and a methodology which takes these effects into account is vital.

4 Conclusion

This paper develops a Bayesian framework to accurately estimate true Sharpe ratios from historical data polluted by selection bias and alpha decay. We show that naive reliance on in-sample performance leads to significant overestimation of future performance. Furthermore, our theoretical and simulation results demonstrate that attempting to correct for these biases using in-sample data alone is insufficient. To correctly identify the model parameters a joint fitting approach utilising the cross-section of both in- and out-of-sample performance is essential.

By providing a principled assessment of true performance, this framework offers a necessary guide to temper overoptimistic estimates, leading to better investment decisions.

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A Extensions, Other Cases & Additional Theory

A.1 A Study on Likelihood Choice

It is generally accepted that financial returns are neither Gaussian nor iid [Con01]. This manifests itself in our own dataset of hedge fund returns as follows. Figure 12 shows the distribution using violin plots of the autocorrelation ϕ, skewness $\gamma_{3,r}$ and excess kurtosis $\gamma_{4,r} - 3$ for the hedge fund returns used in Section 3. We observe that on average, fund returns exhibit little autocorrelation (note the returns are monthly), and generally low skewness and kurtosis. However, even at these levels, the non-Gaussian nature of the returns can affect the distribution of the sample Sharpe ratios, which in turn affect our fitting procedure. Indeed, in our own analysis we found that using the iid Gaussian likelihood in the model caused the optimiser to tend towards extreme values for κ to manipulate the shape of the marginal likelihoods to match the observed data.

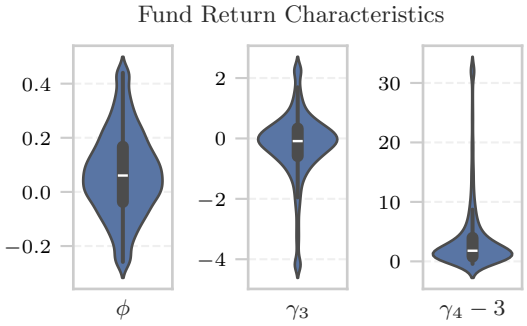


Figure 12: Distributions of autocorrelation ϕ, skewness $\gamma_{3,r}$ and excess kurtosis $\gamma_{4,r} - 3$ for the hedge fund returns used in Section 3.

In Figure 13 we plot the distributions of the in- and out-of-sample Sharpe ratios as a function of the autocorrelation ϕ, skewness $\gamma_{3,r}$ and kurtosis $\gamma_{4,r}$ for the parameters in Table 3. Among the parameters studied, autocorrelation ϕ exerts the most significant influence on the shape of these distributions. Specifically, negative autocorrelation increases the peak density and narrows the OOS distribution, while positive autocorrelation lowers the peak and fattens the tails. Although skewness and kurtosis of the underlying returns have more subtle visual impacts, we found that in certain cases the model only converged to sensible values for the prior once the likelihood was correctly configured to match the statistics of the underlying returns. This suggests that the optimiser can be sensitive to small changes in distribution that are not easily discernible by eye, making the more general likelihood function sometimes necessary for a robust and stable fitting procedure.

Therefore, if fitting with a Gaussian i.i.d. likelihood results in extreme parameter values, we recommend capturing the average autocorrelation of the returns. To maintain computational tractability, a single representative autocorrelation value should be selected for the entire sample rather than estimating a unique value per fund. As discussed in Appendix B.2, the complexity of our calculation is driven by the number of unique likelihood parameterisations. Allowing these parameters to vary by fund would slow the computation significantly, whereas a common parameter ensures most funds share an identical likelihood function.

A.2 Other Types of Selection Bias

For concreteness, the main text focused on correcting for random selection bias. This is the kind studied by Dawid [Daw94] where he declared Bayesian statistics immune to selection bias when computing the posterior (but importantly not when fitting the prior). This form of selection bias is also used by Chen and Zimmermann [CZ20] in their study of publication bias in the asset pricing literature. In this section, we complete the picture by developing the equivalent Bayesian

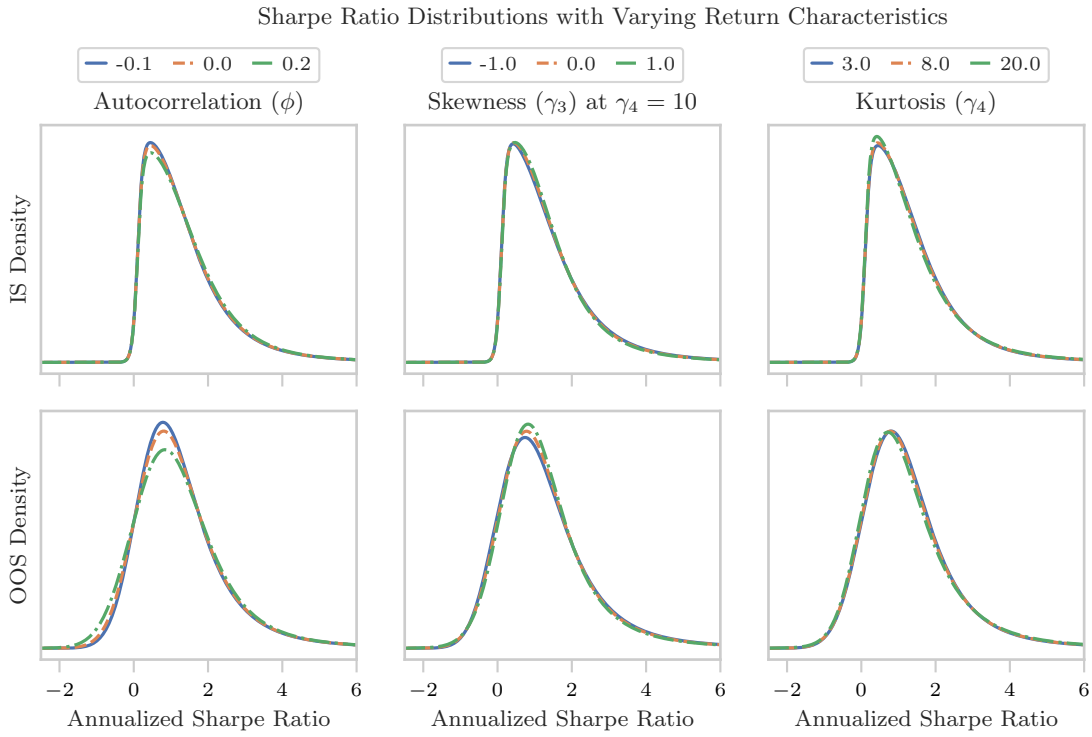


Figure 13: In- and Out-of-Sample Sharpe ratio distributions as functions of the autocorrelation of returns ϕ, skewness of returns $\gamma_{3,r}$ and kurtosis of returns $\gamma_{4,r}$.

Prior	Selection	Sample	Decay
$\text{SR}_0 = 0.35/\sqrt{12}$	$\kappa = 0.1/\sqrt{12}$	$T_{\text{IS}} = 12 \cdot 5$	$\delta = 1$
$\varsigma_0^2 = 3.5/12$	$\ell = 50$	$T_{\text{OOS}} = 12 \cdot 5$	
$\nu = 3$			

Table 3: Prior, selection, sample and decay parameters for the experiments of Appendix A.1.

corrections for the other two forms: fixed and hierarchical bias. We begin by describing fixed and hierarchical selection bias, and then provide the necessary adjustments to the prior, likelihood and posterior for both cases.

A.2.1 Fixed Selection Bias

As with random selection bias, fixed selection bias is best understood through an example. In the random case, we act as a filter on a population: a fund only enters our dataset if its in-sample Sharpe ratio exceeds a certain threshold. While each fund's true skill level is fixed, the collective distribution of funds that survive this filter is skewed upward. This is intuitive: assuming some correlation between past and future performance, picking funds with high in-sample returns successfully weeds out the unskilled managers. In contrast, fixed selection bias distorts the view of a single manager rather than a group. Consider a manager who privately observes their own track record and decides to raise capital only if their performance looks attractive. Their true Sharpe ratio has not changed; they are the same manager they were before the backtest. However, because the data is only revealed conditional on success, the sample is no longer representative of their true performance. We are effectively observing a selectively reported sample rather than a random one, causing our estimate of their skill to be significantly biased.

Hierarchical Selection Bias

Hierarchical selection bias arises from a two-stage process, which is best explained by a quantitative researcher developing a trading strategy. First, the researcher picks a strategy type, such as momentum. This choice corresponds to drawing a family parameter ϑ from a hyperprior $\pi(\vartheta)$, where ϑ represents the mean of the true Sharpe ratio for all momentum strategies. Second, the researcher tests variations within that style, for example 3-, 6-, or 12-month momentum, and selects the best performer. Each variant has its own true Sharpe ratio SR drawn from a conditional prior $\pi(\text{SR} \mid \vartheta)$. Because they all depend on the shared family parameter ϑ, the strategies true Sharpe ratios are correlated. This two-level selection exemplifies hierarchical selection bias, and any form of parameter tuning is an example of this process.

To further clarify, let us make this mathematically precise. We take a location-scale t-distribution for both the hyperprior and conditional prior,

$$\vartheta \sim lst\left(\text{SR}_0, \omega\varsigma_0^2\frac{\nu_\vartheta - 2}{\nu_\vartheta}, \nu_\vartheta\right),$$
$$\text{SR} \mid \vartheta \sim lst\left(\vartheta, (1-\omega)\varsigma_0^2\frac{\nu - 2}{\nu}, \nu\right).$$

The parameter $\omega \in [0, 1]$ allocates the total variance ς_0^2 between the two levels. If $\omega = 1$, all variance is at the family level, the true Sharpe ratio is fixed at $\text{SR} = \vartheta$ for all variants, and we recover the fixed case. If $\omega = 0$, all variance is at the variant level, recovering the random case. The choice of ω therefore models our belief about the efficacy of parameter tuning. A high ω implies that tuning has little effect on the true Sharpe ratio, making the variants highly similar and thus increasing the risk of overfitting. Conversely, a low ω implies that tuning can produce genuinely distinct strategies, which lessens this risk.

More Bias Adjustments

As in the random case, we require the bias adjusted versions of the prior and likelihood for the fixed and hierarchical cases to perform model fitting and inference. The results below adapt those of Yekutieli [Yek12], replacing the original hard threshold with a soft threshold p_{acc}.

In the fixed case, the bias adjusted prior $\pi_B(\text{SR})$ is just $\pi(\text{SR})$, it does not require adjustment. By the definition of fixed selection bias, the act of selection does not affect the true Sharpe ratio, so no adjustment is needed to the prior. This is not the case for the conditional likelihood which does require a bias adjustment, and is the same as the bias adjusted conditional likelihood from the random case, $f_B(\widehat{\text{SR}}_{\text{IS}} \mid \text{SR})$. Therefore, unlike in the random case where the adjustments to the prior and the likelihood cancel when we compute the posterior, this does not occur in the fixed case, and the posterior is a more cautious, bias adjusted posterior.

The hierarchical case interpolates the two, and we denote the bias adjusted unconditional prior by $\pi_H(\text{SR})$. The bias adjusted likelihood is again $f_B(\widehat{\text{SR}}_{\text{IS}} \mid \text{SR})$. We collect the definitions of all the bias adjusted priors and likelihoods in Table 4.

As with the random case, to perform model fitting and inference we simply substitute the correct bias adjusted prior and likelihood into the expressions to compute the marginal likelihood, Equation (2.4), and posterior, Equation (2.5).

	Prior	**Likelihood**
Unbiased	$\pi(\mathrm{SR})$	$f\left(\widehat{\mathrm{SR}}_{\mathrm{IS}} \,\middle|\, \mathrm{SR}\right)$
Random	$\pi_B(\mathrm{SR})$	$f_B\left(\widehat{\mathrm{SR}}_{\mathrm{IS}} \,\middle|\, \mathrm{SR}\right)$
Fixed	$\pi(\mathrm{SR})$	$f_B\left(\widehat{\mathrm{SR}}_{\mathrm{IS}} \,\middle|\, \mathrm{SR}\right)$
Hierarchical	$\pi_H(\mathrm{SR})$	$f_B\left(\widehat{\mathrm{SR}}_{\mathrm{IS}} \,\middle|\, \mathrm{SR}\right)$

$$
\begin{aligned}
\pi_B(\mathrm{SR}) &= \frac{\mathbb{P}(\mathrm{acc} \mid \mathrm{SR})\pi(\mathrm{SR})}{\mathbb{P}\left(\mathrm{acc}\right)}, \\
f_B\left(\widehat{\mathrm{SR}}_{\mathrm{IS}} \,\middle|\, \mathrm{SR}\right) &= \frac{p_{\mathrm{acc}}\left(\widehat{\mathrm{SR}}_{\mathrm{IS}}\right) f\left(\widehat{\mathrm{SR}}_{\mathrm{IS}} \,\middle|\, \mathrm{SR}\right)}{\mathbb{P}(\mathrm{acc} \mid \mathrm{SR})}, \\
\pi_H(\mathrm{SR}) &= \mathbb{P}(\mathrm{acc} \mid \mathrm{SR}) \int_{\vartheta'} \frac{\pi(\vartheta')\pi\left(\mathrm{SR} \mid \vartheta'\right)}{\mathbb{P}(\mathrm{acc} \mid \vartheta')} \,\mathrm{d}\vartheta', \\
\mathbb{P}(\mathrm{acc} \mid \mathrm{SR}) &= \int_{\widehat{\mathrm{SR}}'_{\mathrm{IS}}} p_{\mathrm{acc}}\left(\widehat{\mathrm{SR}}'_{\mathrm{IS}}\right) f\left(\widehat{\mathrm{SR}}'_{\mathrm{IS}} \,\middle|\, \mathrm{SR}\right) \mathrm{d}\widehat{\mathrm{SR}}'_{\mathrm{IS}}, \\
\mathbb{P}\left(\mathrm{acc}\right) &= \int_{\mathrm{SR}'} \pi(\mathrm{SR}')\mathbb{P}\left(\mathrm{acc} \,\middle|\, \mathrm{SR}'\right) \mathrm{dSR}', \\
\mathbb{P}(\mathrm{acc} \mid \vartheta) &= \int_{\mathrm{SR}'} \pi\left(\mathrm{SR}' \,\middle|\, \vartheta\right)\mathbb{P}\left(\mathrm{acc} \,\middle|\, \mathrm{SR}'\right) \mathrm{dSR}'.
\end{aligned}
$$

Table 4: Bias adjusted priors and likelihoods.

B Implementation & Robustness

B.1 The Necessity of Joint Fitting, Alpha Decay & Parameter Recovery

As described in Section 2.2, we select a symmetric prior as the empirically observed distributions of out-of-sample Sharpe ratios are typically rather symmetric.

A skewed distribution of in-sample Sharpe ratios, viewed in isolation, is ambiguous. It could arise from a genuinely skewed prior (e.g., a Gamma distribution) or from selection bias applied to a symmetric one (e.g., a t distribution). As Figure 14 demonstrates, these two scenarios can produce nearly indistinguishable marginal likelihoods for the observed data. Without the out-of-sample evidence, we cannot tell them apart. But in the empirical study in Section 3, we observe a symmetric distribution in the out-of-sample Sharpe ratios, validating our choice. This section expands on that by showing how fitting to in- and out-of-sample Sharpe ratios is key to identify selection bias, and additionally shows why including alpha decay is key for model fitting.

While enough in-sample data allows for the accurate recovery of parameters related to selection bias, we often lack sufficient samples. Further, we want the best possible estimate of out-of-sample performance, which requires accounting for both selection bias and alpha decay (as discussed in Section 2.3). Fitting our model jointly to in- and out-of-sample Sharpe ratios addresses both problems.

To illustrate, we simulate 10,000 in- and out-of-sample Sharpe ratio pairs from a known distribution (Table 5) with hierarchical selection bias and perform three model fits: one using only in-sample data; a joint fit using both data sets; and a joint fit with the decay parameter fixed at $\delta = 1$. For all three we use a Gaussian iid likelihood. The results are in Figure 15 and Table 6. Fitting to in-sample data alone recovers most parameters well but cannot identify δ. (As an aside,

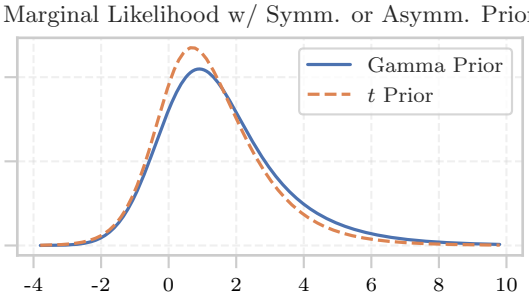


Figure 14: Indistinguishable marginal likelihoods: an unbiased gamma prior, $\Gamma(1, 10)$ versus biased selection with a t prior, $lst(-0.07, 0.02, 7.58)$ with selection parameters $\kappa = 0, \ell = 28$ (see (2.1)).

the hierarchical model is the most difficult to fit; fixing ω helps significantly where possible.[9]) In contrast, the joint fit with a flexible decay parameter recovers all parameters well, successfully distinguishing selection bias from alpha decay by using the shape of the in-sample distribution and the location of the out-of-sample distribution. If we perform the joint fit but fix $\delta = 1$, the model converges to a poor fit with extreme parameter values, as the model is unable to explain the difference between the in- and out-of-sample performance by selection bias alone.

This simulation shows why the alpha decay parameter δ is a structural requirement for the model. By allowing for a systematic shift in performance between periods, it prevents the selection parameters from being forced to capture the entire performance gap. While δ is clearly needed to recover the ground truth in simulations, its inclusion is equally vital for real-world data; without it, the model often yields non-physical selection bias estimates and nonsensical, over-fitted parameters in an attempt to reconcile the in-sample and out-of-sample distributions.

Prior	Selection	Sample	Decay
$\mathrm{SR}_0 = 0.5/\sqrt{252}$	$\kappa = 1/\sqrt{252}$	$T_{\mathrm{IS}} = 252 \cdot 3$	$\delta = 0.25$
$\varsigma_0^2 = 0.5/252$	$\ell = 100$	$T_{\mathrm{OOS}} = 252 \cdot 3$	
$\nu = 10$			
$\nu_\vartheta = 10$			
$\omega = 0.5$			

Table 5: Prior, selection, sample and decay parameters for the experiments of Appendix B.1.

While the above test with 10,000 samples recovered the model's parameters almost perfectly, practical applications rarely permit using such a large dataset. To investigate the effect of sample size, we therefore generated a new set of 10,000 samples with random selection bias using the parameters from Table 7. From this population, we drew 1,000 subsamples of 100, 500, 1,000 and 5,000 in- and out-of-sample Sharpe ratio pairs. We then fitted the model to these subsamples using two methods: first, with in-sample Sharpe ratios only, and second, with both in- and out-of-sample Sharpe ratios. Again, we use a Gaussian iid likelihood for both.

Figure 16 plots the Kullback–Leibler (KL) divergence between the fitted and true models, with lower values indicating a better fit. We clearly see that fitting with both in- and out-of-sample data provides a significantly better fit than fitting with in-sample data alone. The in-sample only method requires 5,000 samples to match the quality that the joint method achieves with just 100.

As KL divergence is not particularly intuitive, Figure 17 shows the relative errors of the fitted parameters themselves. Using both in- and out-of-sample data not only reduces the variance of

[9] For example, if using the hierarchical model to model overfitting in algorithmic trading strategies, you may have a prior belief on the amount of influence that tuning parameters can have on the true Sharpe ratio of the resulting strategy. By selecting an ω value which gives the correct variance to the SR prior, you can model this.

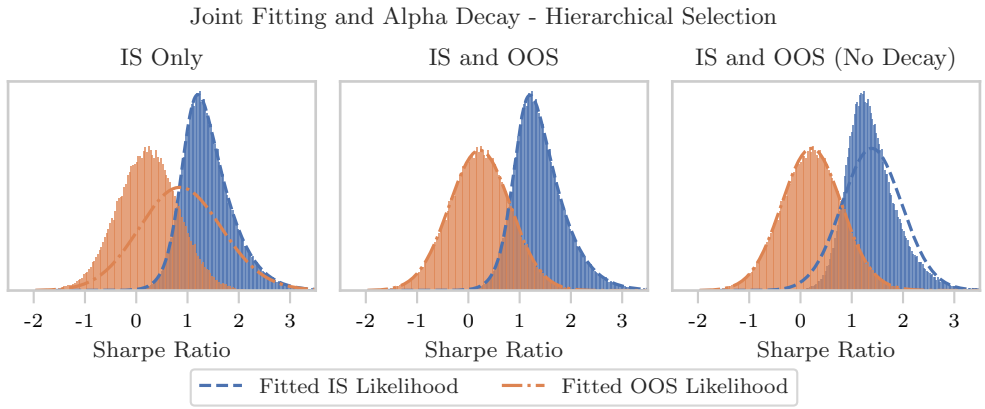


Figure 15: Comparing fitting with in-sample data only, in- and out-of-sample data, and in- and out-of-sample data without allowing for alpha decay.

	IS Only	**IS + OOS**	**IS + OOS No Decay**
SR_0	0.59	0.60	0.19
ς_0^2	0.48	0.47	0.01
ν	7.03	7.20	2.50
ν_ϑ	14.75	13.38	194.96
ω	0.62	0.59	0.01
κ	1.00	1.00	9.98
ℓ	100.73	100.32	55.96
δ	1.0000	0.2516	1.0000

Table 6: Fitted parameters from fitting with in-sample data only, in- and out-of-sample data, and in- and out-of-sample data without allowing for alpha decay.

the estimates, but also removes the significant bias observed in the in-sample only method. The prior's kurtosis parameter ν proves particularly difficult to recover from small samples under both methods, but improves quickly with more samples when using the joint method.

Prior	**Selection**	**Sample**	**Decay**
$\mathrm{SR}_0 = 0.5/\sqrt{252}$	$\kappa = 1/\sqrt{252}$	$T_{\mathrm{IS}} = 252 \cdot 3$	$\delta = 1$
$\varsigma_0^2 = 0.5/252$	$\ell = 100$	$T_{\mathrm{OOS}} = 252 \cdot 3$	
$\nu = 10$			

Table 7: Prior, selection, sample and decay parameters for the subsampling experiments using the random selection bias of Appendix B.1.

Ultimately, accurate parameter recovery matters only if it yields better estimates of the true Sharpe ratio SR. To test this, for each in-sample Sharpe ratio we compute the posterior mean estimate $\mathbb{E}[\mathrm{SR} \mid \widehat{\mathrm{SR}}_{\mathrm{IS}}]$ using the parameters from the in-sample-only fit and, separately, from the joint fit. As a benchmark, we also compute the posterior mean using the true data-generating parameters. We then calculate the Mean Absolute Error (MAE) between these estimates and the true Sharpe ratio for each simulated data point.

Figure 18 plots the average of these MAEs over our 1,000 subsamples. We observe that, on average, fitting with in- and out-of-sample data yields parameters sufficiently close to the

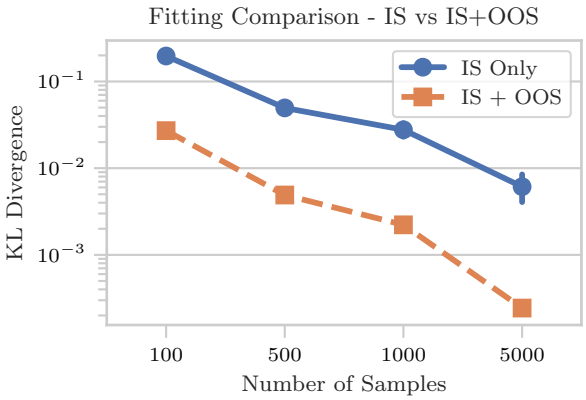


Figure 16: Kullback–Leibler divergence between the fitted and true models. The plot shows how fit quality varies with sample size when using only in-sample data versus using both in- and out-of-sample data.

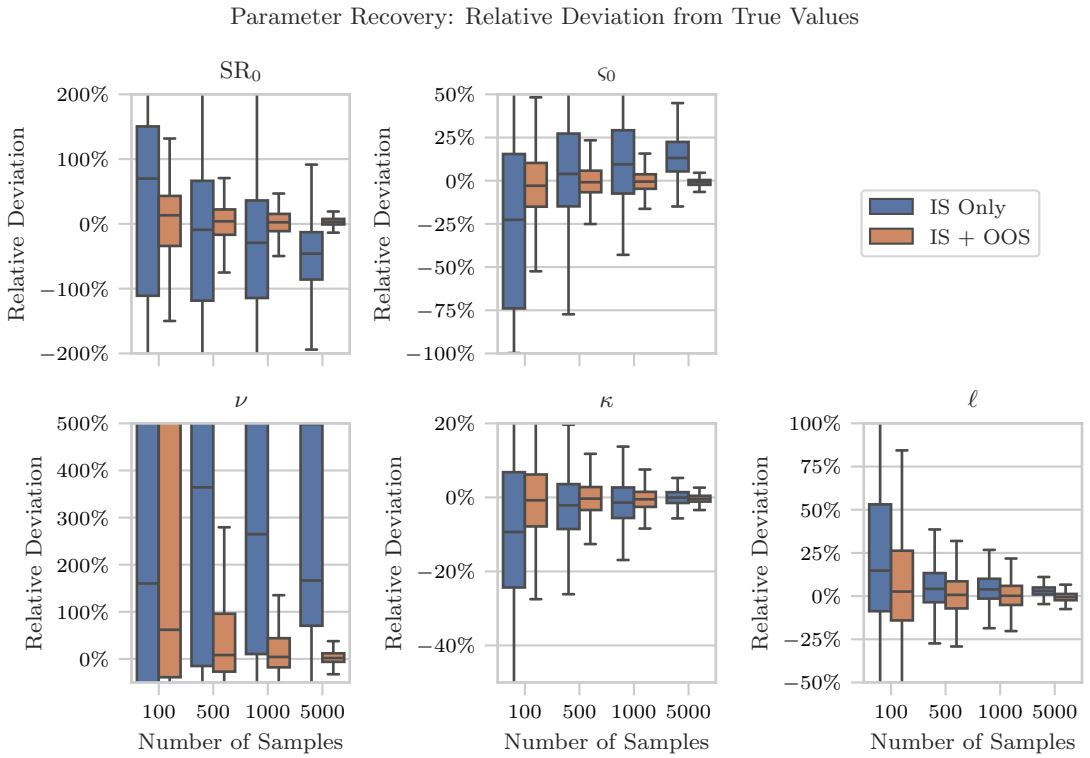


Figure 17: Relative errors in recovered parameters for the two fitting methods. The plots show how the bias and variance of the parameter estimates change with the number of samples used for fitting.

true parameters that inference using them is almost identical. Fitting with in-sample data alone, however, produces a much higher error, up to 0.15 for small datasets, a significant amount when the true Sharpe ratios are of order 1.

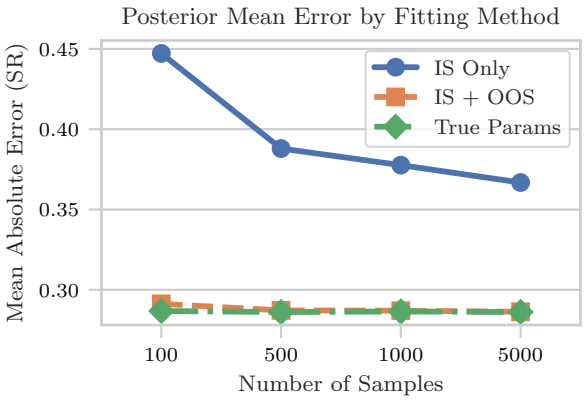


Figure 18: Average Mean Absolute Error of the posterior mean $\mathbb{E}[\mathrm{SR} \mid \widehat{\mathrm{SR}}_{\mathrm{IS}}]$ vs the true Sharpe ratio SR per fit using in-sample only or in- and out-of-sample data.

B.2 Numerical Implementation Details

To use the model, one must fit its parameters via MLE and then compute the posterior (2.5). Both cases require computing a nested integral in the marginal likelihood (2.4) (up to a quadruple integral in the hierarchical case, see Table 4) as quickly as possible. Speed is particularly important for the fitting procedure, with potentially thousands of function evaluations required for convergence. The calculations must also be accurate and robust, especially against extreme parameter values, to prevent the optimiser from converging to a problematic fit, for example where the likelihoods explode due to numerical inaccuracy.

These two goals, speed and accuracy, are somewhat at odds. While standard adaptive quadrature methods, such as those in `scipy` [Vir+20], can accurately calculate the likelihood for a single sample, it is not fast enough for our purposes. To overcome this, we developed a specialised numerical package using the `JAX` library [FJL19] that combines several techniques to achieve high performance, accuracy, and robustness.

Using `JAX` provides four main benefits: Just-In-Time (JIT) compilation, GPU computation, easy parallelisation and automatic differentiation. The first three are key to computing the likelihood rapidly, while the last allows us to pass an analytical gradient to the optimiser. For the optimiser, we utilise `iminuit` [Dem+25], a python wrapper of the C++ `Minuit2` library maintained by CERN [JR75]. We found this optimiser was much more successful in navigating the model's difficult loss landscape than those available in `scipy`. Its success stems from its quasi-Newton algorithm, MIGRAD, which uses gradient information to intelligently approximate the local curvature (the Hessian matrix) of the loss surface. Supplying the precise analytical gradients computed by `JAX` allows this algorithm to perform an accurate line search, taking more efficient and reliable steps towards the minimum compared to methods that rely on numerical derivative estimates. This combination proves particularly effective for achieving rapid convergence in the complex, non-convex landscapes typical of statistical fitting problems. We also aid the optimiser by starting it at sensible parameter values; for instance, our initial guess for SR_0 is the mean of the out-of-sample Sharpe ratios.

Instead of adaptive quadrature, we use the Simpson rule for integration [Sim43]. While less sophisticated, this method is significantly faster and is easily compatible with automatic differentiation. However, we do dynamically adapt the integration grid to encompass the probability mass of the distribution being integrated. This approach, which uses the percentage point function (PPF), ensures the grid is always optimally placed to maximise accuracy. This is vital, as a wide, fixed grid would cause numerical issues for small values of ς_0 or values of ω close to zero or one. As `JAX` lacks a PPF for the t-distribution, we implement a minimal algorithm similar to Hill's algorithm [Hil70] to compute the required quantiles. Specifically we use a four-term Cornish-Fisher correction to the normal quantile [FC60]. We then apply a single Halley iteration [Hal94], which

we find sufficient for our needs.

A significant optimisation comes from noting that the most computationally intensive parts of the likelihood (e.g. $\mathbb{P}(\text{acc} \mid \text{SR})$, defined in (2.2)) depend not on the sample Sharpe ratio $\widehat{\text{SR}}_{\text{IS}}$, but on the length of the in-sample period T_{IS}. Since a dataset typically contains only a few unique values of T_{IS}, we can precompute these expensive calculations once for each unique T_{IS}. This allows us to compute the marginal likelihoods for a very large number of samples almost as quickly as for one. The computation time thus scales with $O(N)$, where N is the number of unique T_{IS} values, not the total number of samples. We can often improve this further by rounding the $T_{\text{IS},i}$ values slightly to reduce the number of unique values. Conveniently, `JAX` enables easy parallel batch computation of these expensive components, significantly reducing the computation time per likelihood evaluation.

To avoid numerical underflow or overflow, we perform all calculations in the logarithmic domain. This technique improves numerical accuracy and is crucial when the divisors (e.g. $\mathbb{P}(\text{acc} \mid \text{SR})$) become very small. Without it, the likelihood could become numerically invalid, drawing the optimiser towards a problematic region. To prevent underflow or overflow when integrating our log-domain integrands, we use the log-sum-exp trick, factoring out the maximum value of the integrand before exponentiation. This ensures the optimiser receives stable, finite values even for likelihoods that would otherwise evaluate to zero in standard floating-point precision.

B.3 Hybrid Parameter Estimation

As discussed in Section 2.2, we found that for certain datasets the parameter set maximising the likelihood was "non-physical", i.e. the optimiser reached extreme, mathematically valid values that nevertheless contradict the underlying logic of the model. In particular, when the selection effect is weakly identified (e.g., when $\ell < 2$), the MLE parameter set $\boldsymbol{\Theta}$ typically included an excessively large κ value. For instance, κ could be estimated at twenty times the prior location SR_0. This creates a scenario where the acceptance probability for all observed Sharpe ratios, $p_{\text{acc}}\left(\widehat{\text{SR}}_{\text{IS}} \;\middle|\; \kappa, \ell\right)$, becomes effectively infinitesimal. Such a result implies that the observed strategies are near-impossible events, which does not align with a useful interpretation of the model. We attribute this to a combination of overfitting and the general difficulty of estimating weakly identified parameters in Bayesian inference [Gel+13, Section 4.3][Gel+20, Section 5].

There are several ways to address this. One option is to add an L_1 or L_2 penalty to the optimisation problem, encouraging smaller parameter estimates. Alternatively, a traditional Bayesian approach could impose a weakly informative prior on parameters such as κ, acting as a soft penalisation on extreme values. Both approaches, however, require the choice of hyperparameters governing the degree of penalisation.

Instead, we adopt a hybrid strategy that combines Maximum Likelihood Estimation (MLE) and the Generalised Method of Moments (GMM). This is a form of two-step estimator [NM94, Section 6] that operates effectively as a profile likelihood approach. We split the parameter set $\boldsymbol{\Theta}$ into those optimised via MLE, denoted $\widehat{\boldsymbol{\Theta}}$, and those estimated via GMM, denoted $\widetilde{\boldsymbol{\Theta}}$. At each step of the optimisation, the MLE optimiser (Minuit in our case) proposes a candidate set $\widehat{\boldsymbol{\Theta}}$; we then solve the inner GMM condition to find $\widetilde{\boldsymbol{\Theta}}$ satisfying the method-of-moments condition

$$m\left(\widetilde{\boldsymbol{\Theta}} \;\middle|\; \widehat{\boldsymbol{\Theta}}\right) - \widehat{\mathbf{m}} = \mathbf{0}_K,$$

where $m\left(\widetilde{\boldsymbol{\Theta}} \;\middle|\; \widehat{\boldsymbol{\Theta}}\right) \in \mathbb{R}^K$ denotes the model-implied moments and $\widehat{\mathbf{m}} \in \mathbb{R}^K$ denotes the empirical moments. The resulting loss is then returned to the optimiser, and the cycle repeats. This combines ideas of constrained maximum likelihood estimation [AS58] and profile likelihood [Mv00; CCG14]. In practice, we find this yields more realistic parameter estimates in cases where pure MLE produces extreme values. For example, in Table 12 we see that the model fit using the MLE approach leads to the selection midpoint κ converging to the boundary set by the optimiser (10), while the MLE-GMM approach yields more reasonable parameter values.

To perform the GMM step, we use the `optimistix` library [RLK24] to solve the non-linear

least-squares form:

$$\min_{\widetilde{\boldsymbol{\Theta}}} \left\| m\Big(\widetilde{\boldsymbol{\Theta}} \;\Big|\; \widehat{\boldsymbol{\Theta}}\Big) - \widehat{\mathbf{m}} \right\|_2^2 .$$

We use the Levenberg–Marquardt solver, which is robust for non-linear problems. Because this solver is designed for unconstrained optimisation, i.e. it does not natively prevent parameters from taking impossible values like negative variance, we apply a logistic reparameterisation. This maps the solver's unconstrained internal variable η_i to the desired range $[a_i, b_i]$,

$$\widetilde{\Theta}_i(\eta_i) = a_i + (b_i - a_i)\,\sigma(\eta_i), \qquad \sigma(x) = \frac{1}{1+e^{-x}}.$$

As with MLE, we start each parameter at sensible, data-driven initialisations. Although Levenberg–Marquardt is theoretically more robust than Gauss–Newton, it did not converge well when supplied with very poor initial guesses (we initially used the midpoint of the bounded region before implementing data-driven initialisation). The starting values are

$$\mathrm{SR}_0 = \widehat{\mathbb{E}}\Big[\widehat{\mathrm{SR}}_{\mathrm{OOS}}\Big], \qquad \varsigma_0 = \sqrt{\widehat{\mathbb{V}}\Big[\widehat{\mathrm{SR}}_{\mathrm{OOS}}\Big] - \frac{1}{T_{\mathrm{OOS}}}}, \qquad \delta = \frac{\widehat{\mathbb{E}}\Big[\widehat{\mathrm{SR}}_{\mathrm{OOS}}\Big]}{\widehat{\mathbb{E}}\Big[\widehat{\mathrm{SR}}_{\mathrm{IS}}\Big]},$$

$$\kappa = \frac{\widehat{\mathbb{E}}\Big[\widehat{\mathrm{SR}}_{\mathrm{OOS}}\Big] + \widehat{\mathbb{E}}\Big[\widehat{\mathrm{SR}}_{\mathrm{IS}}\Big]}{2}, \qquad \ell = 5, \qquad \nu = 5,$$

where $\widehat{\mathbb{E}}, \widehat{\mathbb{V}}$ denote the sample mean and variance respectively. If out-of-sample data is unavailable, in-sample data is used in its place, and ℓ and ν are started at uninformative initialisations due to difficulty in fitting these parameters via GMM. In practice, we recommend applying this methodology primarily to κ, δ and SR_0.

For a chosen profile set $\widetilde{\boldsymbol{\Theta}}$ we include only the moments needed to inform that parameter. The mapping is shown in Table 8. In practice the dataset often contains many unique pairs $(T_{\mathrm{IS}}, T_{\mathrm{OOS}})$. To match empirical moments (computed over all samples) to analytical moments (which depend on T_{IS} and T_{OOS}), we compute each moment per pair and weight them by the observed frequencies of the pairs.

Param	$\mathbb{E}[\widehat{\mathrm{SR}}_{\mathrm{IS}}]$	$\mathbb{E}[\widehat{\mathrm{SR}}_{\mathrm{OOS}}]$	$\mathbb{V}[\widehat{\mathrm{SR}}_{\mathrm{IS}}]$	$\mathbb{V}[\widehat{\mathrm{SR}}_{\mathrm{OOS}}]$	$\mathrm{Cov}(\widehat{\mathrm{SR}}_{\mathrm{IS}}, \widehat{\mathrm{SR}}_{\mathrm{OOS}})$	$\gamma_4(\widehat{\mathrm{SR}}_{\mathrm{IS}})$	$\gamma_4(\widehat{\mathrm{SR}}_{\mathrm{OOS}})$
SR_0	✓	✓	×	×	×	×	×
ς_0	×	×	✓	✓	×	×	×
κ	✓	✓	×	×	×	×	×
ℓ	×	×	✓	✓	×	×	×
δ	✓	✓	×	×	✓	×	×
ν	×	×	✓	✓	×	✓	✓

Table 8: Mapping between model parameters and empirical moments used in GMM estimation. When out-of-sample data is unavailable, all OOS moments are omitted. $\gamma_4(\cdot)$ denotes the kurtosis.

The hybrid MLE-GMM procedure stabilises weakly identified selection parameters and preserves interpretability without introducing additional penalisation hyperparameters. In our applications it consistently prevents extreme estimates for κ while retaining good in-sample fit for the likelihood-driven components. We therefore recommend profiling onto moment conditions for κ, δ and SR_0, leaving the remaining parameters to be determined by MLE.

B.4 Effect of Correlated Sharpe Ratios

To see why independent samples are crucial for calibrating the prior, consider a simple model where returns are correlated through shared exposure to a market factor (for related discussion of

cross-strategy dependence see [Lóp19; LL19; PLZ26]). We model the returns $r_{i,t}$ as a linear combination of this market factor $r_{M,t}$ and an idiosyncratic component $\epsilon_{i,t}$ representing idiosyncratic performance:

$$r_{i,t} = \rho r_{M,t} + (1-\rho)\epsilon_{i,t},$$

where ρ controls the loading on the market, and the idiosyncratic returns are assumed to be independent. For $\rho = 0$, the returns are independent, and we can continue as normal, but for $\rho > 0$ the returns become correlated through their shared exposure to the market factor.

This correlation affects the cross-sectional distribution of sample Sharpe ratios used to fit the prior. Specifically, if the market performs unusually well (or poorly) relative to its true Sharpe ratio, the distribution of sample Sharpe ratios will be pushed upwards (or downwards), causing us to misestimate the location of the prior if we proceed naively. Additionally, because the returns are correlated, a naive estimate of the prior's variance will be too low; for high ρ, it can even yield negative estimates.

The true Sharpe ratio SR_i depends on the market and idiosyncratic components. Assuming the idiosyncratic variance is identical across assets, $\sigma^2_{\epsilon,i} = \sigma^2_\epsilon$, the moments of a single return $r_{i,t}$ imply:

$$\begin{aligned}
\mathbb{E}[r_{i,t}] &= \rho\mu_M + (1-\rho)\mu_{\epsilon,i}, \\
\mathbb{V}[r_{i,t}] &= \rho^2\sigma^2_M + (1-\rho)^2\sigma^2_\epsilon, \\
\mathrm{SR}_i &= \frac{\rho\mu_M + (1-\rho)\mu_{\epsilon,i}}{\sqrt{\rho^2\sigma^2_M + (1-\rho)^2\sigma^2_\epsilon}}.
\end{aligned}$$

Since μ_M is constant across assets, the cross-sectional variance $\varsigma_0^2 = \mathbb{V}[\mathrm{SR}_i]$ depends solely on the variation in idiosyncratic means $\mu_{\epsilon,i}$. We parameterise the idiosyncratic Sharpe ratios $\mathrm{SR}_{\epsilon,i} = \mu_{\epsilon,i}/\sigma_\epsilon$ with mean $\mathrm{SR}_{\epsilon,0}$ and variance $\varsigma^2_{\epsilon,0}$, implying $\mathbb{V}[\mu_{\epsilon,i}] = \sigma^2_\epsilon\varsigma^2_{\epsilon,0}$. Applying the variance operator to SR_i yields:

$$\varsigma_0^2 = \left(\frac{1-\rho}{\sqrt{\rho^2\sigma^2_M + (1-\rho)^2\sigma^2_\epsilon}}\right)^2 \sigma^2_\epsilon\varsigma^2_{\epsilon,0}.$$

Simplifying this expression provides the final moments:

$$\mathrm{SR}_0 = \frac{\rho\sigma_M\mathrm{SR}_M + (1-\rho)\sigma_\epsilon\mathrm{SR}_{\epsilon,0}}{\sqrt{\rho^2\sigma^2_M + (1-\rho)^2\sigma^2_\epsilon}}, \qquad \varsigma_0^2 = \frac{(1-\rho)^2\sigma^2_\epsilon\varsigma^2_{\epsilon,0}}{\rho^2\sigma^2_M + (1-\rho)^2\sigma^2_\epsilon},$$

where $\mathrm{SR}_M = \mu_M/\sigma_M$ is the market Sharpe ratio.

A naive way to estimate these parameters from a cross-section of N sample Sharpe ratios, $(\widehat{\mathrm{SR}}_i)_{i\in[1,N]}$, uses the method of moments. We treat the observed cross-sectional variance as the sum of the true parameter variance and the average estimation noise variance. To isolate the true variance, we subtract the noise component C_{Lo}:

$$\widehat{\mathrm{SR}}_{0,N} = \widehat{\mathbb{E}}\left[\left(\widehat{\mathrm{SR}}_i\right)_{i\in[1,N]}\right], \qquad \text{and} \qquad \widehat{\varsigma}^2_{0,N} = \widehat{\mathbb{V}}\left[\left(\widehat{\mathrm{SR}}_i\right)_{i\in[1,N]}\right] - C_{Lo},$$

where $\widehat{\mathbb{E}}[\cdot], \widehat{\mathbb{V}}[\cdot]$ represent the sample mean and variance of the sample Sharpe ratios $(\widehat{\mathrm{SR}}_i)_{i\in[1,N]}$. We have assumed the returns are iid across time with no excess skewness or kurtosis to derive the correction term C_{Lo}, which accounts for the finite-sample variance of an individual sample Sharpe ratio, as given by Lo [Lo02, Equation 9],

$$C_{Lo} = \frac{1}{T_{\mathrm{IS}}}\left(1 + \frac{1}{2}\widehat{\mathrm{SR}}^2_{0,N}\right).$$

These estimators perform poorly as they ignore cross-sectional correlation. We can instead construct "market-aware" estimators using observable market data. For the variance, we scale the

correction factor to remove only the idiosyncratic portion of the estimation noise, since systematic error does not contribute to cross-sectional dispersion. Indeed, from the expression for ς_0^2 above,

$$\varsigma_0^2 = \frac{(1-\rho)^2\sigma_\epsilon^2}{\rho^2\sigma_M^2 + (1-\rho)^2\sigma_\epsilon^2}\,\varsigma_{\epsilon,0}^2,$$

so the idiosyncratic share of total variance is $\frac{(1-\rho)^2\sigma_\epsilon^2}{\rho^2\sigma_M^2+(1-\rho)^2\sigma_\epsilon^2}$. We therefore subtract the same fraction of Lo's correction:

$$\widehat{\varsigma}_{0,A}^2 = \widehat{\mathbb{V}}\left[\left(\widehat{\mathrm{SR}}_i\right)_{i\in[1,N]}\right] - C_A, \qquad \text{with} \quad C_A = \frac{(1-\widehat{\rho})^2\widehat{\sigma}_\epsilon^2}{\widehat{\rho}^2\widehat{\sigma}_M^2 + (1-\widehat{\rho})^2\widehat{\sigma}_\epsilon^2} C_{Lo},$$

where $\widehat{\sigma}_M^2$ is the sample variance of market returns, $\widehat{\sigma}_\epsilon^2$ is the sample variance of idiosyncratic returns and $\widehat{\rho}$ an estimate for ρ. These parameters are readily estimable by regressing individual returns on the observed market returns.

For SR_0 we are not so fortunate. While a market aware correction can be calculated,

$$\widehat{\mathrm{SR}}_{\mathrm{IS},0,A} = \widehat{\mathrm{SR}}_{\mathrm{IS},0,N} - \frac{\rho\sigma_M(\widehat{\mathrm{SR}}_M - \mathrm{SR}_M)}{\sqrt{\rho^2\sigma_M^2 + (1-\rho)^2\sigma_\epsilon^2}},$$

it depends on the true market Sharpe ratio, SR_M, which is unknown. Using our best estimate, $\widehat{\mathrm{SR}}_M$, reduces the correction term to zero, leaving us with the naive estimator.

To demonstrate these effects we perform a straightforward simulation. We generate 1,000 market paths and 10,000 idiosyncratic return series of length $T_{\mathrm{IS}} = 252 \cdot 3$ from the distributions

$$r_{M,t} \sim \mathcal{N}(\mathrm{SR}_M \cdot \sigma, \sigma^2), \qquad \epsilon_{i,t} \sim \mathcal{N}(\mathrm{SR}_{\epsilon,i} \cdot \sigma, \sigma^2),$$

where to reduce the number of parameters we have set the market and idiosyncratic volatilities to be equal $\sigma_M = \sigma_\epsilon = \sigma$, the distribution of idiosyncratic Sharpe ratios is $\mathrm{SR}_{\epsilon,i} \sim \mathcal{N}(\mathrm{SR}_{\epsilon,0}, \varsigma_{\epsilon,0}^2)$ and the parameters are set as:

$$\mathrm{SR}_M = 0.75/\sqrt{252}, \qquad \sigma = 0.1/\sqrt{252},$$
$$\mathrm{SR}_{\epsilon,0} = 0, \qquad \varsigma_{\epsilon,0}^2 = 0.5/252.$$

For each market path, we construct a set of $N = 10,000$ returns and estimate SR_0 and ς_0^2 using the naive and market-aware methods. The outcome is shown in Figure 19. We see that as ρ increases, the distribution of errors, $\widehat{\mathrm{SR}}_0 - \mathrm{SR}_0$, in our estimate of SR_0 increases, but across all the market paths, the naive estimate is unbiased. For ς_0^2, we see that the naive estimate $\widehat{\varsigma}_{0,N}$ yields impossible negative estimates for the variance of the distribution of SR_i. But with the corrected estimator $\widehat{\varsigma}_{0,A}$ we perfectly estimate the true variance, and in fact the error per market path is very small, due to the large number of returns and reasonably long time series used to estimate the inputs to the corrected estimator.

Ultimately, this was the simplest possible case, and once we introduce selection bias the process of yielding good estimates for the prior parameters would become much more complex, if not intractable. This difficulty, combined with the inability to produce a market-aware correction for the prior's location, suggests the most robust approach is to work with the idiosyncratic component directly. Should the overall Sharpe ratio be required, the systematic contribution can be added back to the corrected estimate of idiosyncratic performance. We can express the overall Sharpe ratio in terms of the component Sharpe ratios,

$$\mathrm{SR}_i = \frac{\rho\mathrm{SR}_M\sigma_M + (1-\rho)\mathrm{SR}_{\epsilon,i}^*\sigma_{\epsilon,i}}{\sqrt{\rho^2\sigma_M^2 + (1-\rho)^2\sigma_{\epsilon,i}^2}},$$

where $\mathrm{SR}_{\epsilon,i}^*$ denotes our corrected estimate for the idiosyncratic Sharpe ratio, and, as previously discussed, the remaining parameters are reliably estimated directly from the data.

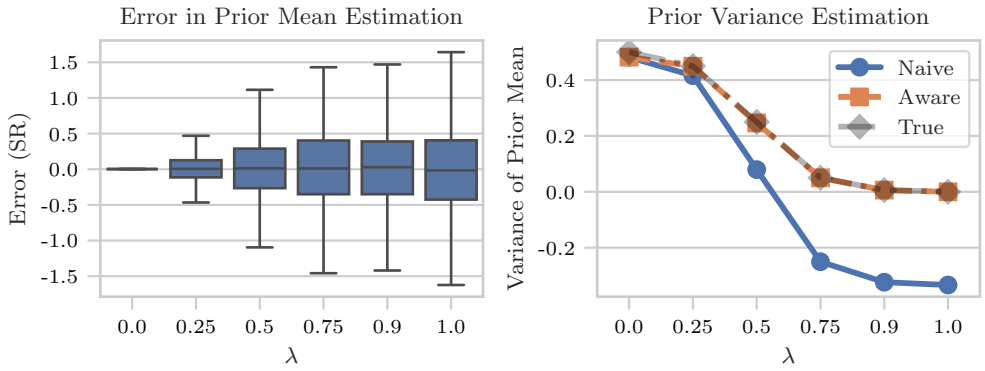


Figure 19: The impact of correlation, controlled by ρ, on the estimation of SR_0 and ς_0^2.

C Supplementary Material for Section 3

C.1 Filter Criteria

To filter down the initial large dataset into a set of suitable funds for our approach we apply the following criteria to the dataset, yielding a final total of 7,303 funds:

1. Maximum AUM must be $>$ \$20m,
2. At least 24 months of returns,
3. At least 6 months of returns in both pre- and post-listing periods,
4. Monthly volatility must be $> 0.5\%$,
5. Annualised Sharpe ratio must be < 10,
6. All monthly returns must be within $[-90\%, 500\%]$,
7. De-duplicate highly similar funds: for any pair with absolute correlation $> 95\%$, retain the fund with the longer history.

These filters are set to exclude very small hedge funds, potential outliers and funds with insufficient history.

C.2 Deriving Independent Returns

As described in Section 2.3, we require the samples used to fit the Bayesian model to be sufficiently independent to avoid issues with estimating the parameters correctly, see Appendix B.4. For hedge fund returns, this poses an immediate issue, as many funds are relatively correlated through exposure to shared risk factors. To resolve this, we split the returns for each fund into a "systematic" part and a "non-systematic" part.

We define systematic returns as the returns earned through passive (as opposed to active) exposure to known risk factors, and define all other returns as non-systematic. Specifically, we allow factor timing to be considered skill on the part of the fund manager, in addition to returns earned through manager specific non-factor explainable returns. To estimate these components we fit a single regression model for each fund,

$$r_{i,t} = \underbrace{\alpha_i + \boldsymbol{\beta}_i^{\text{Timing}} \mathbf{F}_t^2}_{\text{Non-Systematic Part}} + \underbrace{\boldsymbol{\beta}_i^{\text{Passive}} \mathbf{F}_t}_{\text{Systematic Part}} ,$$

where we have taken the approach of Treynor–Mazuy [TM66] by adding the squared factor returns to the regression to capture factor timing skill.

For our factors, we take a combination of the Fung–Hsieh seven factor model [FH04], a collection of global factors Barth et al. [Bar+23], and the factor set proposed by Chen et al. [Che+25], yielding a total of 20 factors. Once we add the squared factors and the intercept, this is a total of 41 possible parameters per regression.

This rules out two approaches. First, we cannot use simple OLS, as funds with short histories often have relatively few data points. Similarly, we do not take a walkforward approach for the same reason as we wish to use as much data as possible for our regression and later analysis. However, by not using a walkforward approach we do introduce lookahead bias, as the non-systematic component represents perfectly hedged returns that would be impossible to capture in reality. Relatedly, in our later analysis we assume investors may trade in any fund in any amount, ignoring lockups, closures to new capital, and other constraints.

Our intention, however, is to provide a proof of concept: it is possible to fit our model to data exhibiting selection bias and realistic statistical effects, such as autocorrelation and fat tails. We therefore advise the reader to interpret Sharpe ratios derived herein with a grain of salt. The central aim is to demonstrate that the Bayesian method outperforms naive benchmarks (which also benefit from these same biases).

Returning to our model estimation problem, we take a similar approach to [Che+25], and estimate the parameters for each fund using an adaptive LASSO. This neatly resolves our issues of identifying a well fitting model for each fund.

Specifically, for each fund, we first fit a ridge regression using `sklearn` [Ped+11], tuning the degree of regularisation via Leave-One-Out Cross-Validation. We then use these coefficients for the penalty weights of the adaptive LASSO step, where the adaptive LASSO is performed using `asgl` [MAL21]. To tune the degree of regularisation in the adaptive LASSO, we minimise the extended regularized information criterion (ERIC) which has been shown to outperform more traditional information criteria (AIC/BIC) for adaptive LASSO [HWF15].

We find that this procedure works well, and yields reasonable parameter sets across the dataset. In particular, by examining which factors each fund loads on, we see an expected pattern: the majority of funds load on either the global or US market factor. In Figure 20 we see that more than 40% of funds load on the global market factor, and more than 30% of funds load on the US market factor, with 65% of funds loading on at least one market factor.

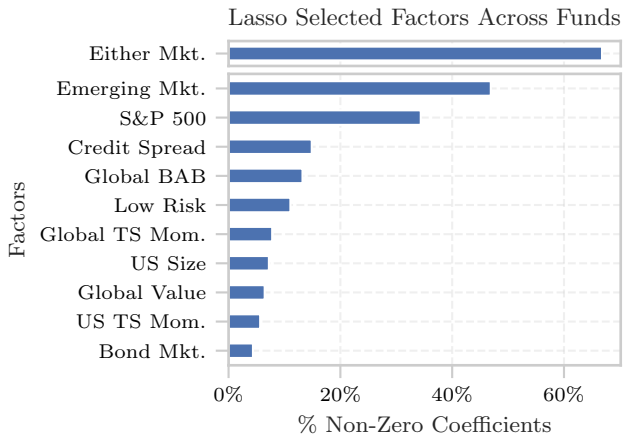


Figure 20: Proportion of funds with non-zero loading on the top 10 factors, and the proportion of funds that load on either of the two market factors. BAB denotes "Betting Against Beta" [FP14] and TS Mom. denotes "Time Series Momentum" [MOP12].

Importantly for our use case, we find this procedure significantly reduces the average correlation between funds, with the mean correlation reducing from 28% for the total returns to 8% for the non-systematic returns, and the distribution of correlations significantly improving, as can be seen in Figure 21.

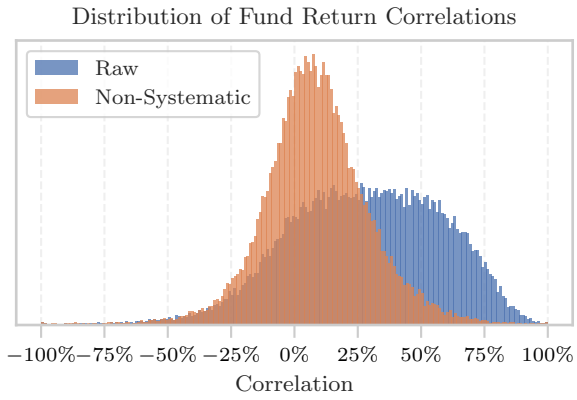


Figure 21: Pairwise correlations between fund returns, before and after removal of systematic factor contributions.

What is the impact of this procedure on the Sharpe ratios of the funds? Figure 22 shows the distributions of pre- and post-listing Sharpe ratios for the raw and non-systematic returns. On average, the annualised pre-listing Sharpe ratio falls from about 1.35 on the raw returns to 1.05 after hedging, so the impact on pre-listing performance is noticeable but relatively muted, with a small shift of the mass of the distribution to the left and the overall shape largely preserved. By contrast, the mean post-listing Sharpe ratio falls from around 0.45 to 0.25 once we remove the systematic component, and the dispersion of post-listing Sharpe ratios is greatly increased. Quantitatively, the average reduction in Sharpe from pre- to post-listing is about 67% for the raw returns and 76% for the hedged returns, indicating that the deterioration in performance is larger once we strip out systematic exposures. This suggests that pre-listing returns tend to be less systematic, while post-listing returns are more so, which is consistent with the idea that funds scale up after listing and therefore take on more systematic risk due to capacity constraints on non-systematic strategies.

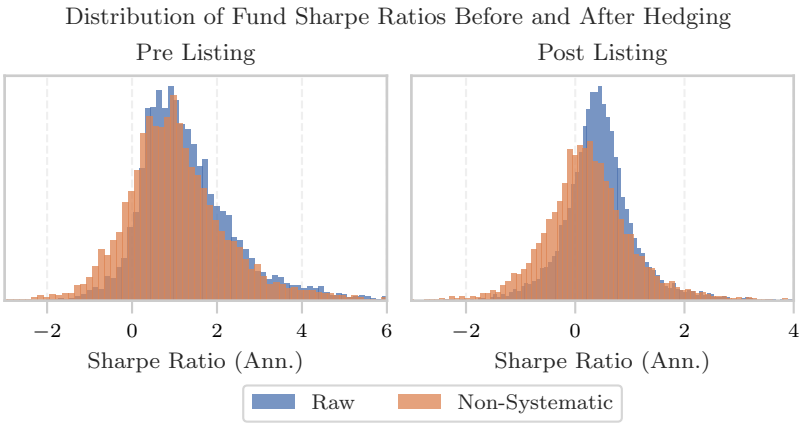


Figure 22: Distribution of Sharpe ratios before and after removal of systematic factor contributions, split into the pre- and post-listing periods for each fund.

The impact of systematic vs non-systematic returns is further illustrated in Figure 23, which plots the cumulative performance of inverse volatility weighted portfolios using both return types. The weights are computed by taking the inverse volatility of each fund over its entire history and normalised such that the weights each month sum to 1, with the final performance normalised to unit volatility. We see that the performance divides into three periods: pre September 2001,

between September 2001 and July 2008, and post July 2008. Pre-2008, the non-systematic portfolio significantly outperforms the raw portfolio, but post 2008 slightly underperforms. In the first period, the non-systematic portfolio yields an annualised Sharpe ratio of 5.0, falling to 2.9 in the second period and finally 0.6 in the post-2008 period. By comparison, the raw portfolio has Sharpe ratios of 3.1, 2.4 and 0.7 respectively. This confirms that the non-systematic alpha which drove early hedge fund outperformance has decayed more aggressively than the systematic components.

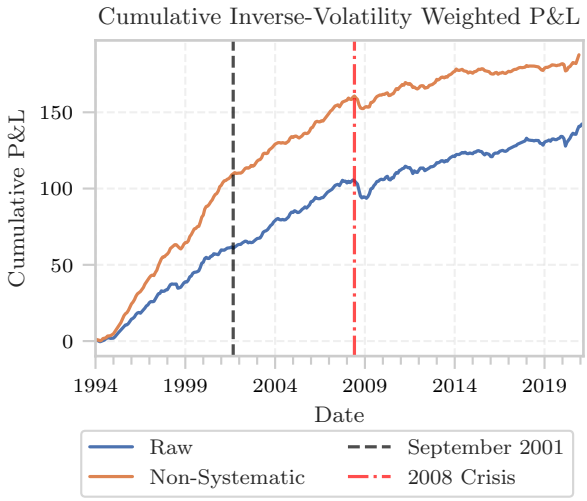


Figure 23: Cumulative P&L of inverse volatility weighted combination of raw and non-systematic returns.

C.3 Fitting Approach

For the non-walkforward analysis, we employed the hybrid MLE-GMM approach detailed in Appendix B.3. We found that a pure MLE approach often converged to non-physical regions, where κ diverged to implausibly large values (e.g., $\kappa = 10$ for the first and second periods). This necessitated the hybrid approach, which identifies parameters of reasonable magnitude that still fit the data well. For example, for the first period, the MLE fitted model has a negative log likelihood (NLL) of 891. The MLE-GMM fitted models have NLLs of 894, 898, and 918 when profiling out κ, (κ, δ), or $(\kappa, \delta, \mathrm{SR}_0)$ respectively, a negligible difference in visual fit.

We ran the fitting procedure using $\{\kappa\}$, $\{\kappa, \delta\}$, or $\{\kappa, \delta, \mathrm{SR}_0\}$ as the GMM profile set. We selected these parameters as they proved reliable for profiling, unlike ℓ, which is difficult to fit via GMM. Our experiments identified $\{\kappa, \delta\}$ as the most "stable" set; that is, it yielded the parameters closest (in L^2 distance) to $\bar{\Theta}$, the mean of the vectors derived from the three profile sets. Consequently, we use $\{\kappa, \delta\}$ as the profile set for the MLE-GMM analysis.

C.4 Fund Clustering

In addition to the returns being suitably independent, we also require that the samples used to fit the Bayesian model are representative of the samples we wish to perform inference on. Naturally, hedge funds are a reasonable proxy for other hedge funds, so one approach is to simply fit the model to all the funds in our dataset, which is the approach taken in the main text. But we can also consider that not all funds behave the same with regard to pre- and post-listing performance. Therefore, we would like to somehow group similar funds together, and investigate if these groups behave differently within our model.

To do this, we cluster our funds together in a two-step process. First, we compute an extensive list of metrics (see Table 9), which we then pass through the factor analysis of mixed data (FAMD) method to reduce the categorical and quantitative variables to a lower dimensional representation

(FAMD is similar to PCA, but allows qualitative in addition to quantitative variables, we use the `prince` library [Hal]), selecting the first 34 components, which explain around 80% of the total variance, as we see in Figure 24.

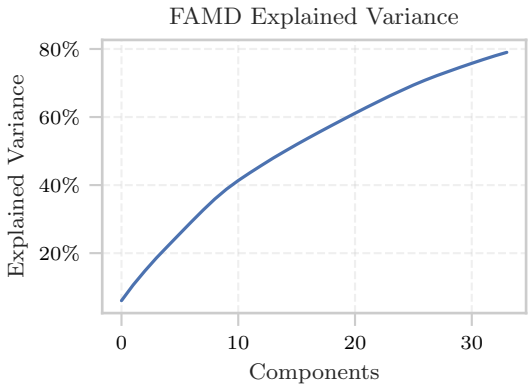


Figure 24: Explained variance by number of components for the FAMD analysis.

Fund Attributes	**Performance**	**Factor Exposure**
• Firm & fund domicile • Strategy style • Fees & high-water mark provision • Lockup, notice period, redemption frequency • Leverage (indicator) and log leverage • Launch, peak & close log AUM • Initial, listing & close year • Sample length: pre-listing, post-listing, total	• Pre-, post-listing & overall Sharpe ratio • Mean, standard deviation, skewness and kurtosis • Maximum drawdown • Replication ratio (OOS/IS SR) • Stability of Sharpe ("Sharpe-of-Sharpe")	• Overall loading magnitude: $\log L_2$ of non-timing loadings; $\log L_2$ of timing loadings; ratio of non-timing to timing loadings • Signed exposures to systematic factors (with squared terms for timing)

Table 9: Qualities supplied to FAMD, grouped by theme. All performance qualities are computed using both the raw and non-systematic returns.

We then perform hierarchical clustering using Ward linkage [War63], and select the number of clusters which maximises the silhouette score [Rou87]. For our dataset, this is 26 clusters, seen in Figure 25.

Many of these clusters are too small for our approach; without sufficient data, the Bayesian parameter estimates would be unreliable. To resolve this, we compute centroids for each cluster and merge those with fewer than 300 funds into the nearest cluster exceeding that threshold. This results in nine final clusters: three large and six small. Figure 26 displays the distribution of fund counts across clusters.

To identify which features are important in cluster allocation, we fit a boosting model to predict the cluster of each fund using the original mixed categorical and quantitative metrics (specifically, we use `catboost` [Pro+19] which natively supports categorical variables). We can then compute SHAP values [LL17; Lun+20] for the model to identify which original features are most important in cluster allocation. Figure 27 plots the features SHAP values in order of importance. Reassuringly, we observe that the fund style is the most important feature, followed by domicile, liquidity terms, leverage, vintage and fees.

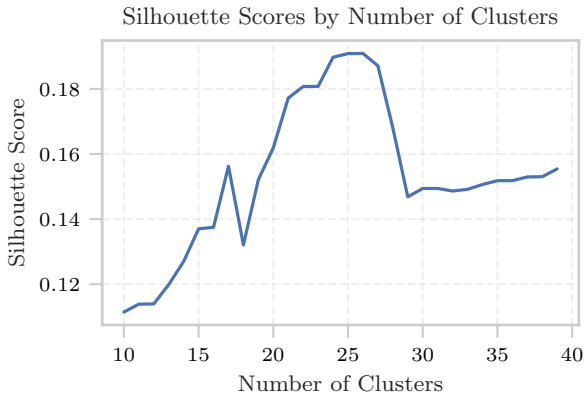


Figure 25: Silhouette score by number of clusters.

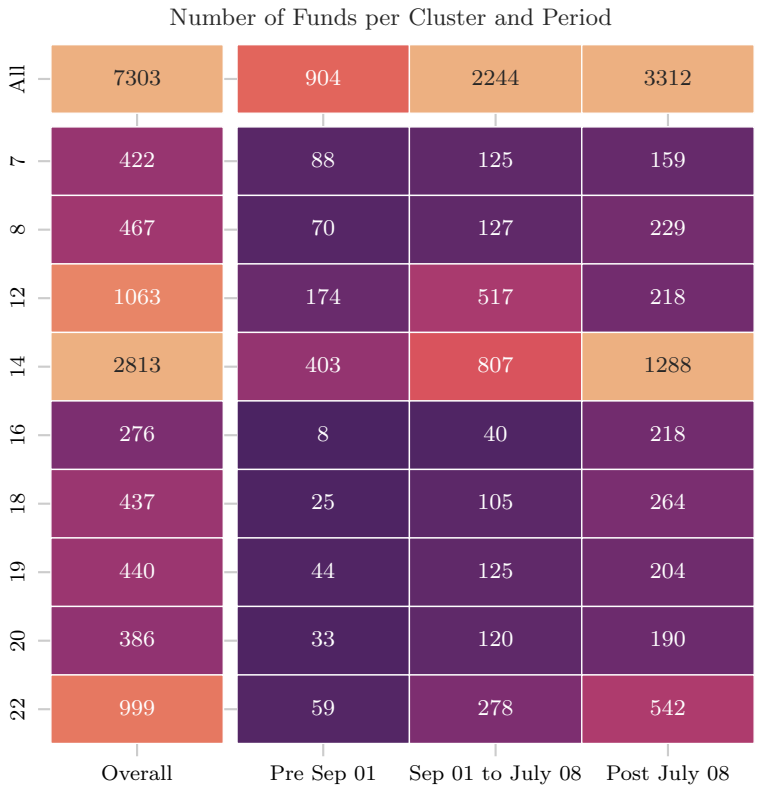


Figure 26: Number of funds per cluster, and per period

As we only have a relatively small number of final clusters, we can easily check the composition of each cluster and identify roughly what each cluster corresponds to. Table 10 shows the breakdown of each cluster by investment style and fund domicile. Most clusters consist primarily of a single style, while three clusters contain a mix of strategies, with equity being the dominant style in these clusters, as can be seen in Figure 28.

As our primary focus is the pre-listing and post-listing performance for each fund, we should check if the clusters have different distributions, as if they simply had the same distribution we could fit one model and use it for all funds. In short, they do indeed have quite different distributions. Figure 29 shows the distributions of pre- and post-listing annualised Sharpe ratios for each cluster.

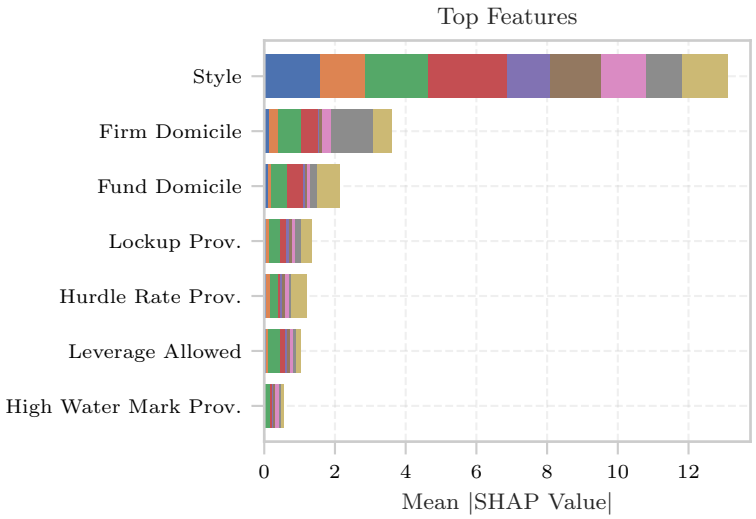


Figure 27: Top features for cluster selection, sorted by mean |SHAP Value|.

Cluster	Style	Fund Domicile
7	Event Driven	Caribbean/North America
8	Relative Value (RV)	Caribbean/NA/EU
12	Fund of Funds (FoF)	Caribbean/North America
14	Equity + Multi-Strat	Caribbean/North America
16	Credit	Caribbean/NA/EU
18	Macro	Caribbean/NA/EU
19	CTA	Caribbean/NA/EU
20	Equity + FoF + CTA + RV	Caribbean/EU
22	Equity + FoF	Europe

Table 10: Cluster composition. Note most clusters are primarily US firm domicile, bar cluster 22 which is European.

As discussed in the main text, we split the data into three periods defined by the regimes observed in Appendix C.2: pre-September 2001, September 2001 to July 2008, and post-July 2008. The first two periods have insufficient sample sizes for cluster-specific fitting. Consequently, we analyse cluster-level fits only for the full time series and the post-2008 period.

We report the parameter estimates per cluster in the overall and post 2008 periods in Table 11. We can draw a few inferences directly from the estimated parameter estimates. Firstly, the amount of alpha decay is quite high across all clusters, with δ generally below 0.6. Secondly, the selection effect varies across clusters, but is generally quite soft, with $\ell < 10$. Thirdly, the distribution of true Sharpe ratios is estimated to have fat tails across almost all clusters, with ν typically < 6.

Finally, we can look at the posterior estimates $\mathbb{E}[\mathrm{SR}_{\delta,i}|\widehat{\mathrm{SR}}_{\mathrm{IS},i}, \widehat{\mathrm{SR}}_{\mathrm{OOS},i}]$. These reflect our best Bayesian guess for the true Sharpe ratio for each fund, which we can decompose into a haircut due to Bayesian shrinkage and selection bias (these two effects are intertwined, as previously discussed) and alpha decay. Figure 30 shows this decomposition. For each fund we compute the posterior estimate for the true Sharpe ratio $\mathbb{E}[\mathrm{SR}_{\delta,i}|\widehat{\mathrm{SR}}_{\mathrm{IS},i}, \widehat{\mathrm{SR}}_{\mathrm{OOS},i}]$, and then apply the decay factor δ. We then compute the haircut vs the pre-listing return per fund, $\mathbb{E}[\mathrm{SR}_{\delta,i}]/\widehat{\mathrm{SR}_{\delta,i}}$ and $\delta\mathbb{E}[\mathrm{SR}_{\delta,i}]/\widehat{\mathrm{SR}}_{\mathrm{IS},i}$, and plot the medians of these values. We observe that the split between initial shrinkage and alpha decay varies significantly by cluster, with some clusters being dominated by alpha decay (for example, Cluster 7, Event driven funds) while others are primarily shrinkage dominated with

Whole Sample						
	SR_0	ς_0	ν	κ	ℓ	δ
All Funds	-0.07	1.00	3.44	1.73	4.34	0.39
Cluster 7	0.83	0.97	3.25	4.12	2.82	0.40
Cluster 8	0.31	1.14	5.78	6.48	2.19	0.53
Cluster 12	0.55	1.49	3.23	0.80	2.55	0.37
Cluster 14	0.28	0.66	4.09	3.60	3.39	0.41
Cluster 16	0.17	1.04	3.64	5.20	1.98	0.58
Cluster 18	0.00	0.73	14.25	3.60	2.85	0.54
Cluster 19	0.52	0.83	3.84	-0.24	4.32	0.30
Cluster 20	-0.35	0.69	4.31	2.32	4.37	0.61
Cluster 22	-0.17	0.73	2.86	2.88	2.47	0.49
Post 2008						
	SR_0	ς_0	ν	κ	ℓ	δ
All Funds	-0.03	0.73	5.21	4.12	2.50	0.55
Cluster 7	0.35	0.59	5.96	4.64	3.43	0.51
Cluster 8	-0.10	0.97	6.10	6.03	2.81	0.61
Cluster 12	0.31	1.39	4.71	-1.28	12.32	0.60
Cluster 14	-0.03	0.87	2.26	1.39	4.77	0.56
Cluster 16	0.35	2.36	2.43	-1.04	7.08	0.56
Cluster 18	-0.28	0.62	9.85	2.70	2.81	0.73
Cluster 19	0.35	0.73	3.93	6.27	1.15	0.26
Cluster 20	-0.17	0.69	3.71	1.70	3.27	0.65
Cluster 22	-0.24	0.52	2.94	5.65	1.21	0.83

Table 11: Posterior parameter estimates by cluster and period. SR_0, ς_0 and κ are reported in annualised terms, divide by $\sqrt{12}$ to yield the original parameter estimates.

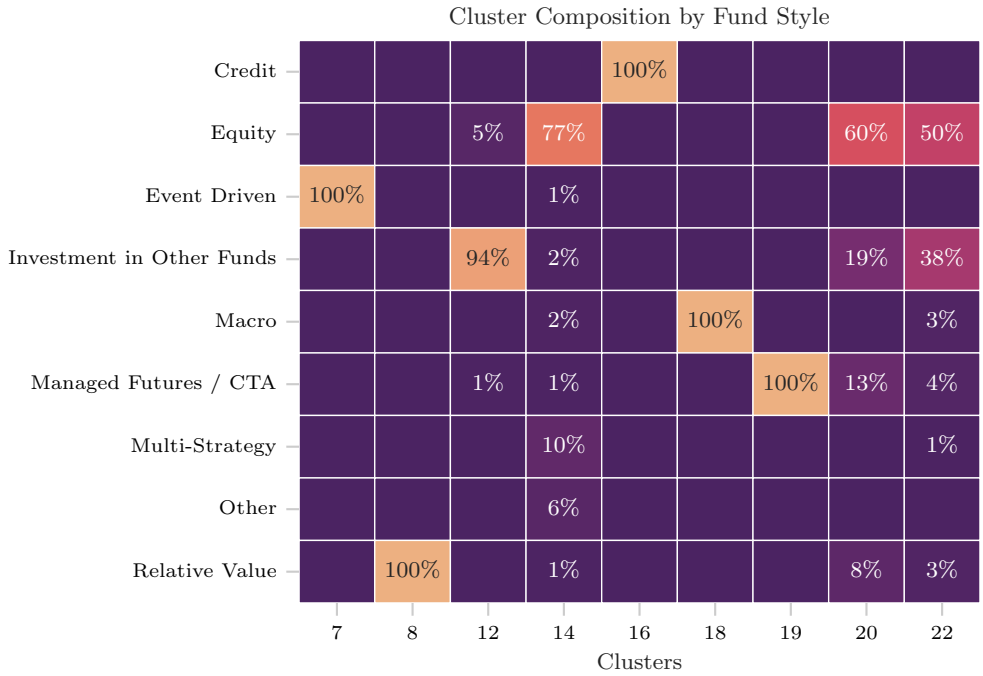


Figure 28: Cluster composition by style.

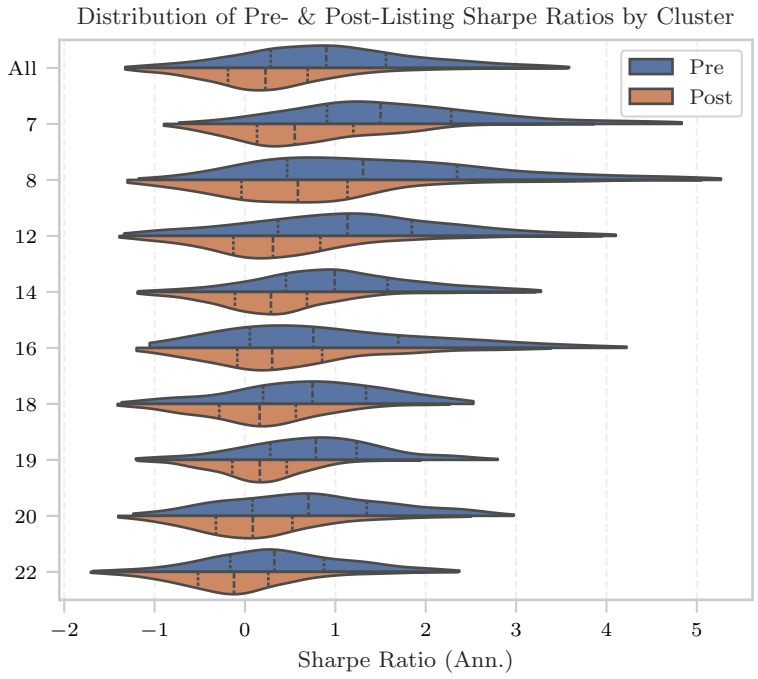


Figure 29: Pre- and Post-Listing Sharpe ratio distribution by cluster.

relatively little alpha decay (for example, Cluster 20). This variation in parameter estimates across fund types suggests that a global model may mask important style-specific dynamics. Consequently, we compare using these cluster-level posterior distributions vs a global model, identifying marginal improvements in performance.

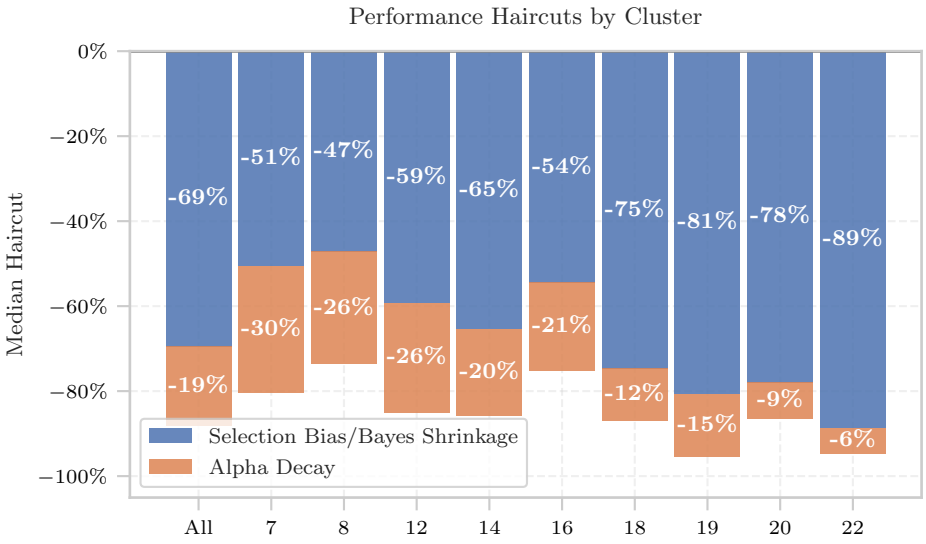


Figure 30: Median pre- to post-listing Sharpe ratio haircuts by cluster, decomposed into selection bias and alpha decay.

C.5 Comparator Models & Metrics

C.5.1 Point Prediction & Model Calibration Metrics

We start by defining $\mathrm{SR}_i^{\mathrm{pred}}$ as the prediction for the out-of-sample Sharpe ratio $\widehat{\mathrm{SR}}_{\mathrm{OOS},i}$ implied by the model. For the various Bayesian models, this is the alpha decayed posterior mean of the true Sharpe ratio,

$$\mathrm{SR}_i^{\mathrm{pred}} := \delta\, \mathbb{E}\Big[\mathrm{SR}_i \;\Big|\; \widehat{\mathrm{SR}}_{\mathrm{IS},i}\Big],$$

and for the James–Stein approach it is simply the James–Stein estimator $\widehat{\mathrm{SR}}_{\mathrm{IS},i}^{\mathrm{JS}}$, as defined in Appendix C.5.2. As the most naive estimator, we can simply take the in-sample Sharpe ratio directly $\widehat{\mathrm{SR}}_{\mathrm{IS},i}$.

The first measure of prediction quality is the mean squared error (MSE), which we compute for both our predictions $\mathrm{SR}_i^{\mathrm{pred}}$ and our naive estimate $\widehat{\mathrm{SR}}_{\mathrm{IS},i}$,

$$\mathrm{MSE} := \frac{1}{N}\sum_{i=1}^{N}\Big(\mathrm{SR}_i^{\mathrm{pred}} - \widehat{\mathrm{SR}}_{\mathrm{OOS},i}\Big)^2, \quad \mathrm{MSE}_{\mathrm{base}} := \frac{1}{N}\sum_{i=1}^{N}\Big(\widehat{\mathrm{SR}}_{\mathrm{IS},i} - \widehat{\mathrm{SR}}_{\mathrm{OOS},i}\Big)^2,$$

with lower values indicating that the prediction better captures the central tendency of out-of-sample Sharpe ratios. To aid interpretation, we report the percentage improvement in MSE relative to the baseline,

$$\mathrm{Improvement} := 1 - \frac{\mathrm{MSE}}{\mathrm{MSE}_{\mathrm{base}}}.$$

However, this disregards one of the main benefits of Bayesian approaches: the quantification of uncertainty. Specifically, a model which is confident but wrong should be penalised more than a model which correctly identifies its own uncertainty. To assess this we follow [Gel+13, Ch. 7] and use the log predictive density to measure probabilistic predictive accuracy.

For each fund we compute the predictive density of the out-of-sample Sharpe ratio at its observed value $\widehat{\mathrm{SR}}_{\mathrm{OOS},i}$. This is the posterior predictive density, obtained by marginalising the out-of-sample likelihood over the posterior distribution of the fund's true Sharpe ratio:

$$f\Big(\widehat{\mathrm{SR}}_{\mathrm{OOS},i} \;\Big|\; \widehat{\mathrm{SR}}_{\mathrm{IS},i}\Big) = \int_{\mathrm{SR}_i} f\Big(\widehat{\mathrm{SR}}_{\mathrm{OOS},i} \;\Big|\; \mathrm{SR}_i\Big)\, f\Big(\mathrm{SR}_i \;\Big|\; \widehat{\mathrm{SR}}_{\mathrm{IS},i}\Big)\, \mathrm{d}\mathrm{SR}_i,$$

from which we form the log pointwise predictive density (LPPD) contribution

$$\mathrm{LPPD}_i \coloneqq \log f\Big(\widehat{\mathrm{SR}}_{\mathrm{OOS},i} \;\Big|\; \widehat{\mathrm{SR}}_{\mathrm{IS},i}\Big),$$

and thus compute the mean LPPD (MLPPD)

$$\mathrm{MLPPD} \coloneqq \frac{1}{N}\sum_{i=1}^{N}\mathrm{LPPD}_i.$$

Higher values of MLPPD indicate that the predictive distribution assigns more probability mass to the realised post-listing Sharpe ratios.

To provide a probabilistic benchmark (in the same sense that we use SR_i as a naive benchmark for the MSE) we also construct a baseline predictive distribution by treating the in-sample Sharpe ratio as a point estimate for the true Sharpe ratio and use Lo's approximation for the sampling distribution of Sharpe ratios [Lo02] to yield

$$\widehat{\mathrm{SR}}_{\mathrm{OOS},i} \mid \widehat{\mathrm{SR}}_{\mathrm{IS},i} \sim \mathcal{N}\left(\widehat{\mathrm{SR}}_{\mathrm{IS},i}, \frac{1+\frac{1}{2}\widehat{\mathrm{SR}}_{\mathrm{IS},i}^{\,2}}{T_{\mathrm{OOS},i}}\right).$$

Let $\mathrm{LPPD}_i^{\mathrm{base}}$ denote the corresponding log pointwise predictive density contribution from this baseline, and let

$$\mathrm{MLPPD}_{\mathrm{base}} \coloneqq \frac{1}{N}\sum_{i=1}^{N}\mathrm{LPPD}_i^{\mathrm{base}}.$$

When presenting results we report the difference

$$\Delta\mathrm{MLPPD} \coloneqq \mathrm{MLPPD} - \mathrm{MLPPD}_{\mathrm{base}},$$

so that larger values of ΔMLPPD indicate an improvement over the naive baseline in terms of probabilistic predictive accuracy.

As a complementary diagnostic, we examine the calibration of the predictive distributions. For each fund we compute the probability integral transform (PIT)

$$u_i \coloneqq F_{\widehat{\mathrm{SR}}_{\mathrm{OOS},i}\mid\widehat{\mathrm{SR}}_{\mathrm{IS},i}}\big(\widehat{\mathrm{SR}}_{\mathrm{OOS},i}\big),$$

where $F_{\widehat{\mathrm{SR}}_{\mathrm{OOS},i}\mid\widehat{\mathrm{SR}}_{\mathrm{IS},i}}$ is the predictive cumulative distribution function of $\widehat{\mathrm{SR}}_{\mathrm{OOS},i}$ conditional on $\widehat{\mathrm{SR}}_{\mathrm{IS},i}$. If the model is correctly specified and well calibrated, then the PIT values $\{u_i\}$ should behave approximately as independent draws from a Uniform$(0,1)$ distribution [Gel+13, Ch. 6].

To assess deviations from uniformity we compute the Kolmogorov–Smirnov statistic

$$\mathrm{KS} \coloneqq \sup_{u\in[0,1]}\Big|\widehat{F}(u) - u\Big|,$$

where $\widehat{F}$ is the empirical distribution function of the PIT values. From this we compute the associated p-value under the null hypothesis of uniformity. Smaller values of KS and larger p-values are consistent with adequate model calibration.

For comparison, we compute the same calibration diagnostics for the naive baseline predictive distribution. Let

$$u_{i,\mathrm{base}} \coloneqq F^{\mathrm{base}}_{\widehat{\mathrm{SR}}_{\mathrm{OOS},i}\mid\widehat{\mathrm{SR}}_{\mathrm{IS},i}}\big(\widehat{\mathrm{SR}}_{\mathrm{OOS},i}\big)$$

denote the PIT values obtained from the baseline predictive CDF, and let $\mathrm{KS}_{\mathrm{base}}$ be the associated Kolmogorov–Smirnov statistic. In our summary plots we report the difference

$$\Delta\mathrm{KS} \coloneqq \mathrm{KS}_{\mathrm{base}} - \mathrm{KS},$$

so that larger values of ΔKS correspond to a reduction in the KS distance relative to the naive baseline (i.e., better calibration).

C.5.2 Comparator Models

We apply the metrics above to a set of comparator models that are designed to disentangle the roles of shrinkage, selection bias and alpha decay; we utilise these models as baseline comparisons against our Bayesian approach in Section 3.

James–Stein shrinkage As a robust benchmark we use the James–Stein estimator [JS60; EM75; Jor86], which performs cross-sectional shrinkage without explicitly modelling selection bias or decay. The estimator is given by

$$\widehat{\mathrm{SR}}^{\mathrm{JS}}_{\mathrm{IS},i} \coloneqq \overline{\mathrm{SR}} + \eta\left(\widehat{\mathrm{SR}}_{\mathrm{IS},i} - \overline{\mathrm{SR}}\right),$$

where $\overline{\mathrm{SR}}$ denotes the grand mean of the in-sample Sharpe ratios $\frac{1}{N}\sum_{i=1}^{N}\widehat{\mathrm{SR}}_{\mathrm{IS},i}$, and we use positive-part shrinkage [Bar70],

$$\eta = \max\left(0, 1 - \frac{(N-3)\overline{\sigma}^2}{\sum_{j=1}^{N}(\widehat{\mathrm{SR}}_{\mathrm{IS},j} - \overline{\mathrm{SR}})^2}\right),$$

where $\overline{\sigma}^2$ is the average estimation variance of the in-sample Sharpe ratios, calculated as the arithmetic mean of the individual variances derived using Lo's approximation [Lo02]:

$$\overline{\sigma}^2 \coloneqq \frac{1}{N}\sum_{i=1}^{N}\frac{1 + \frac{1}{2}\widehat{\mathrm{SR}}^2_{\mathrm{IS},i}}{T_{\mathrm{IS},i}}.$$

Here, each term $\frac{1+\frac{1}{2}\widehat{\mathrm{SR}}^2_{\mathrm{IS},i}}{T_{\mathrm{IS},i}}$ approximates $\mathbb{V}(\widehat{\mathrm{SR}}_{\mathrm{IS},i}|\mathrm{SR}_i)$ under iid Gaussian returns, with $\widehat{\mathrm{SR}}_{\mathrm{IS},i}$ used as a plug-in estimate for SR_i.

For the predictive density used to compute MLPPD and PIT/KS metrics, we take $\widehat{\mathrm{SR}}^{\mathrm{JS}}_{\mathrm{IS},i}$ as a point estimator for SR_i and use the standard sample distribution for Sharpe ratios, again using Lo's approximation,

$$\widehat{\mathrm{SR}}_{\mathrm{OOS},i}|\widehat{\mathrm{SR}}^{\mathrm{JS}}_{\mathrm{IS},i} \sim \mathcal{N}\left(\widehat{\mathrm{SR}}^{\mathrm{JS}}_{\mathrm{IS},i}, \frac{1 + \frac{1}{2}\left(\widehat{\mathrm{SR}}^{\mathrm{JS}}_{\mathrm{IS},i}\right)^2}{T_{\mathrm{OOS},i}}\right).$$

Unbiased Bayesian Models The first set of Bayesian comparators do not take selection bias into consideration, so that pre- vs post-listing differences are driven entirely by shrinkage and alpha decay. To fit these models we simply proceed by MLE as in Section 2.3, but using the unbiased marginal likelihood

$$f(\widehat{\mathrm{SR}}_{\mathrm{IS},i}, \widehat{\mathrm{SR}}_{\mathrm{OOS},i}) = \int_{\mathrm{SR}}^{\prime} f\left(\widehat{\mathrm{SR}}_{\mathrm{IS},i} \,\middle|\, \mathrm{SR}'\right) f\left(\widehat{\mathrm{SR}}_{\mathrm{OOS},i} \,\middle|\, \delta\mathrm{SR}'\right)\pi(\mathrm{SR}')\,\mathrm{dSR}',$$

rather than the bias aware marginal likelihood we use for the standard model.

To further isolate the importance of alpha decay we fit two models, one where we fix $\delta = 1$ in the optimiser, and a second model where we allow δ to be optimised along with the other parameters.

Bias aware Bayesian models. These are the full models introduced in the main text, which explicitly account for selection bias and alpha decay. We consider three variations:

- *Random IS only.* The model is fit using only the in-sample (pre-listing) Sharpe ratios. This primarily functions to show that model estimation is difficult using the in-sample only data; as we see in the main text, this model collapses to a poor fit.

- *Random.* The baseline Bayesian model using both pre- and post-listing Sharpe ratios for the fitting process.
- *Random with clusters.* As a minor illustration of why fitting with similar strategies (in terms of performance) is a useful endeavour, we fit a model to each of the clusters created in Appendix C.4. We will observe that this can provide minor improvements in model calibration.

C.6 Walkforward Analysis

We have also conducted a walkforward style analysis which avoids the lookahead bias in the fitting of the model parameters Θ. To do this, we walk forward a year at a time and use only the returns up to that year-end to fit the model, i.e., compute the pre- and post-listing Sharpe ratios for all funds listed before the end of that year. We then use these fitted parameters to compute the posterior and predictive distributions for all the funds listed the following year, using only their pre-listing returns to estimate the posterior (as we did above). In Figure 37 we plot the evolution of the fitted parameters over time.

The issue with this walkforward approach lies in that there is a step change difference in the performance of funds pre- and post 2008. This both violates our assumptions of stationarity, and yields estimation issues whereby the parameters fit using, say, data from 1996 to 2009 are "stale" relative to the prevailing regime. To resolve this, we start our analysis in either 1996 or 2009 (Jan 31st), and end it in either 2008 or 2020 (December 31st), truncating the returns for each fund to this period. This yields three separate walkforward samples, corresponding to 1996 to 2008, 1996 to 2020 (our entire sample) and 2009 to 2020.

As above, we compare the estimated posterior and predictive distribution using the pre-listing Sharpe ratio to the post-listing Sharpe ratio for each fund, and compute the MSE, MLPPD and K–S statistic improvements relative to the naive baseline of the raw pre-listing Sharpe ratio. The outcome of this analysis is shown in Figure 31. We have excluded the by cluster fit in the interest of computation time, but otherwise include the same models as in the above analysis.

From an MSE perspective, the James–Stein estimator remains unchanged, as expected. However, the unbiased model without alpha decay and the bias adjusted model fitted to in-sample-only data suffer significantly. Crucially, the unbiased model with alpha decay continues to perform well, and the full bias adjusted model dominates across metrics and walkforward periods.

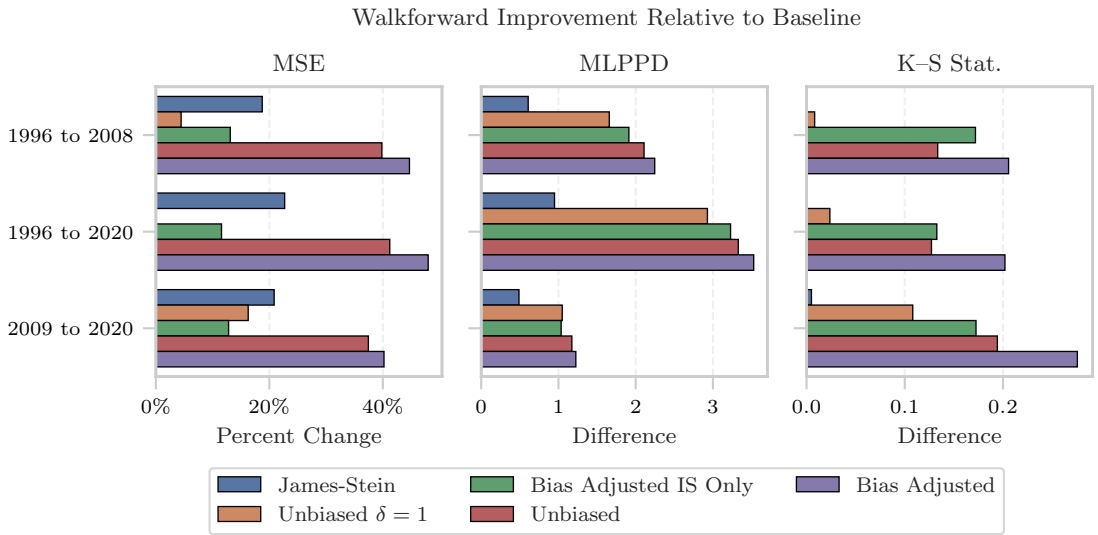


Figure 31: Improvement in statistical metrics relative to a naive baseline when fitted with a walkforward approach.

C.7 Additional Figures & Tables

This section provides an overview of the relationship between AUM and listing date (Figure 32), AUM trends over time (Figure 33), changing trends in the hedge fund universe (e.g. the growth of Credit AUM post 2008, Figure 34), the amount of listed vs unlisted AUM over time (with the caveat that it only covers funds that have listed by the end of the dataset, Figure 35) and the distribution of pre- and post-listing period lengths (Figure 36). Finally, we provide a comparison of fitted model parameters across time periods in Table 12 and evaluate their stability via walkforward analysis in Figure 37, highlighting the shift in selection intensity and alpha decay in the post-2008 era.

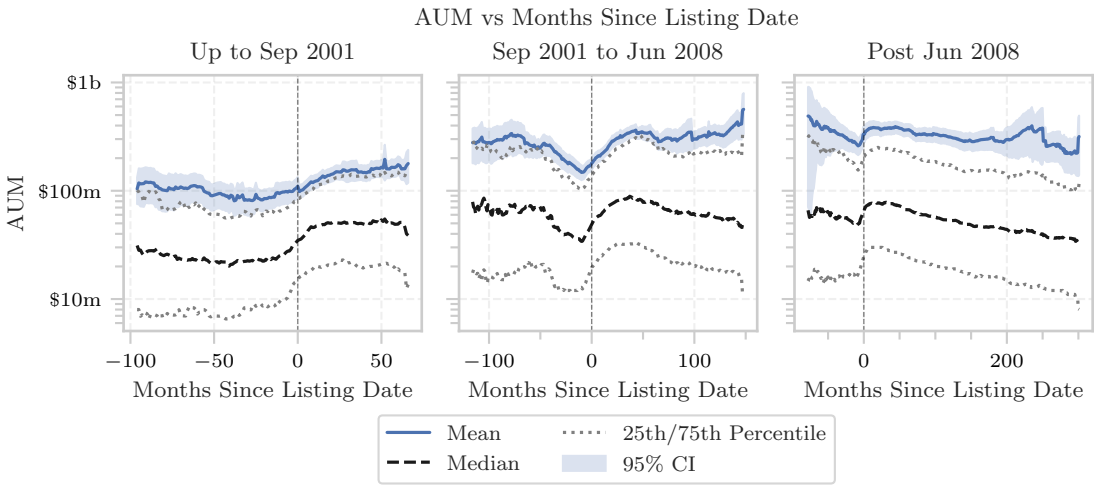


Figure 32: Average AUM in USD vs months pre- or post-listing in commercial databases.

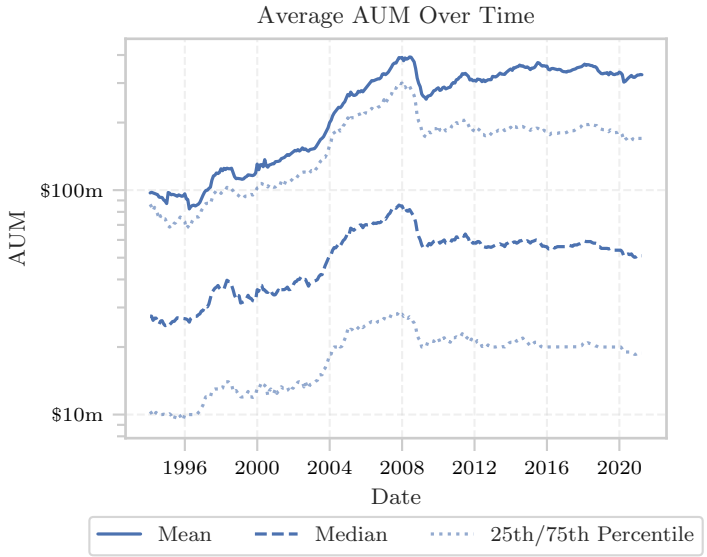


Figure 33: Average AUM in USD over time.

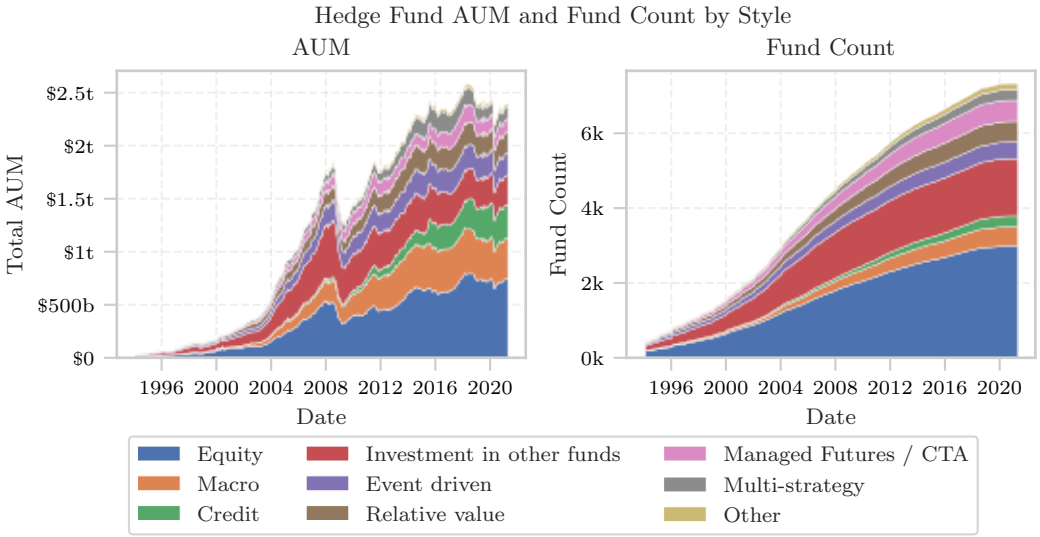


Figure 34: Total AUM in USD, and total fund count, by style over time.

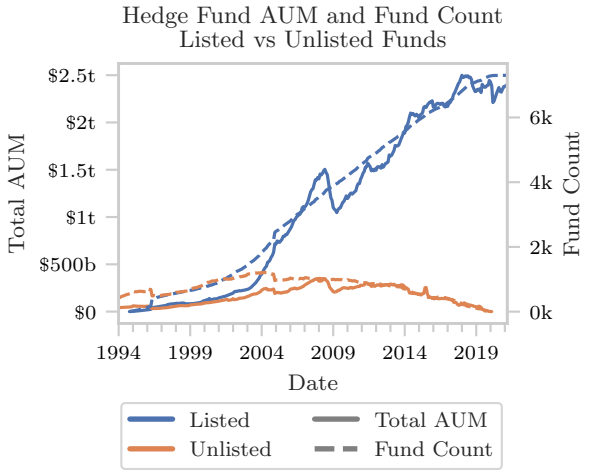


Figure 35: Total AUM in USD, and total fund count, by listing status over time.

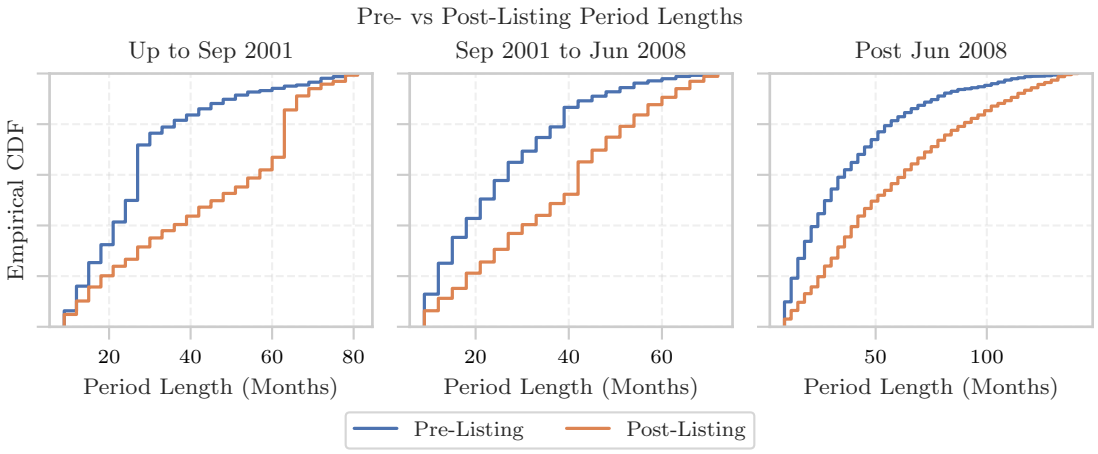


Figure 36: Cumulative distribution of period lengths.

Model / fit type	Param	Up to Sep 2001	Sep 2001 to July 2008	Post July 2008
Unbiased $\delta = 1$ (MLE)	SR_0	0.36 (1.26)	0.25 (0.87)	0.08 (0.27)
	ς_0	0.38 (1.32)	0.30 (1.03)	0.23 (0.78)
	ν	2.59	2.56	3.00
	κ	–	–	–
	ℓ	–	–	–
	δ	1.00	1.00	1.00
Bias Adjusted IS only (MLE)	SR_0	0.24 (0.84)	0.20 (0.69)	0.02 (0.08)
	ς_0	0.29 (1.01)	0.30 (1.03)	0.22 (0.77)
	ν	3.59	3.26	4.11
	κ	2.55 (8.85)	3.89 (13.48)	3.33 (11.52)
	ℓ	1.69	0.85	1.70
	δ	1.00	1.00	1.00
Unbiased (MLE)	SR_0	0.42 (1.47)	0.29 (1.01)	0.12 (0.41)
	ς_0	0.43 (1.50)	0.33 (1.15)	0.32 (1.12)
	ν	2.58	2.64	2.78
	κ	–	–	–
	ℓ	–	–	–
	δ	0.77	0.77	0.54
Bias Adjusted (MLE)	SR_0	0.26 (0.89)	0.24 (0.83)	-0.01 (-0.03)
	ς_0	0.25 (0.87)	0.27 (0.95)	0.22 (0.75)
	ν	4.36	3.04	3.95
	κ	9.65 (33.44)	10.00 (34.64)	1.69 (5.87)
	ℓ	1.83	0.60	2.13
	δ	0.91	0.83	0.68
Bias Adjusted (MLE-GMM κ, δ)	SR_0	0.30 (1.05)	0.27 (0.93)	-0.01 (-0.05)
	ς_0	0.30 (1.04)	1.79 (6.22)	0.21 (0.73)
	ν	3.26	2.01	5.21
	κ	2.33 (8.07)	-0.07 (-0.25)	1.19 (4.12)
	ℓ	1.24	0.86	2.50
	δ	0.87	0.68	0.55

Table 12: Fitted parameters by model and period. For SR_0, ς_0 and κ, annualised equivalents are shown in parentheses. A dash (–) indicates that the parameter is not identified in the corresponding model.

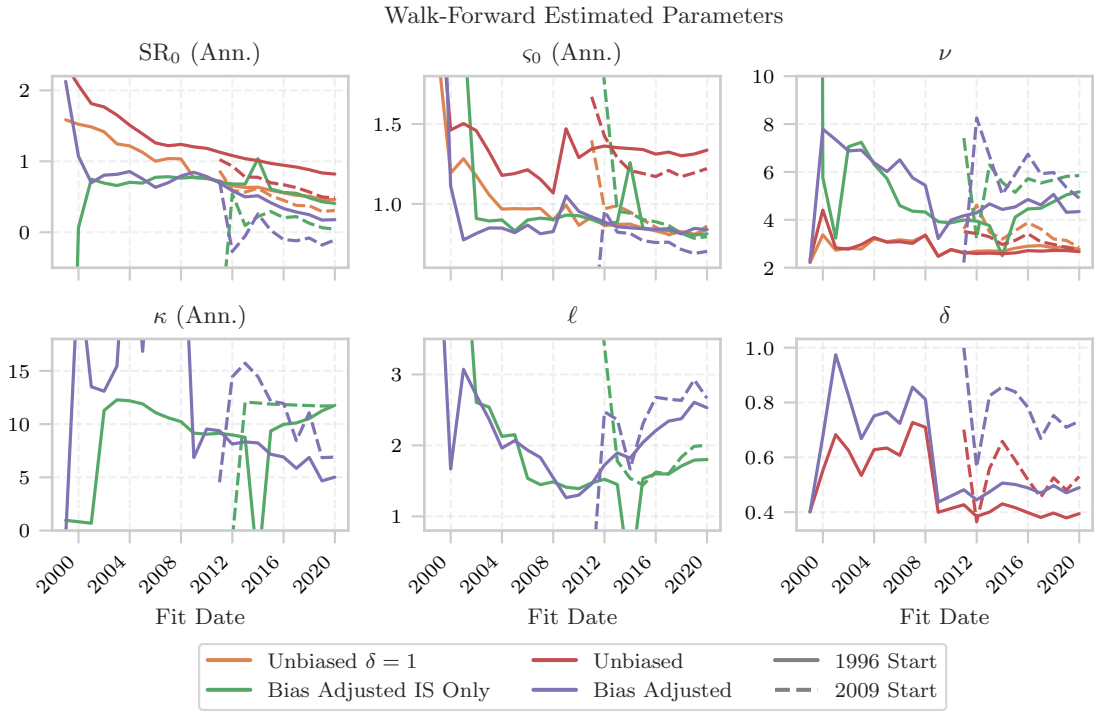


Figure 37: Fitted parameters for the walkforward analysis. Solid lines show the fitted parameters for the 1996 start date, and dashed lines for the 2009 start date. The y-axis is truncated for each plot for clarity.

D Illustrative Examples

The following examples are designed to build intuition for how selection bias distorts the in-sample Sharpe ratio, and how different parameters drive this effect. The results are intended to be illustrative, rather than prescriptive. The model parameters for this section are based on our empirical analysis.

The prior we use here is somewhat pessimistic, as we assume the distribution of true Sharpe ratios is centred around zero. In many scenarios this would likely be shifted to have a more positive midpoint, but the pessimistic prior is useful to demonstrate how severe the issue of selection bias can be. The goal, as ever, is for researchers to apply this framework to their own data to yield insights specific to their strategies, but this section's aim is to provide valuable intuition into how the selection process biases backtests.

Unless otherwise stated, we use the parameters in Table 13 for all the examples in this section, and always use a Gaussian IID likelihood. We assume returns are measured daily with 252 trading days per year, and the parameters reflect this. For visualisations and discussions, however, we will work in annualised scales, as this is the scale most familiar to readers. For example, SR, $\boldsymbol{\theta}$ or κ shown as 1 represents a daily value of $1/\sqrt{252}$. A useful first step is visualising what the prior and likelihoods look like for these parameter values.

Prior	Selection	Sample
$\mathrm{SR}_0 = 0$	$\kappa = 1/\sqrt{252}$	$T_{\mathrm{IS}} = 252 \cdot 3$
$\varsigma_0^2 = 0.5/252$	$\ell = 100$	$T_{\mathrm{OOS}} = 252 \cdot 3$
$\nu = 10$		
$\nu_\vartheta = 10$		
$\omega = 0.9$		

Table 13: Prior, selection and sample parameters for the experiments of Appendix D.

Figure 38 shows the density of the prior for the true Sharpe ratio in the unbiased, hierarchical and random cases. For the hierarchical and random cases, the figure shows the bias adjusted prior. We see that the unbiased prior is centred around SR_0 and the fixed prior is identically distributed, while the hierarchical and random priors have been shifted upwards by the selection process. As we allocate the majority of the variance to the fixed ($\boldsymbol{\theta}$) level of the prior, the hierarchical prior remains close to the unbiased one. As $\omega \to 0$, it would converge towards the random case as the total variance becomes fully allocated to the random SR layer.

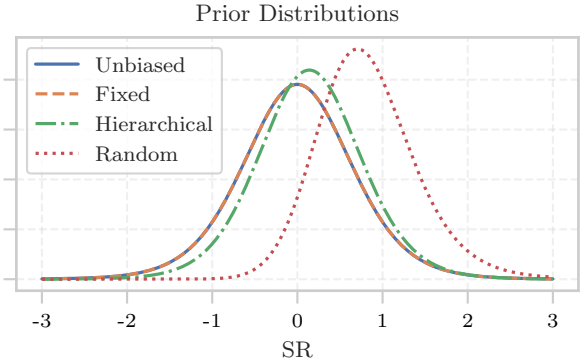


Figure 38: Prior distribution of true Sharpe ratios under unbiased, fixed, hierarchical and random selection.

Looking at the hierarchical prior a touch closer, Figure 39 plots the density of the hyperprior

on $\boldsymbol{\theta}$ and the conditional prior for SR given $\boldsymbol{\theta} = 1$. These densities are not bias adjusted, as the intention is simply to show how much the true Sharpe ratio can vary once $\boldsymbol{\theta}$ is fixed. A relatively small amount of change is possible; three standard deviations of the conditional prior correspond to about 0.66 in annualised Sharpe. This represents a realistic scenario where a researcher finds a core strategy and then tweaks it to yield incremental improvements.

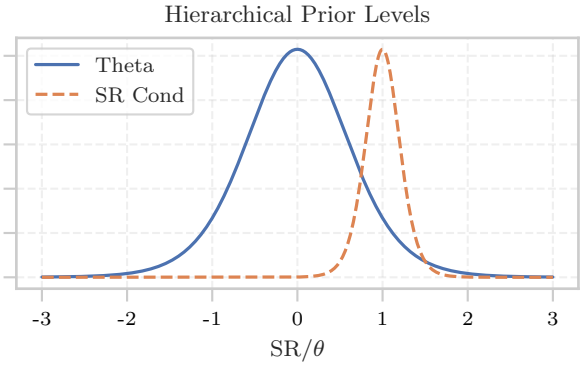


Figure 39: Densities for the hyperprior and conditional prior of the hierarchical case. Note these are not bias adjusted.

Figure 40 shows the marginal likelihoods for the in- and out-of-sample Sharpe ratios under the four cases: unbiased, fixed, random, and hierarchical. In-sample, the likelihoods for the fixed, hierarchical, and random models are shifted substantially upwards. The random model yields the highest Sharpe ratios; because ω is close to 1, the distributions for the fixed and hierarchical models are nearly identical. Out-of-sample, however, the likelihoods for the unbiased, fixed, and hierarchical models are centred around zero, reflecting the prior. In contrast, the random model's likelihood is skewed positive, as the selection process influences the true Sharpe ratio upwards.

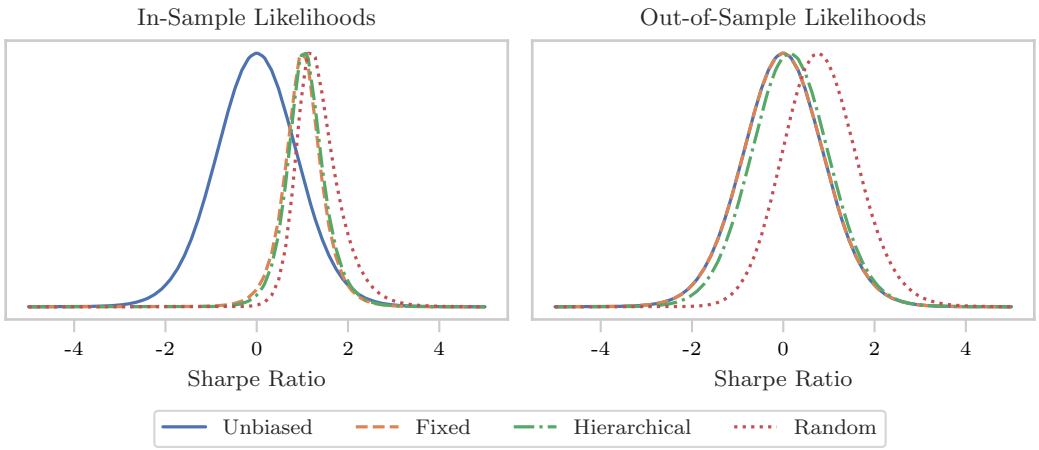


Figure 40: Densities for the in- and out-of-sample Sharpe ratios in the unbiased, fixed, hierarchical and random cases.

To demonstrate how the different selection bias types affect the posterior, we fix $\widehat{\mathrm{SR}}_{\mathrm{IS}} = 1.5$ and $T_{\mathrm{IS}} = 252 \cdot 3$. The resulting posteriors are shown in Figure 41. First, Bayesian regularisation shrinks all posteriors towards the prior. The evidence (backtest length) is insufficient to support a true Sharpe ratio as high as the in-sample estimate, so the posterior is naturally more cautious. Second, the fixed and hierarchical posteriors are even lower, while the random posterior is identical to the unbiased posterior. The intuition is that for the random case, the selection process shifts

both the prior and the likelihood upwards, and the posterior is thus unbiased. In the fixed and hierarchical cases however, the selection process has little or no influence on the true Sharpe ratio, forcing a downward adjustment to the posterior.

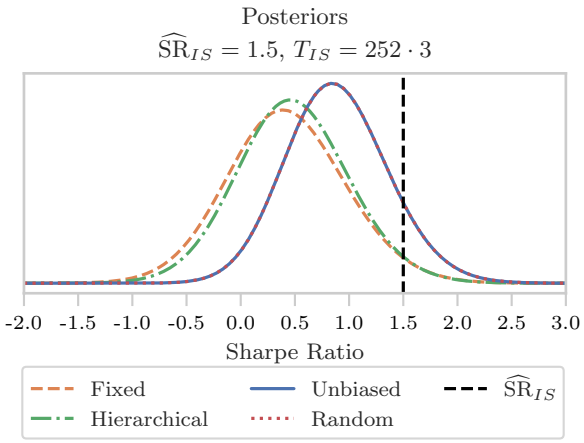


Figure 41: Posteriors for the fixed, hierarchical, random and unbiased cases; $\widehat{\mathrm{SR}}_{\mathrm{IS}} = 1.5$, $T_{\mathrm{IS}} = 252 \cdot 3$.

How does this change for varying $\widehat{\mathrm{SR}}_{\mathrm{IS}}$ and T_{IS}? In Figure 42 we plot the replication ratio $\mathbb{E}[\mathrm{SR} \mid \widehat{\mathrm{SR}}_{\mathrm{IS}}]/\widehat{\mathrm{SR}}_{\mathrm{IS}}$ (the ratio of the posterior mean to the in-sample Sharpe ratio) for the unbiased/random cases, which are equal, and the fixed case. The hierarchical case is omitted because for $\omega = 0.9$ it is very similar to the fixed case. In the unbiased/random cases, the replication ratio rises slightly with the in-sample Sharpe ratio, but increases primarily with the length of the backtest T_{IS}. In the fixed (and hierarchical) case, the ratio depends strongly on the distance between the sample Sharpe ratio and the selection threshold. With $\kappa = 1$, an in-sample Sharpe ratio just above this level warrants scepticism; it is more likely to be a low true Sharpe ratio that had an unusually high in-sample realisation. As the in-sample Sharpe ratio moves further from the threshold, this possibility diminishes, and it becomes more likely that the true Sharpe ratio that yielded a large in-sample Sharpe ratio is itself large, and therefore less distorted by selection bias. As in the unbiased/random case, longer backtests mitigate this effect by reducing the measurement error in the sample Sharpe ratio.

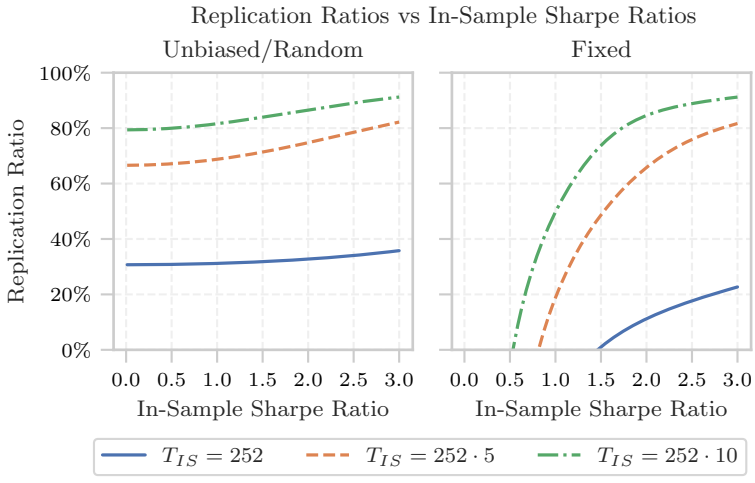


Figure 42: Replication ratios for unbiased/random selection bias vs fixed selection bias.

Next we examine how the variance split affects the replication ratio in the hierarchical case. Figure 43 shows that as the variance split ω approaches one, allocating more variance to the fixed ($\boldsymbol{\theta}$) level of the prior, the replication ratio decreases. The intuition is straightforward: if our selection process has less influence on the true Sharpe ratio, a high in-sample result is likely driven more by overfitting than by genuine improvement. We should therefore reduce our expectations for out-of-sample performance.

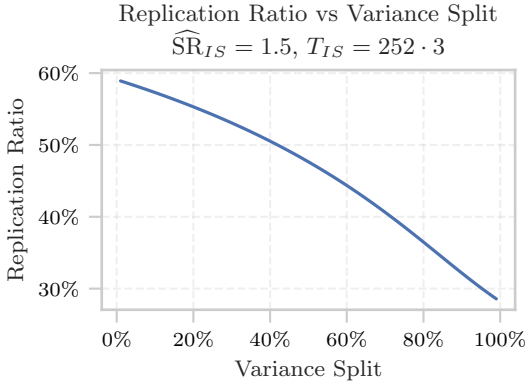


Figure 43: Replication ratio $\mathbb{E}[\mathrm{SR} \mid \widehat{\mathrm{SR}}_{\mathrm{IS}}]/\widehat{\mathrm{SR}}_{\mathrm{IS}}$ by variance split ω.

Rather than vary the observed in-sample Sharpe ratio, we can also examine how the replication ratio responds to changes in the prior and the selection bias parameters, as shown in Figure 44 and Figure 45 respectively. In each plot, we vary only one parameter while holding the others fixed to their values in Table 13.

In Figure 44, we plot the random and fixed cases as we vary the prior parameters. We omit the unbiased case, which is equivalent to the random case, and the hierarchical case, which is an interpolation of the two. In the fixed case, not only is the replication ratio lower across the board, but its response to changing prior parameters is also different. As the prior mean SR_0 increases towards the in-sample Sharpe ratio ($\widehat{\mathrm{SR}}_{\mathrm{IS}} = 1.5$), the gap between the fixed and random cases closes. A higher prior mean for the true Sharpe ratio lessens the impact of selection bias, causing the posterior in the fixed case to converge towards the random one. Additionally, and naturally, as SR_0 increases we see that Bayesian shrinkage decreases, and once $\mathrm{SR}_0 > \widehat{\mathrm{SR}}_{\mathrm{IS}}$ it pulls our estimate of the true Sharpe ratio above the in-sample value. Relatedly, but to a lesser degree, as the variance of the prior ς_0^2 increases we see that the replication ratio increases. This is caused by a higher likelihood of observing large true Sharpe ratios, and therefore our fairly conservative in-sample Sharpe ratio $\widehat{\mathrm{SR}}_{\mathrm{IS}} = 1.5$ becomes more probable. However, as the mean SR_0 is still less than κ, there is still a degree of selection bias, and thus the fixed replication ratio is much lower than the random one.

In Figure 45, we plot only the fixed case, as the replication ratio for the unbiased/random case is unaffected by the selection bias parameters. As in Figure 42, we observe that the replication ratio depends on the distance between the threshold κ and the in-sample Sharpe ratio $\widehat{\mathrm{SR}}_{\mathrm{IS}}$. As κ increases the replication ratio decreases, becoming negative for large κ. It is possible to have $\kappa > \widehat{\mathrm{SR}}_{\mathrm{IS}}$ because the selection threshold with $\ell = 100$ is "soft". As this threshold gets sharper (i.e., as ℓ increases), the replication ratio falls rapidly. This reflects the increasing strictness of the selection procedure. For large ℓ any strategy with $\widehat{\mathrm{SR}}_{\mathrm{IS}} < \kappa$ is discarded. Thus, the selection bias effect becomes stronger, and observed Sharpe ratios just above this threshold are more likely to be lucky than genuinely significant.

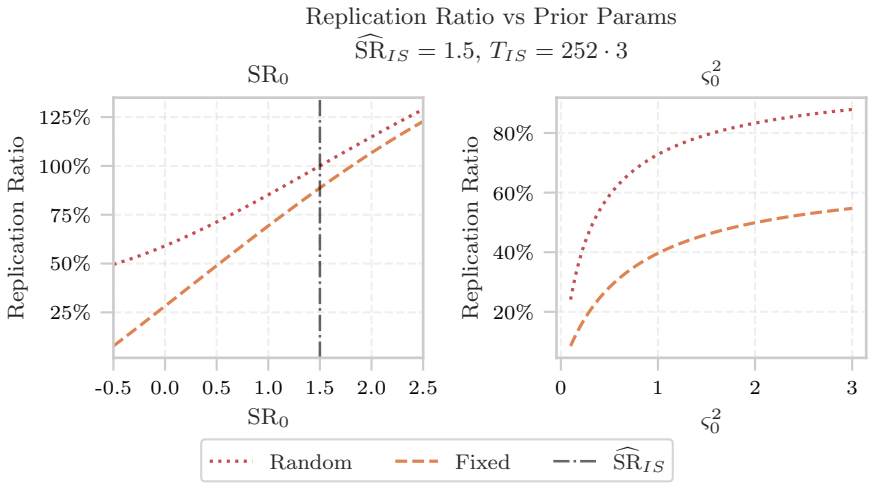


Figure 44: Replication ratio by SR_0 and ς_0^2.

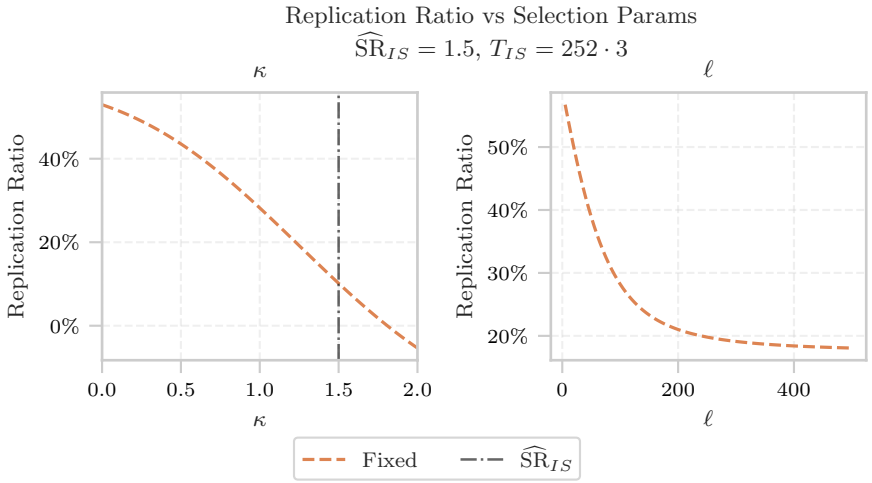


Figure 45: Replication ratio by κ and ℓ.

E Comparison to Chen and Zimmermann

Our approach is conceptually related to the study of Chen and Zimmermann [CZ20] (CZ), who develop a model-implied estimate of publication bias in the asset pricing literature. Here, we explore the theoretical differences between our models and assess how these differences affect fitting and inference when applied to the predictor data they kindly provide.

The primary distinction is one of objective. CZ seek to estimate the magnitude of publication bias under a structural model, and use only in-sample data for this goal. We aim to produce the most accurate forecast of an investment's future performance. This goal motivates our use of all available data, both biased in-sample and unbiased out-of-sample returns, and informs the key methodological differences that follow.

First, our model explicitly incorporates alpha decay. This allows us to fit in-sample and out-of-sample data jointly, disentangling the distinct effects of selection bias and performance degradation over time. CZ acknowledge alpha decay but treat it as a residual between their bias adjusted estimates and the observed post-publication return. They choose to use in-sample data alone, arguing that out-of-sample periods are clouded by economic effects that would corrupt a pure

measure of publication bias. By modelling alpha decay as a parameter, we can account for these effects directly. This joint-fitting also yields improved parameter estimates, a significant benefit for smaller datasets like the one here, with roughly 200 predictors (Appendix B.1).

Second, we focus on different performance metrics. As our work is oriented toward capital allocation, we concentrate on the Sharpe ratio. Accordingly, our likelihood function is built from the asymptotic distribution of the sample Sharpe ratio [Lo02; LW08; Pav21]; the most general form that we use here accommodates non-Gaussian and autocorrelated returns [PLZ26]. CZ, in line with the academic literature they study, focus on mean returns and t-statistics, employing a Normal distribution for the likelihood. While the metrics are related, our choice allows for a more direct and robust analysis of risk-adjusted performance.

Finally, our model addresses a broader set of selection mechanisms, which we explore in Appendix A.2. CZ focus exclusively on random selection, a specific case that is well-suited for their inferential goal but is one of several scenarios our framework can handle.

With the methodological differences established, we can turn to studying their impact on the open-source asset pricing (OSA) dataset, which Chen and Zimmermann kindly maintain [CZ22]. The dataset contains reproduced monthly returns for 212 predictors from the asset pricing literature, of which we select 203 after dropping factors which have time series that end prior to 2023.

The OSA dataset includes returns for a variety of portfolio construction methods. We select the kind used by CZ, the "Original Cuts" dataset, which splits the assets by quantile and equally weights the assets within each quantile. The authors also calculate a long-short portfolio return by simply going long the top (or bottom) quantile, and shorting the bottom (or top) quantile. The dataset also includes the start and end year used by the original paper for each predictor, and the year that paper was published. Using these three dates we can define a few key periods:

- The In-Sample Period: The period used by the original paper for evaluating the predictor,
- The Pre-Publication Period: The period between the end of the in-sample period and the publication date,
- The Post-Publication Period: The period from the publication date till now,
- The Post-Sample Period: The pre-publication and post-publication period combined.

Chen and Zimmermann note that for some factors their reproduced t-statistics are lower than the original paper, for example the predictors by Haugen and Baker [HB96]. Often, the top and bottom quantiles have significantly different returns profiles (in terms of both expected returns and volatility). We found that the t-statistics for these predictors returned to significant levels by taking a Markowitz combination of the top and bottom quantiles. To construct the Markowitz portfolio we compute the mean and covariance of the top and bottom quantiles using the in-sample period, and use the calculated weights to compute the Markowitz Long-Short return for all the periods.

Figure 46 shows the cumulative returns for the cash flow yield variance (VarCF) predictor using CZ's original long-short portfolio, and our Markowitz constructed long-short portfolio, normalised such that the monthly volatility of both returns time series is 1. We see that the Markowitz constructed portfolio performs significantly better than the original, both in-sample and post-sample. The in-sample t-statistic increases from 1.9 to 3.7, and the post-sample t-statistic from 0.3 to 3.1.

In addition to the predictors CZ found problematic (those with lower than expected in-sample t-statistics), we find that the Markowitz construction improves t-statistics for all the predictors, both in- and post-sample. Figure 47 shows these distributions for the original and Markowitz portfolios. As expected, the in-sample t-statistics are significantly higher for the Markowitz portfolios, since they are built using in-sample moments, but the distribution of post-sample t-statistics improves too. The mean in-sample t-statistic increases from 4.0 to 6.2, and the mean post-sample t-statistic increases from 1.7 to 3.4.

Thus, for our comparison we will use the Markowitz constructed portfolios, rather than the original portfolios, as we feel they more accurately capture the true performance of each predictor.

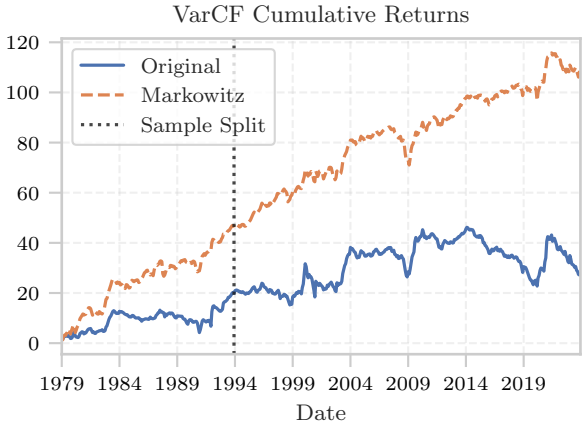


Figure 46: Cumulative long-short returns for the cash flow yield variance predictor [HB96] using the original or Markowitz construction.

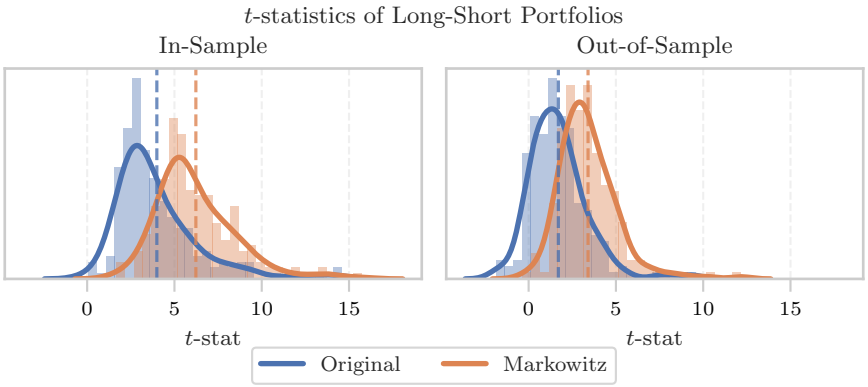


Figure 47: In- and post-sample long-short t-statistics for original and Markowitz portfolios, dashed lines denote means.

Additionally, as our framework is grounded in Sharpe ratios, rather than t-statistics, we will conduct our experiments in terms of Sharpe ratios, but these could be converted back into t-statistics at the end by a simple rescaling according to sample length. This does of course mean any comparison to particular values from [CZ20] will not be an exact apples-to-apples comparison, but we aim to capture the spirit of their model versus our own.

We fit six models. The first three use in-sample data alone, in-sample with pre-publication data, or in-sample with post-sample data. The remaining three are variants where we fix $\mathrm{SR}_0 = 0$, mirroring the approach of [CZ20]. Each model is fitted using a non-Gaussian AR(1) likelihood with moments estimated from the data: autocorrelation $\phi = 0.11$, skewness $\gamma_{3,r} = 0.77$, and kurtosis $\gamma_{4,r} = 15.52$. The parameters are estimated by taking the average across all the predictors, and by using the full time-series available for each predictor.

The fitted parameters are reported in Table 14. For the three models where SR_0 is fitted, its value is stable, varying by only 0.03 (0.1 annualised) around the in-sample estimate of 0.27 (0.92 annualised). Notably, the alpha decay parameter is much lower for the in- and post-sample fit than for the in-sample and pre-publication fit, which aligns with the expectation that the pre-publication period is more insulated from alpha decay. Across all six fits, the selection function remains nearly identical, with $\kappa \approx 0.2$ (0.7 annualised) and $\ell \approx 45$.

The primary difference between fixing SR_0 and fitting it lies in the prior's standard deviation ς_0 and kurtosis ν parameters. When SR_0 is fitted, the prior is narrower ($\varsigma_0 \approx 0.15$) and has fatter

tails ($\nu \approx 5$). In contrast, fixing $\text{SR}_0 = 0$ results in a wider prior ($\varsigma_0 = 0.24$) with little excess kurtosis, due to a large estimated ν. We believe this is because fixing the mean at zero requires a wider prior to capture the bulk of the in-sample distribution; the model then fits a larger ν to reduce kurtosis, preventing the tails from becoming excessively fat.

	IS Only		**IS & Pre**		**IS & OOS**	
	Fit SR_0	$\text{SR}_0 = 0$	Fit SR_0	$\text{SR}_0 = 0$	Fit SR_0	$\text{SR}_0 = 0$
SR_0	0.27	0.00	0.24	0.00	0.30	0.00
ν	5.67	100	4.37	25.83	3.24	100
ς_0	0.15	0.24	0.17	0.24	0.14	0.24
δ	1.00	1.00	0.93	0.94	0.61	0.62
κ	0.19	0.22	0.20	0.22	0.18	0.22
ℓ	45.43	45.10	42.93	44.54	46.64	44.68

Table 14: Fitted parameter values for Appendix E. Note that the parameter values are in monthly terms.

To illustrate the difference between the model fits, we visualise their likelihoods and priors. We select two models for comparison: the in-sample only model with $\text{SR}_0 = 0$, representing CZ's approach, and our model fitted to both in-sample and pre-publication data. We focus for now on the pre-publication Sharpe ratios, rather than the post-sample Sharpe ratios, as they offer a closer comparison to the issue CZ studied.

Figure 48 shows that the in-sample likelihoods are nearly identical. This highlights the difficulty of fitting with in-sample data alone; both models appear to fit the data well, even though we have assumed the $\text{SR}_0 = 0$ for one of the models, which as we will see later appears to be an incorrect assumption. For the pre-publication Sharpe ratios, however, the difference between the fits is marginally more apparent. Fitting to the in-sample and pre-publication data jointly yields a closer fit for these pre-publication ratios; when we look at the post-sample data this is even more apparent. To more clearly observe the effect of the assumption to fix $\text{SR}_0 = 0$, we should look at the priors, which we do next.

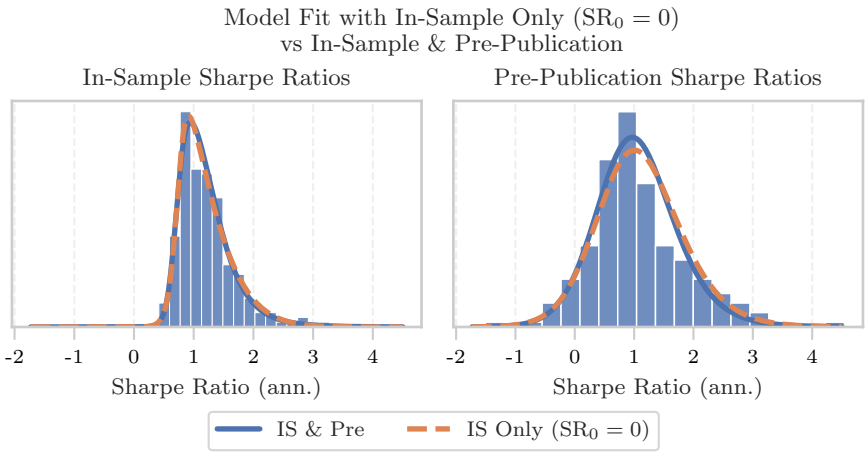


Figure 48: Model fit (marginal likelihoods) for in-sample only model (with $\text{SR}_0 = 0$) versus in- and pre-publication model.

Figure 49 plots the raw prior, $\pi(\text{SR})$, and the bias adjusted prior, $\pi_B(\text{SR})$, for both models (see Appendix A.2). Since the in-sample only model fixes $\text{SR}_0 = 0$, its unbiased prior is centred at zero and is relatively wide to accommodate the observed in-sample Sharpe ratios. The prior for our jointly fitted model, in comparison, is considerably more optimistic. The bias adjusted priors for

both models are, like the likelihoods, very similar, though the jointly fitted prior is slightly more peaked.

This result may seem slightly odd. It is natural that the prior for the Sharpe ratios conditional on acceptance has some reasonable positive expectation ($\mathbb{E}[\pi_B(\text{SR})] = 1.14$ in annualised terms for the in- and pre-publication fitted model). But why is the expectation of the unconditional prior ($\mathbb{E}[\pi(\text{SR})] = 0.84$) quite high? The nuance lies in that this is not the completely unconditional prior of any possible strategy, but rather it is only unconditional of the modelled selection effect. More explicitly, it is still conditional on the initial selection that the researcher implicitly performs from their own economic priors on what they believe could work, known stylised facts, portfolio construction, etc. Naturally this leads to the researcher filtering out many obviously poor strategies, resulting in an overall positive distribution of possible strategies. We also note that the average out-of-sample Sharpe ratio, which we note is subject to not insignificant alpha decay, is 0.62.

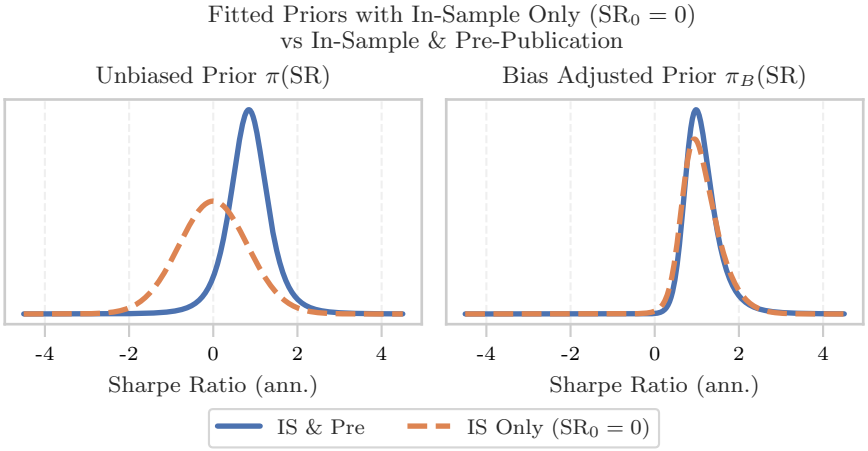


Figure 49: Unbiased and bias adjusted priors for in-sample only model (with $\text{SR}_0 = 0$) versus in- and pre-publication model.

How do these differences affect inference? Figure 50 plots the percentage change from the in-sample Sharpe ratio $\widehat{\text{SR}}_{\text{IS}}$ to the posterior mean $\mathbb{E}[\text{SR} \mid \widehat{\text{SR}}_{\text{IS}}]$: $\mathbb{E}[\text{SR} \mid \widehat{\text{SR}}_{\text{IS},i}]/\widehat{\text{SR}}_{\text{IS},i} - 1$. For the model where SR_0 is fixed at zero, all posterior means are shrunk towards the prior mean, resulting in adjustments of -2% to -34%. In the jointly fitted model, by contrast, the more optimistic prior leads to most posteriors being shrunk, but a few (those with low in-sample Sharpe ratios) are increased, with values ranging from +30% to -32%. This makes clear that the calculated adjustments strongly depend on the fitted prior, and the estimated adjustment per predictor is a combination of selection bias inferred from the fitting procedure, and standard Bayesian shrinkage.

The two models yield average adjustments, $\frac{1}{N}\sum_{i=1}^{N} \mathbb{E}[\text{SR} \mid \widehat{\text{SR}}_{\text{IS},i}]/\widehat{\text{SR}}_{\text{IS},i} - 1$, of -7% and -4% respectively. For comparison, [CZ20, fig. A.5] reports a bias adjustment of -10% for their sample; but as noted, differences in the models and data treatment make this an inexact comparison.[10]

Given the noted importance of the prior mean for the estimated posterior means, and therefore also for estimates of the magnitude of publication bias, we believe it is better to allow the data to fully inform the prior, rather than fix parameters at arbitrary values. But with a sample size of just 200 values to fit the parameters, any particular calculated posteriors and adjustments should be taken with a pinch of salt. Some comfort can be taken following our findings in Appendix B.1 that jointly fitting the data improves parameter estimates, and this is further underscored by the similar values for SR_0 across all three unrestricted models.

Finally, we study the impact of alpha decay using two visualisations. The first, in Figure 51, plots the distributions of in-sample Sharpe ratios $\widehat{\text{SR}}_{\text{IS}}$, posterior means $\mathbb{E}[\text{SR} \mid \widehat{\text{SR}}_{\text{IS}}, \widehat{\text{SR}}_{\text{OOS}}]$, de-

[10] We note that these adjustments are much smaller than those we observed in the hedge fund dataset; however, the purpose of this section is to compare our approach against CZ, rather than hedge funds against anomalies.

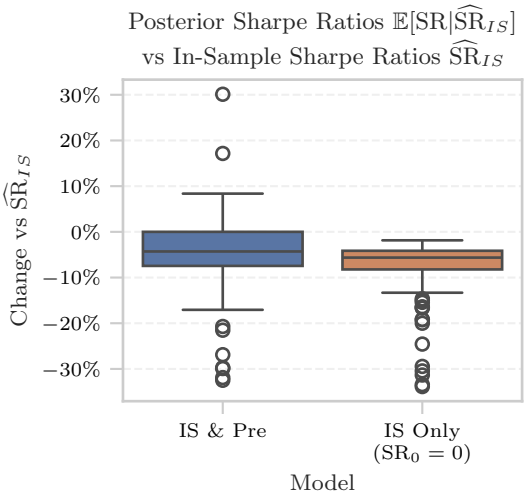


Figure 50: Percent change from the in-sample Sharpe ratio to the posterior mean $\mathbb{E}[\mathrm{SR} \mid \widehat{\mathrm{SR}}_{\mathrm{IS},i}]/\widehat{\mathrm{SR}}_{\mathrm{IS},i} - 1$ for in-sample only model (with $\mathrm{SR}_0 = 0$) versus in- and post-sample model.

cayed posterior means $\mathbb{E}[\mathrm{SR}_\delta \mid \widehat{\mathrm{SR}}_{\mathrm{IS}}, \widehat{\mathrm{SR}}_{\mathrm{OOS}}]$, and the out-of-sample Sharpe ratios $\widehat{\mathrm{SR}}_{\mathrm{OOS}}$. The out-of-sample data is taken to be either the pre-publication or post-sample data. To obtain the best possible estimate for the true Sharpe ratio SR, we fit the model and compute the posterior using the combined in- and out-of-sample data. The second visualisation, in Figure 52, shows a waterfall breakdown of the average adjustment into components: selection bias ($\Delta_{SB} = \mathbb{E}[\mathrm{SR} \mid \widehat{\mathrm{SR}}_{\mathrm{IS}}, \widehat{\mathrm{SR}}_{\mathrm{OOS}}]/\widehat{\mathrm{SR}}_{\mathrm{IS}} - 1$), alpha decay ($\Delta_{AD} = \mathbb{E}[\mathrm{SR}_\delta \mid \widehat{\mathrm{SR}}_{\mathrm{IS}}, \widehat{\mathrm{SR}}_{\mathrm{OOS}}]/\widehat{\mathrm{SR}}_{\mathrm{IS}} - 1 - \Delta_{SB}$) and noise ($\Delta_\epsilon = \widehat{\mathrm{SR}}_{\mathrm{OOS}}/\widehat{\mathrm{SR}}_{\mathrm{IS}} - 1 - \Delta_{SB} - \Delta_{AD}$).

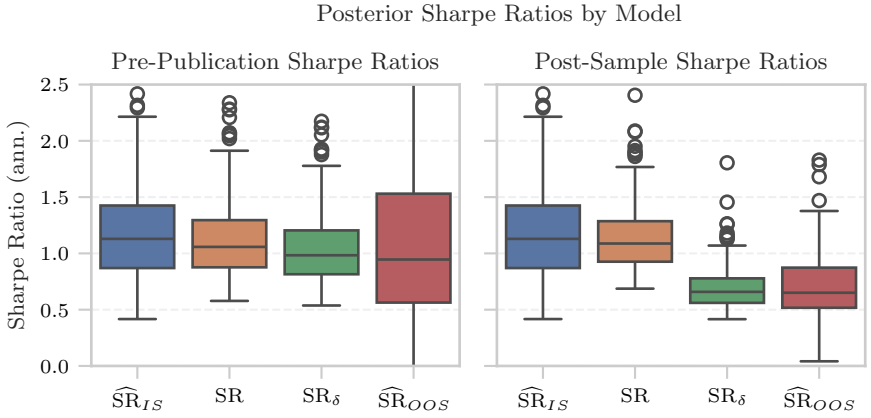


Figure 51: Distributions of in-sample Sharpe ratios $\widehat{\mathrm{SR}}_{\mathrm{IS}}$, posterior means $\mathbb{E}[\mathrm{SR} \mid \widehat{\mathrm{SR}}_{\mathrm{IS}}, \widehat{\mathrm{SR}}_{\mathrm{OOS}}]$, decayed posterior means $\mathbb{E}[\mathrm{SR}_\delta \mid \widehat{\mathrm{SR}}_{\mathrm{IS}}, \widehat{\mathrm{SR}}_{\mathrm{OOS}}]$, and the out-of-sample Sharpe ratios $\widehat{\mathrm{SR}}_{\mathrm{OOS}}$, for the pre-publication and post-sample models.

For both out-of-sample periods, the alpha-decayed Sharpe ratio $\mathbb{E}[\mathrm{SR}_\delta \mid \widehat{\mathrm{SR}}_{\mathrm{IS}}, \widehat{\mathrm{SR}}_{\mathrm{OOS}}]$ better captures the location of the out-of-sample distribution $\widehat{\mathrm{SR}}_{\mathrm{OOS}}$ than the raw Sharpe ratio $\mathbb{E}[\mathrm{SR} \mid \widehat{\mathrm{SR}}_{\mathrm{IS}}, \widehat{\mathrm{SR}}_{\mathrm{OOS}}]$. As expected, the distribution of posterior estimates is also much narrower than that of the out-of-sample Sharpe ratios. For the pre-publication period, the average adjustment decomposes into approximately 5% from selection bias (or Bayesian shrinkage) and 7% from

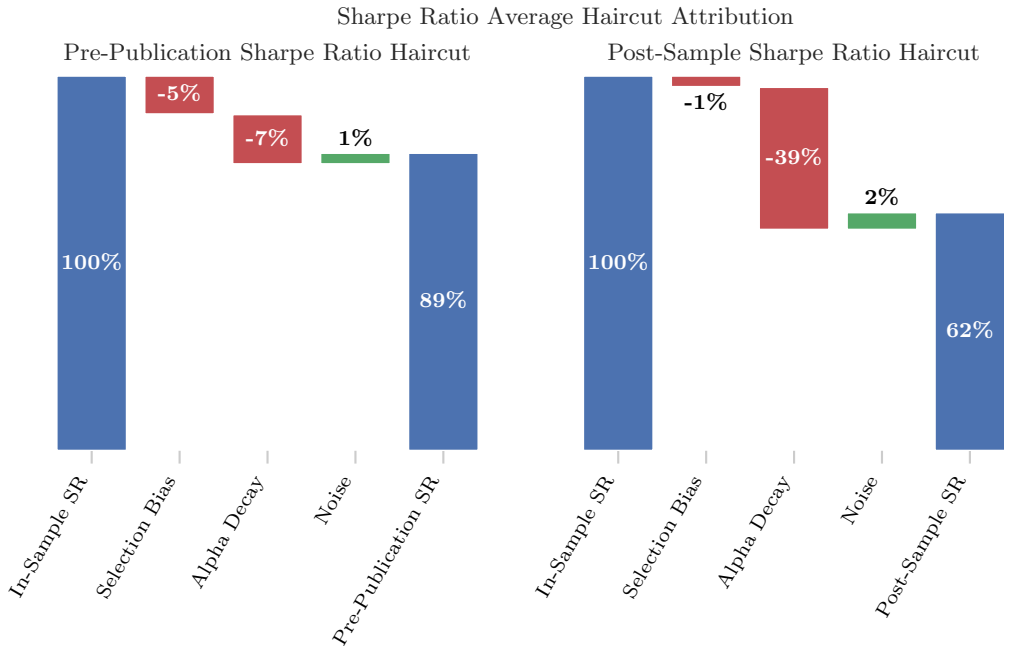


Figure 52: Attribution of average adjustments into selection bias ($\Delta_{SB} = \mathbb{E}[\text{SR} \mid \widehat{\text{SR}}_{\text{IS}}, \widehat{\text{SR}}_{\text{OOS}}]/\widehat{\text{SR}}_{\text{IS}} - 1$), alpha decay ($\Delta_{AD} = \mathbb{E}[\text{SR}_\delta \mid \widehat{\text{SR}}_{\text{IS}}, \widehat{\text{SR}}_{\text{OOS}}]/\widehat{\text{SR}}_{\text{IS}} - 1 - \Delta_{SB}$) and noise ($\Delta_\epsilon = \widehat{\text{SR}}_{\text{OOS}}/\widehat{\text{SR}}_{\text{IS}} - 1 - \Delta_{SB} - \Delta_{AD}$) for the pre-publication and post-sample models.

alpha decay; noise adds back 1%, yielding a final out-of-sample performance of 90% of the in-sample figure. For the post-sample data, we attribute only 1% to selection bias but 39% to alpha decay, with noise adding back 2% for a final performance of 62% of the in-sample result.

These figures however are highly conditional on the estimated parameters $\boldsymbol{\Theta}$. Fixing the prior's location SR_0 at a lower value, as CZ do for instance, would attribute more of the haircut to selection bias and less to alpha decay. A dataset with thousands of predictors might yield more definitive parameter estimates, but with only 200 samples, the haircut attributions presented here should be interpreted with caution.